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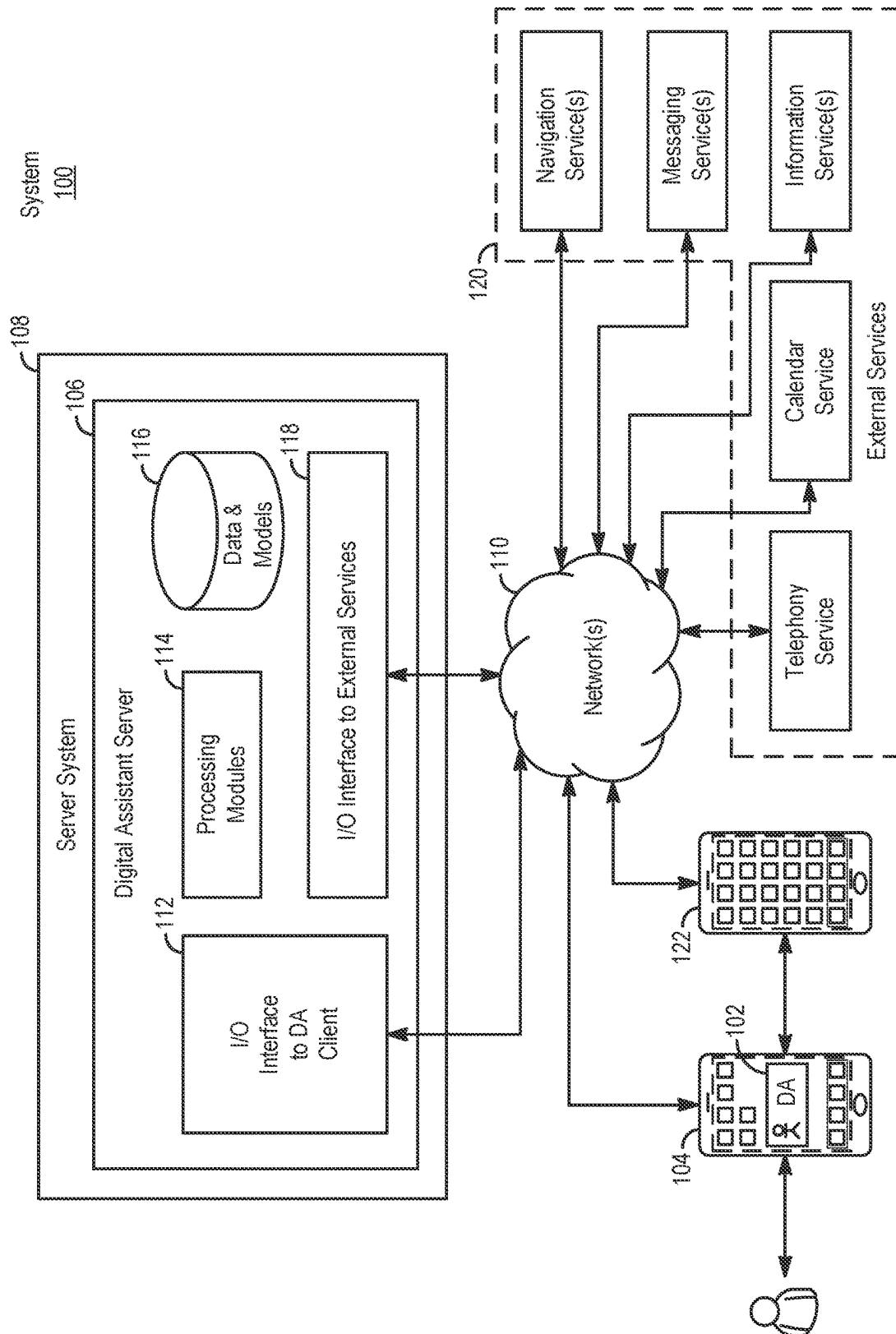


FIG. 1

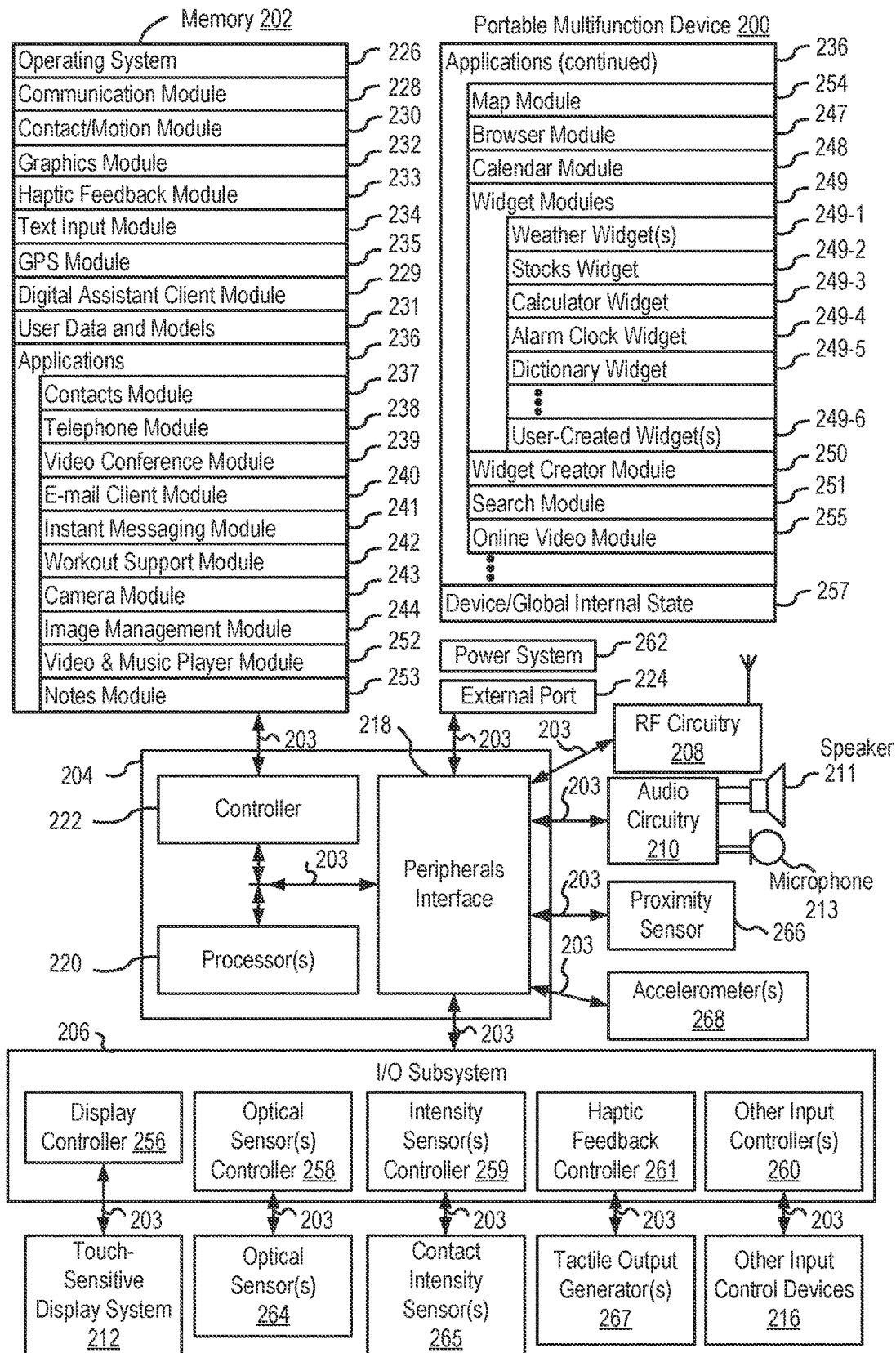
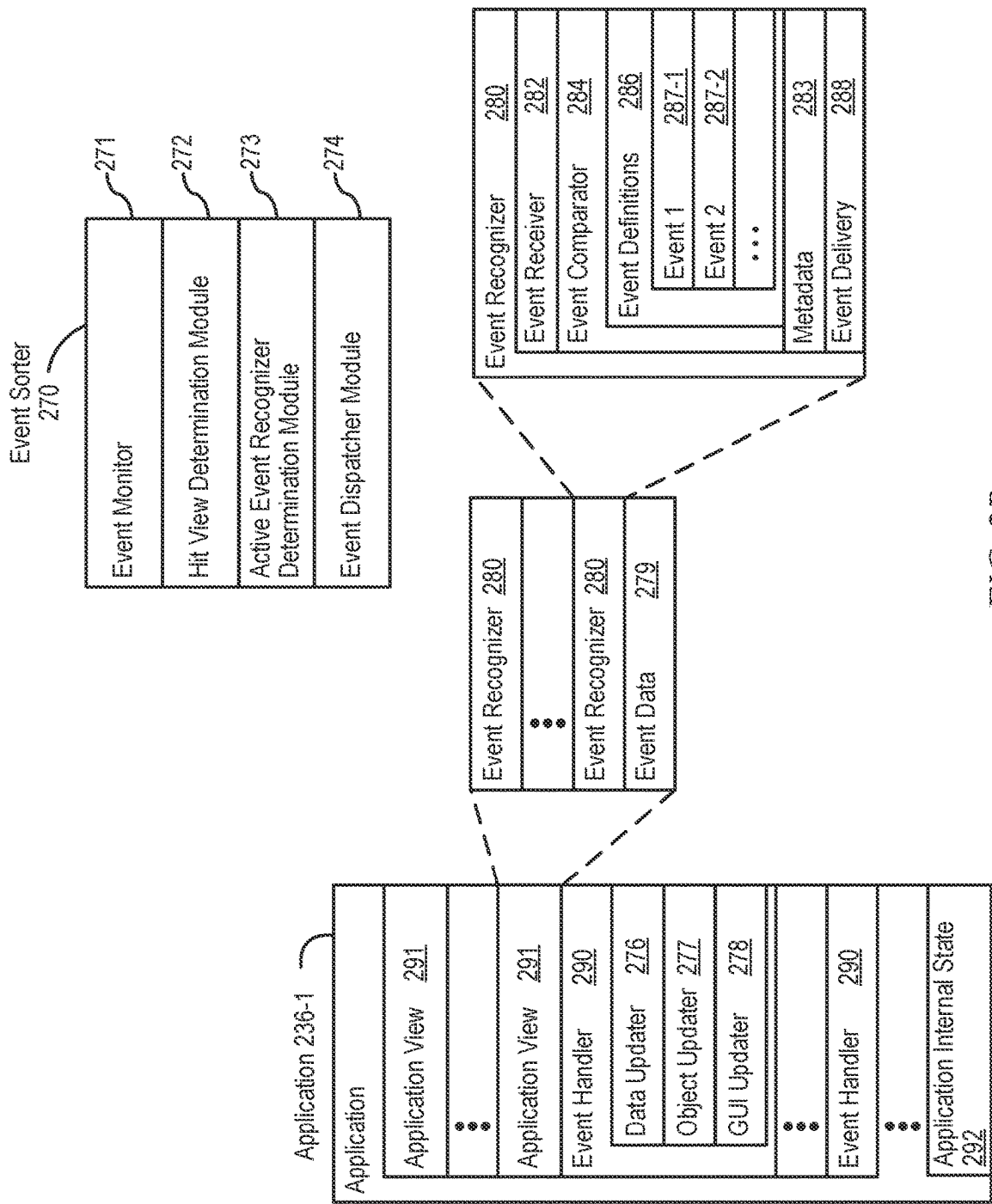


FIG. 2A



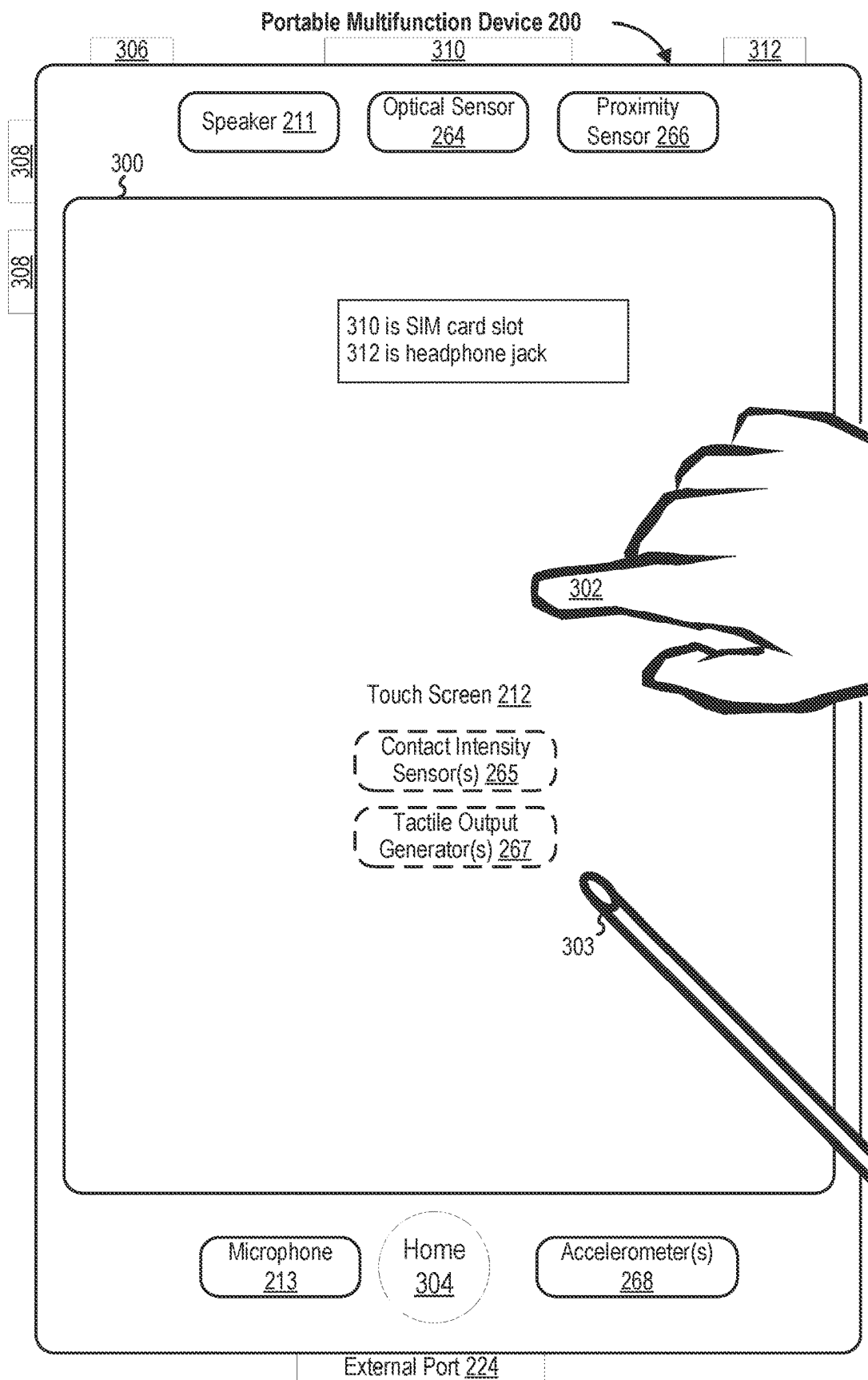


FIG. 3

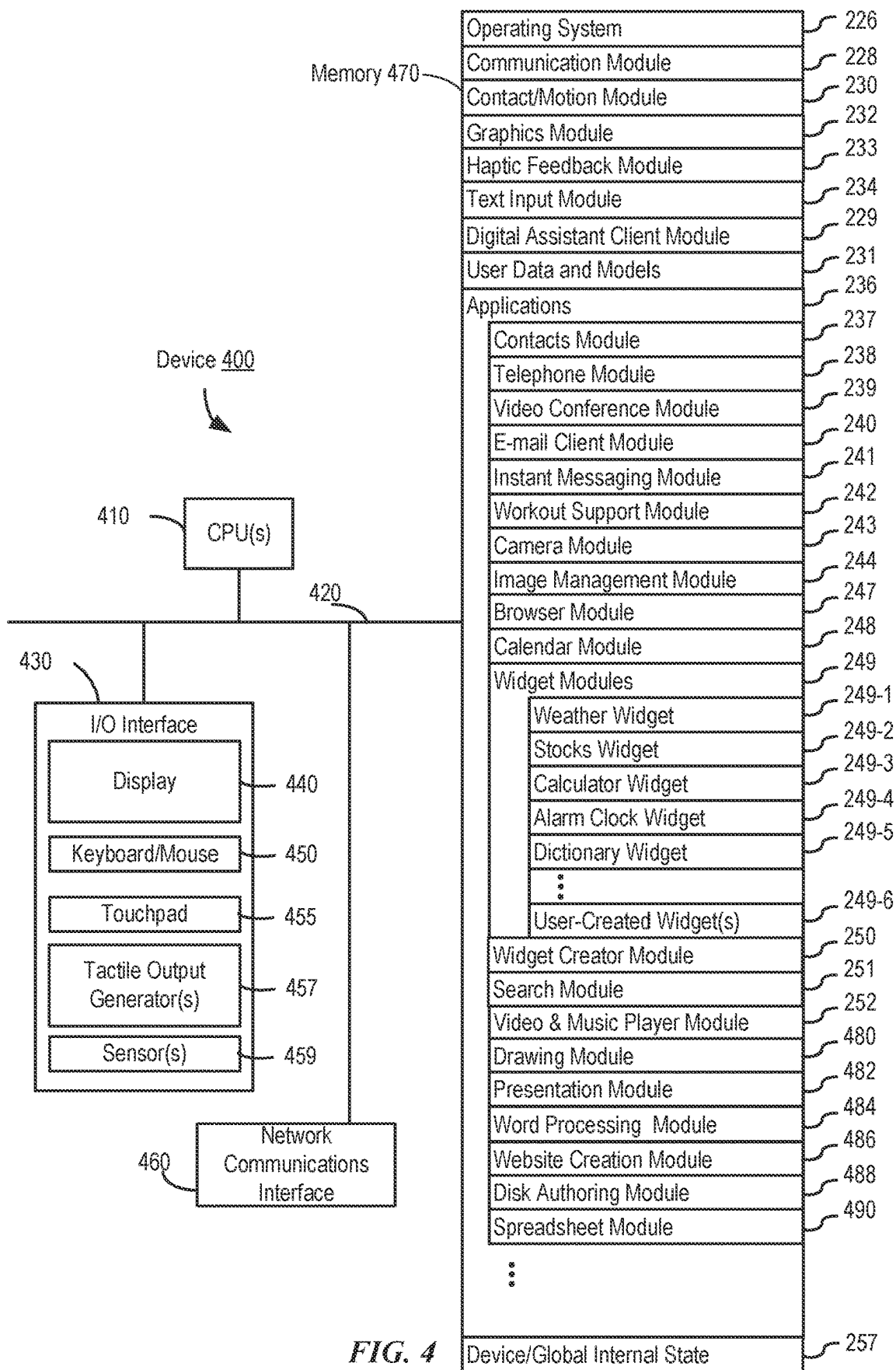


FIG. 4

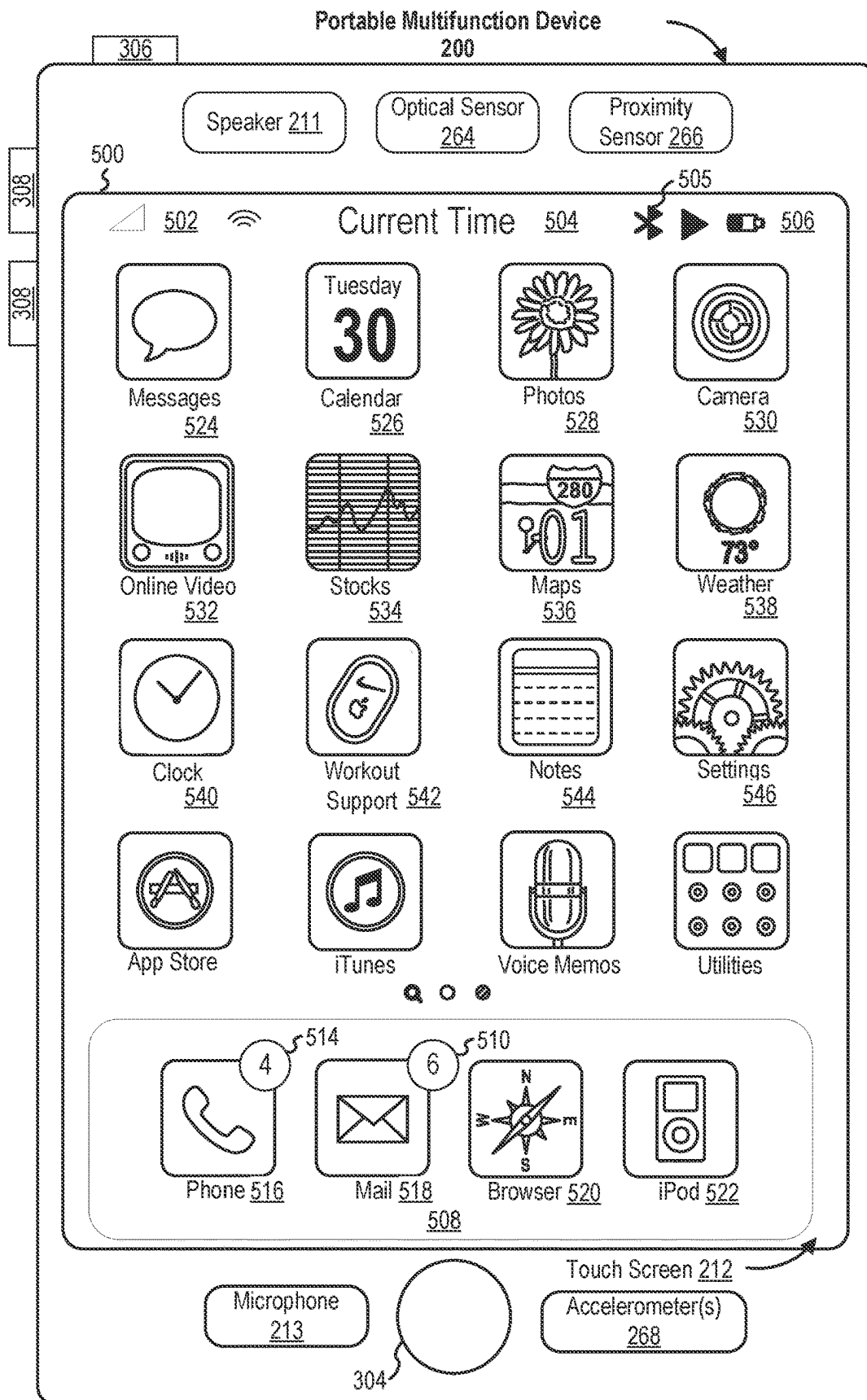


FIG. 5A

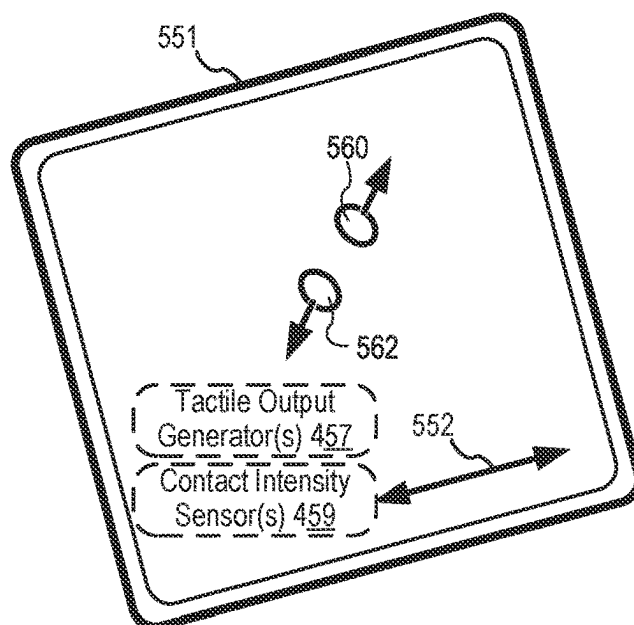
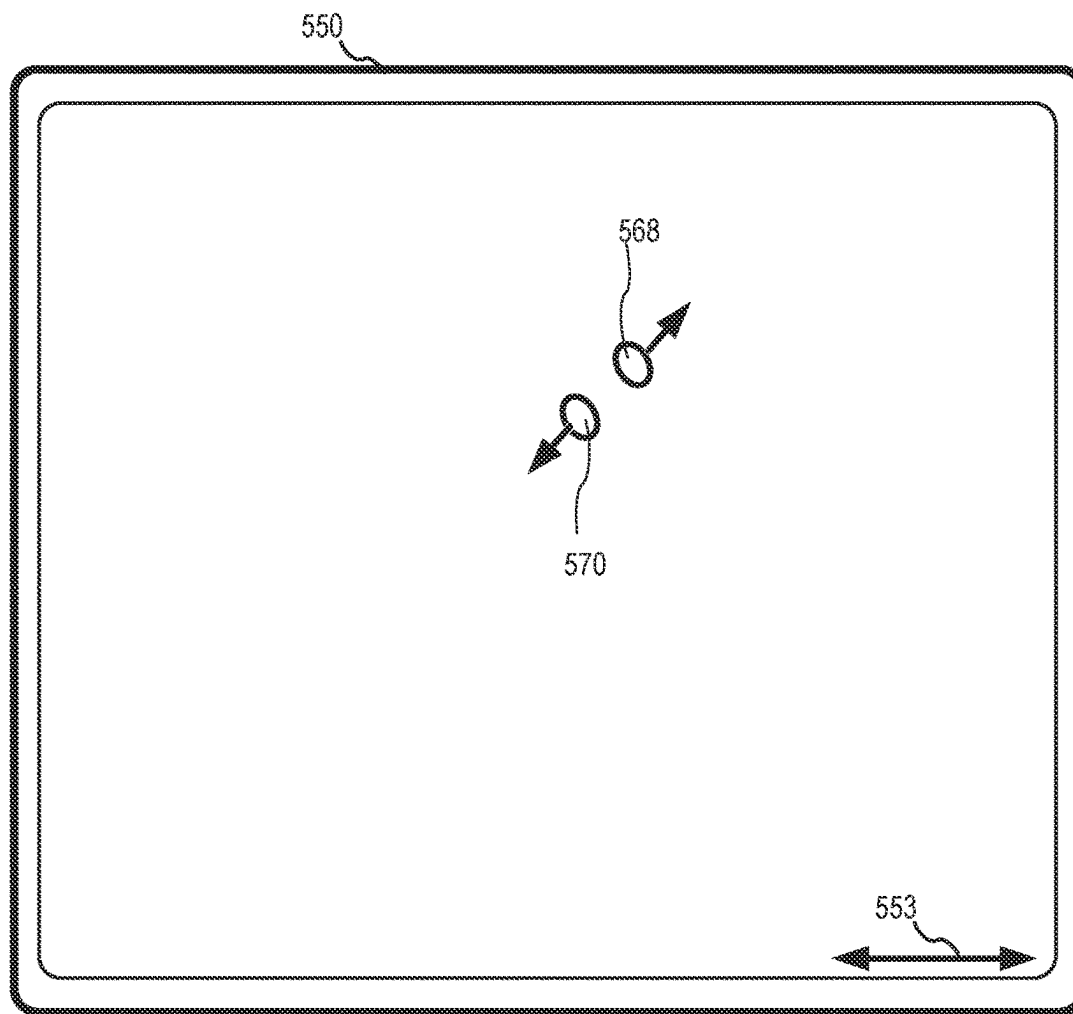
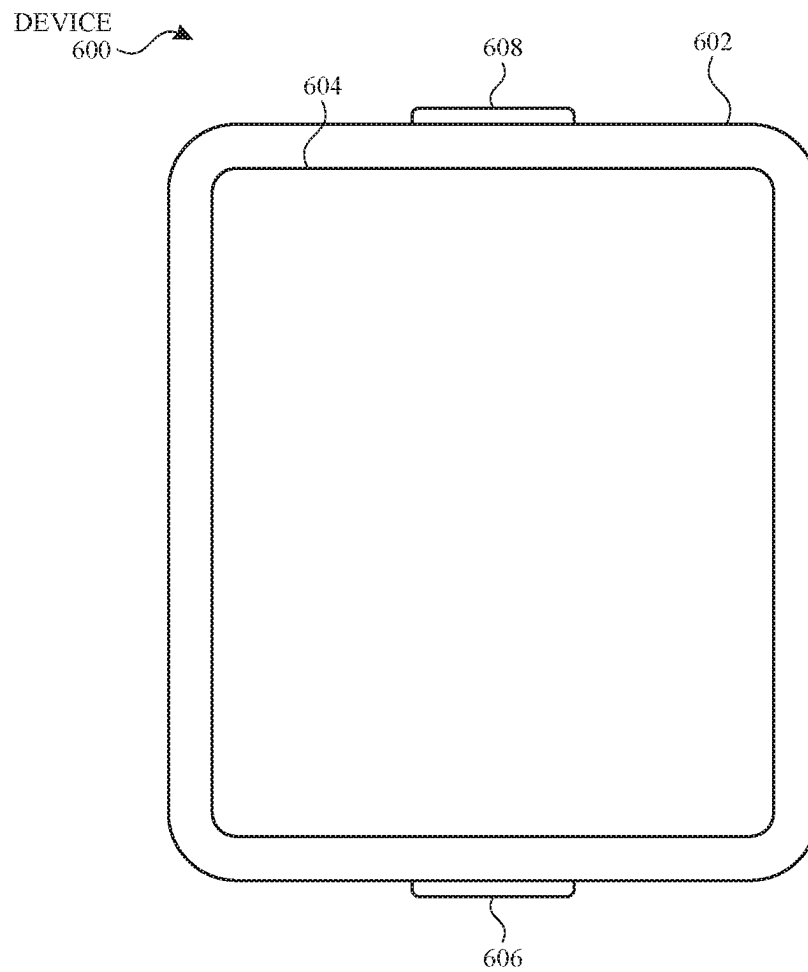


FIG. 5B

*FIG. 6A*

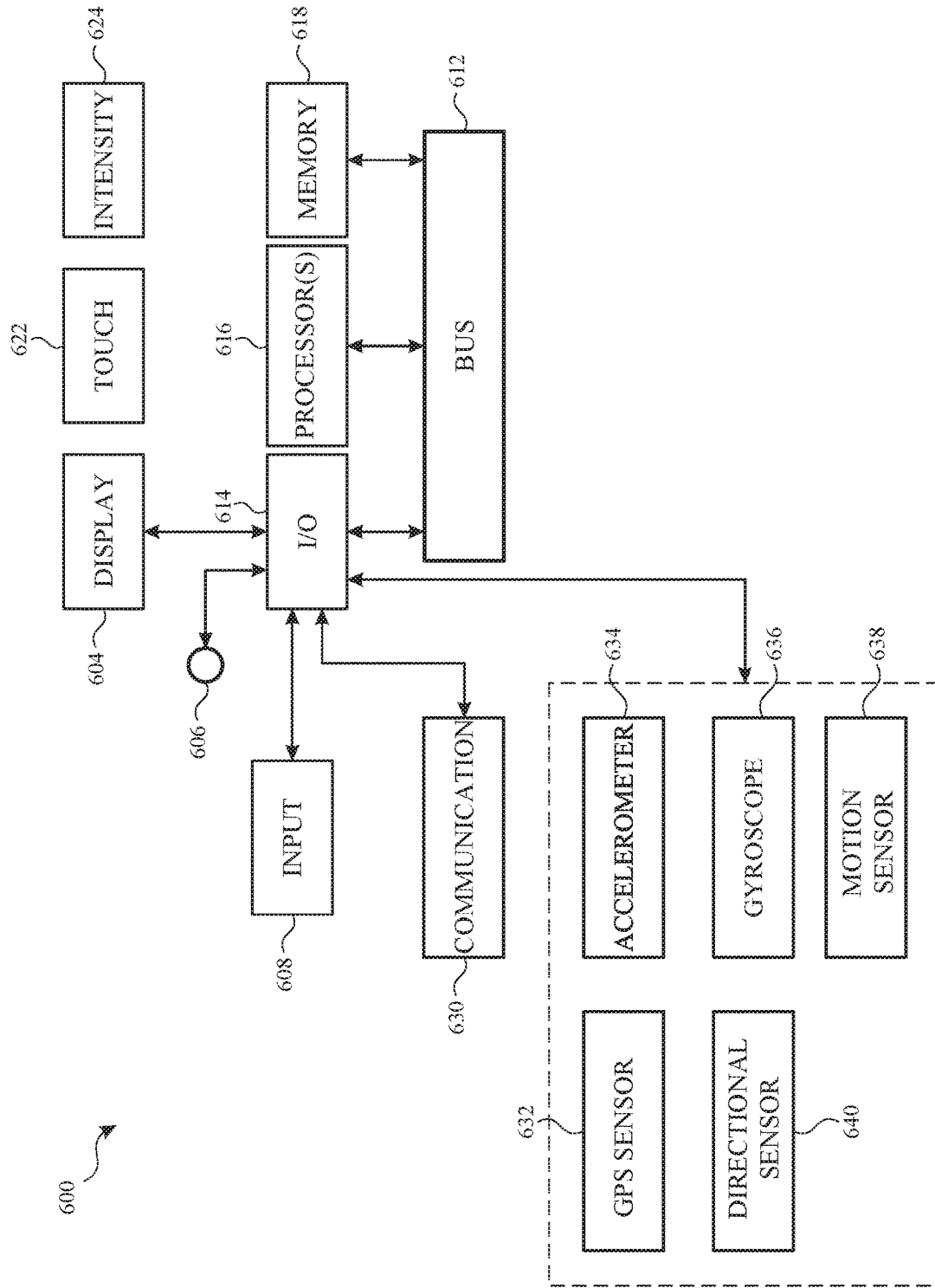


FIG. 6B

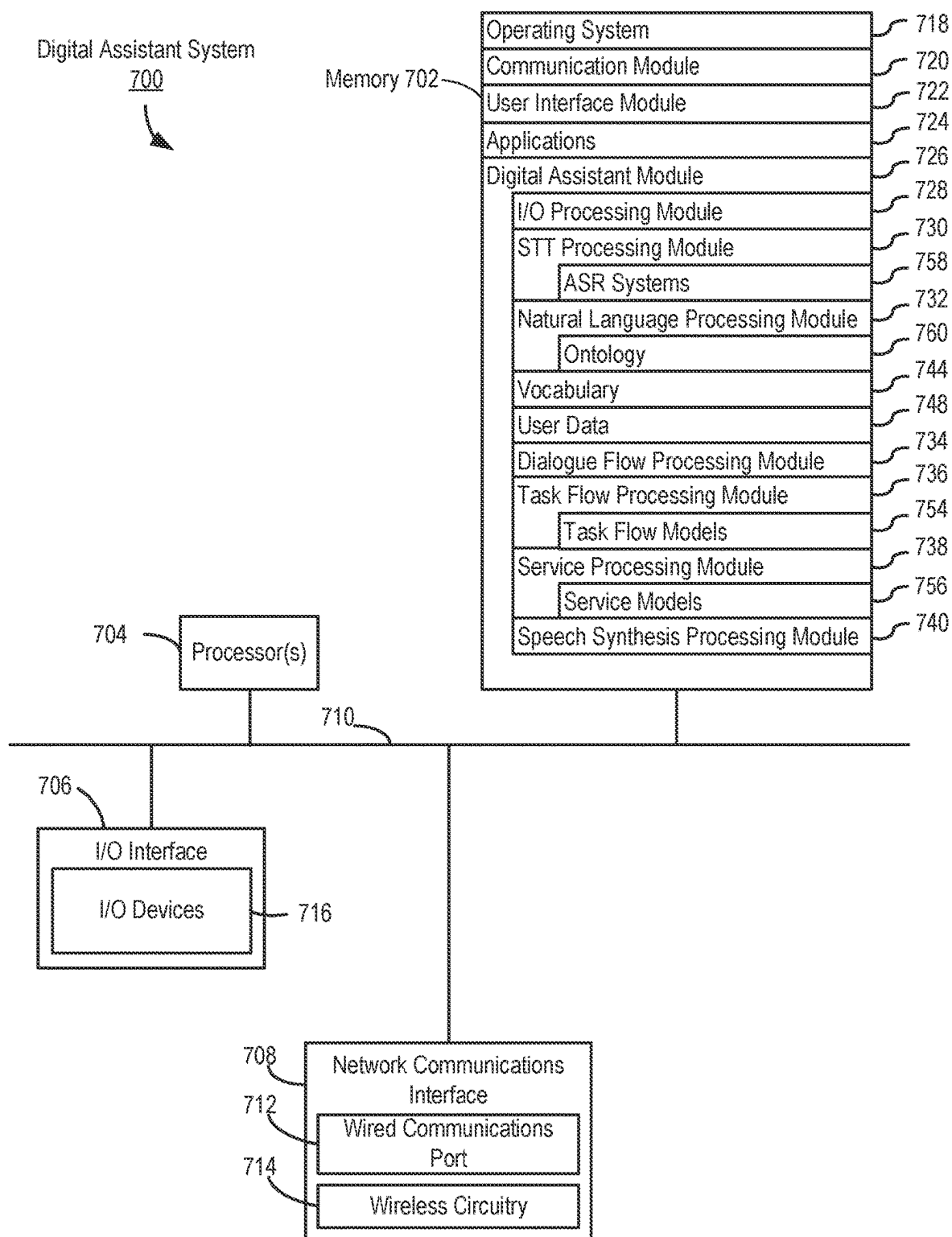


FIG. 7A

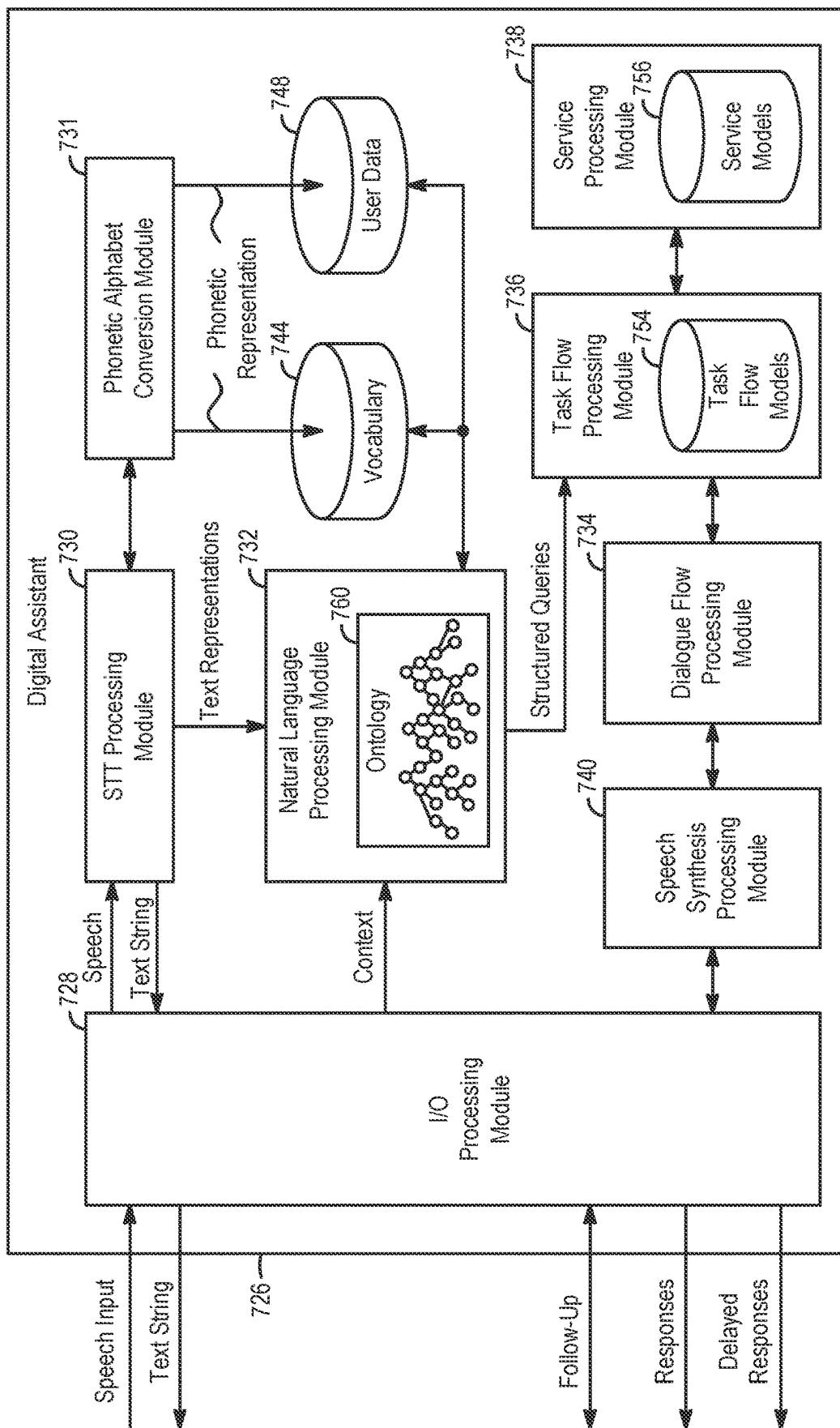


FIG. 7B

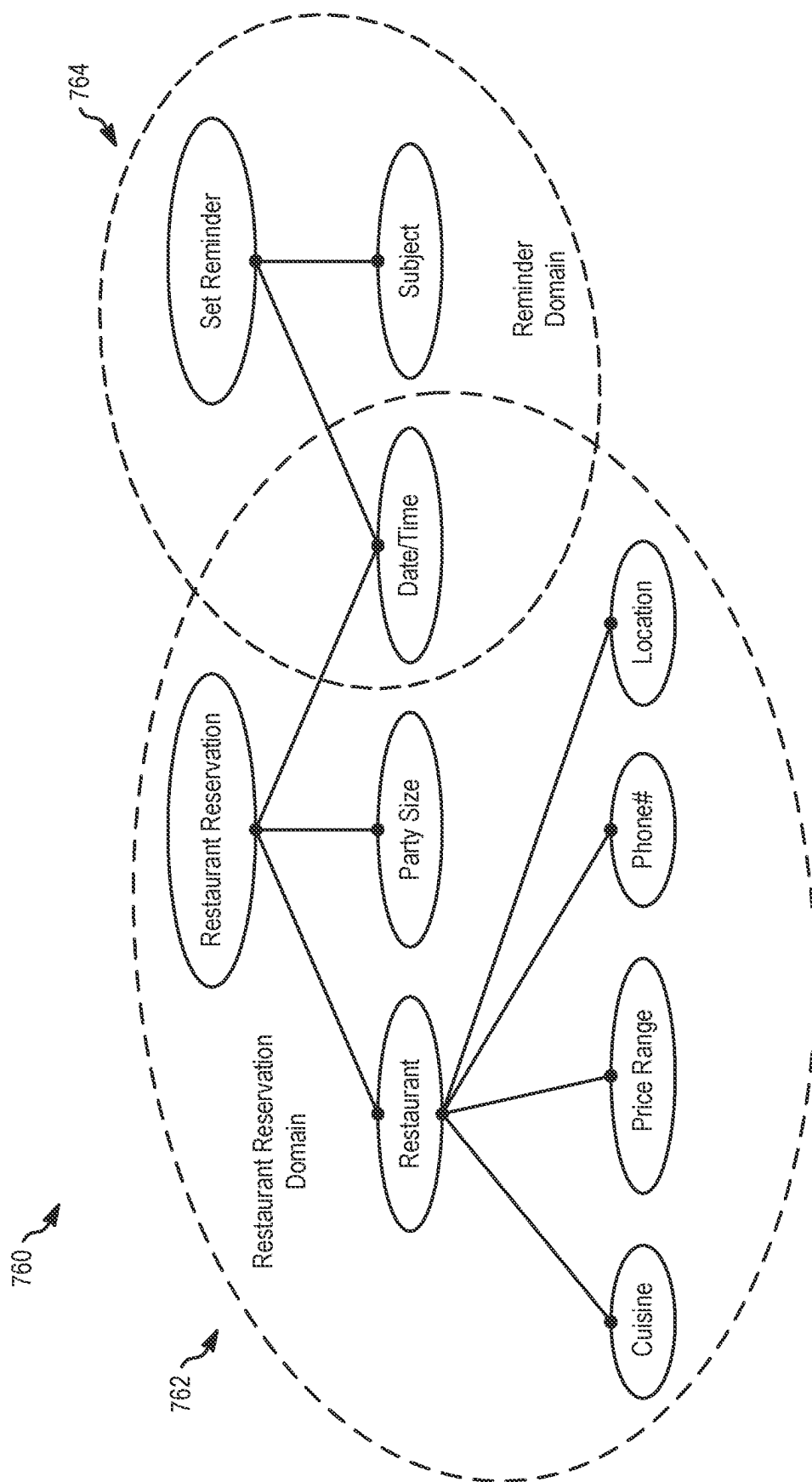


FIG. 7C

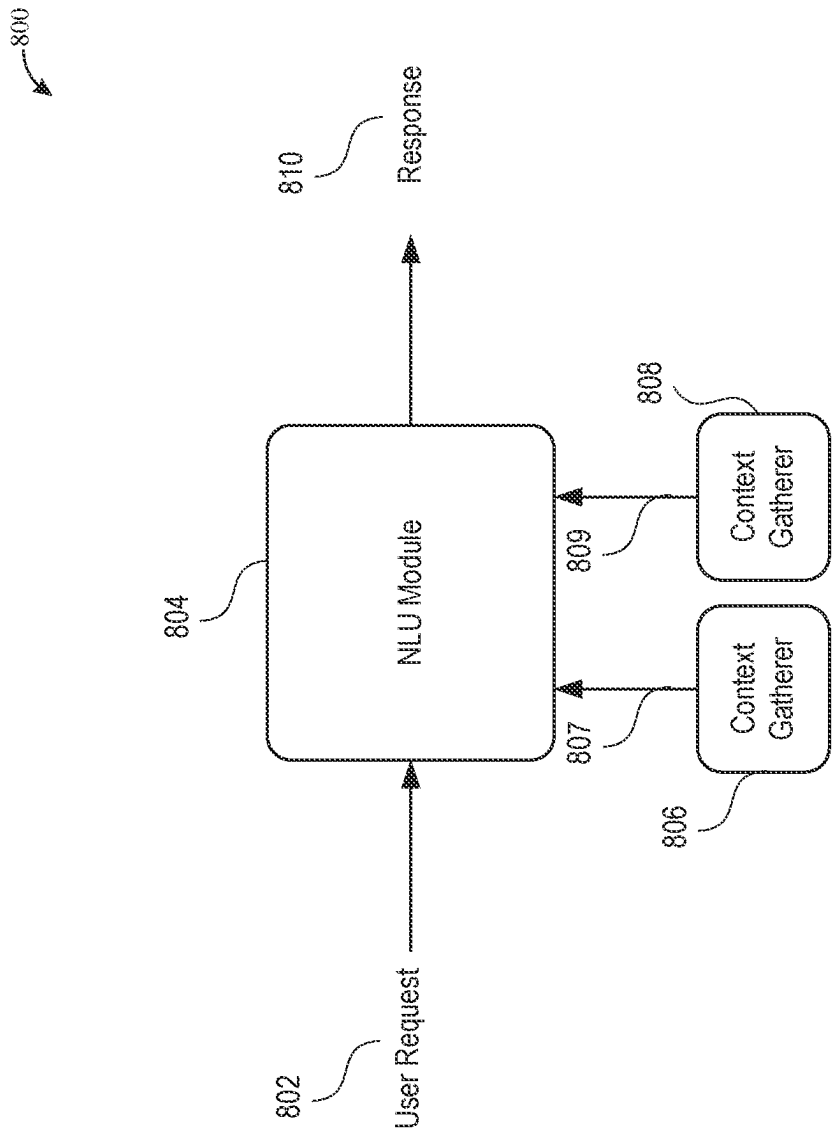


FIG. 8

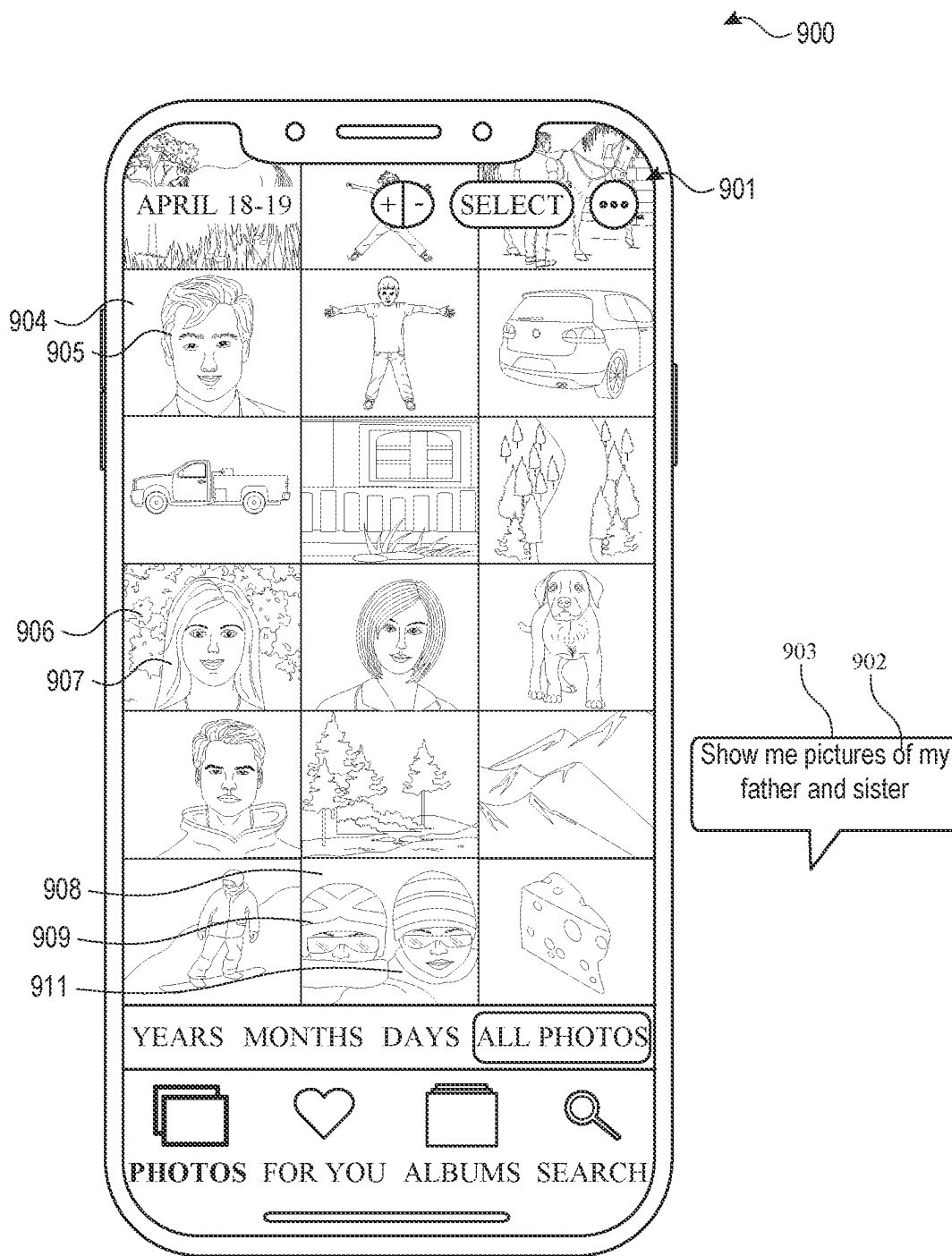


FIG. 9A

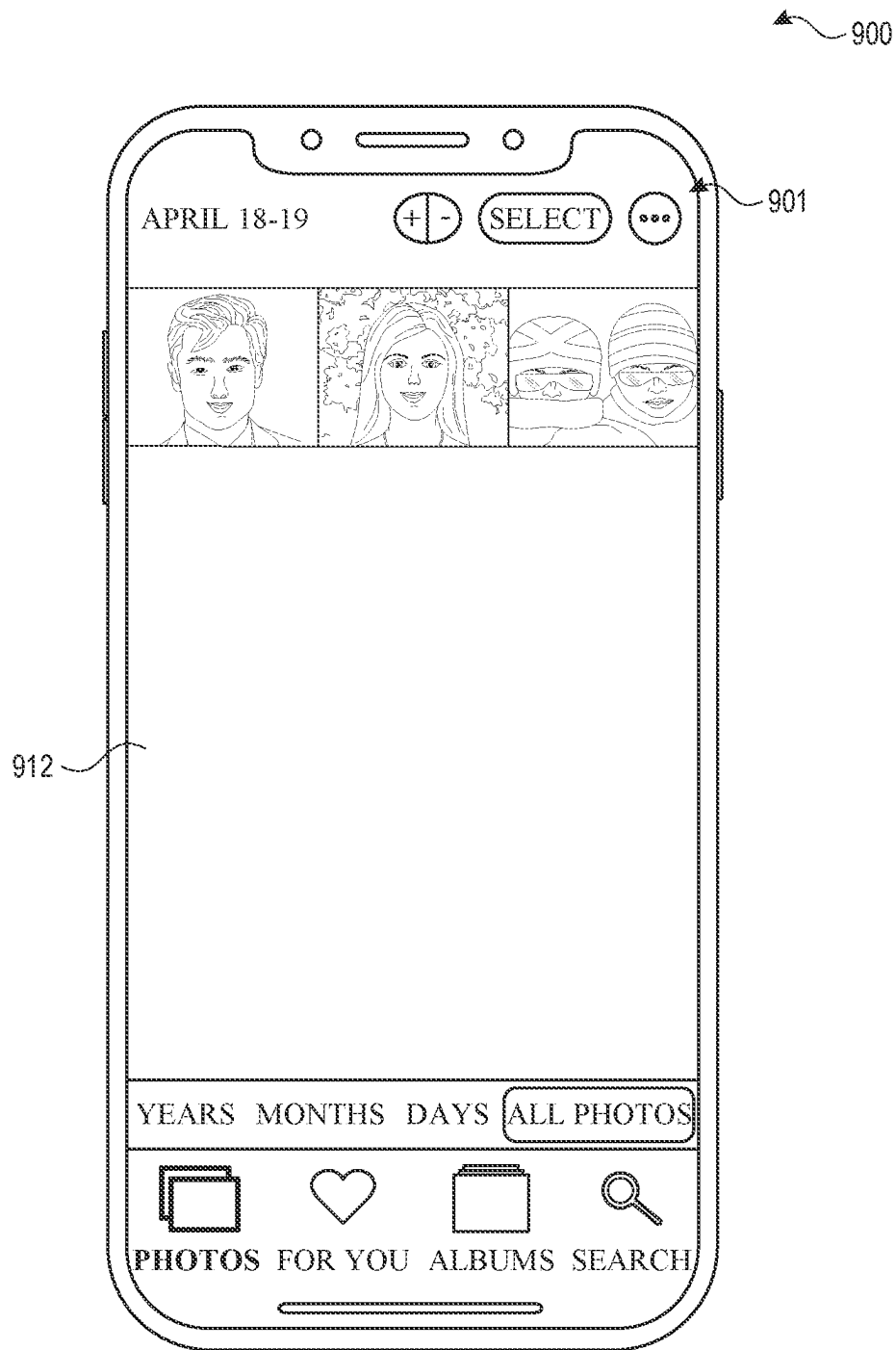


FIG. 9B

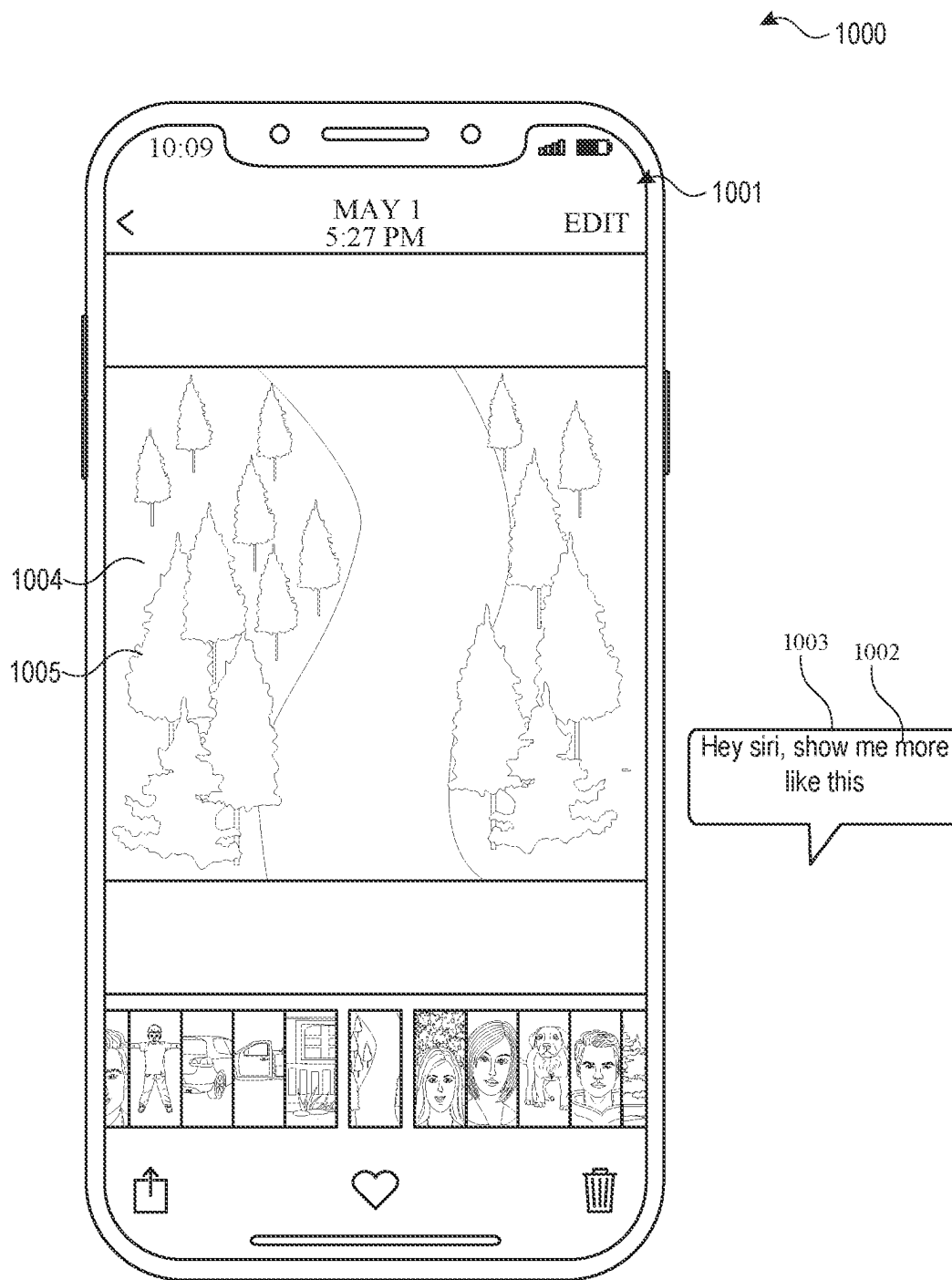


FIG. 10A

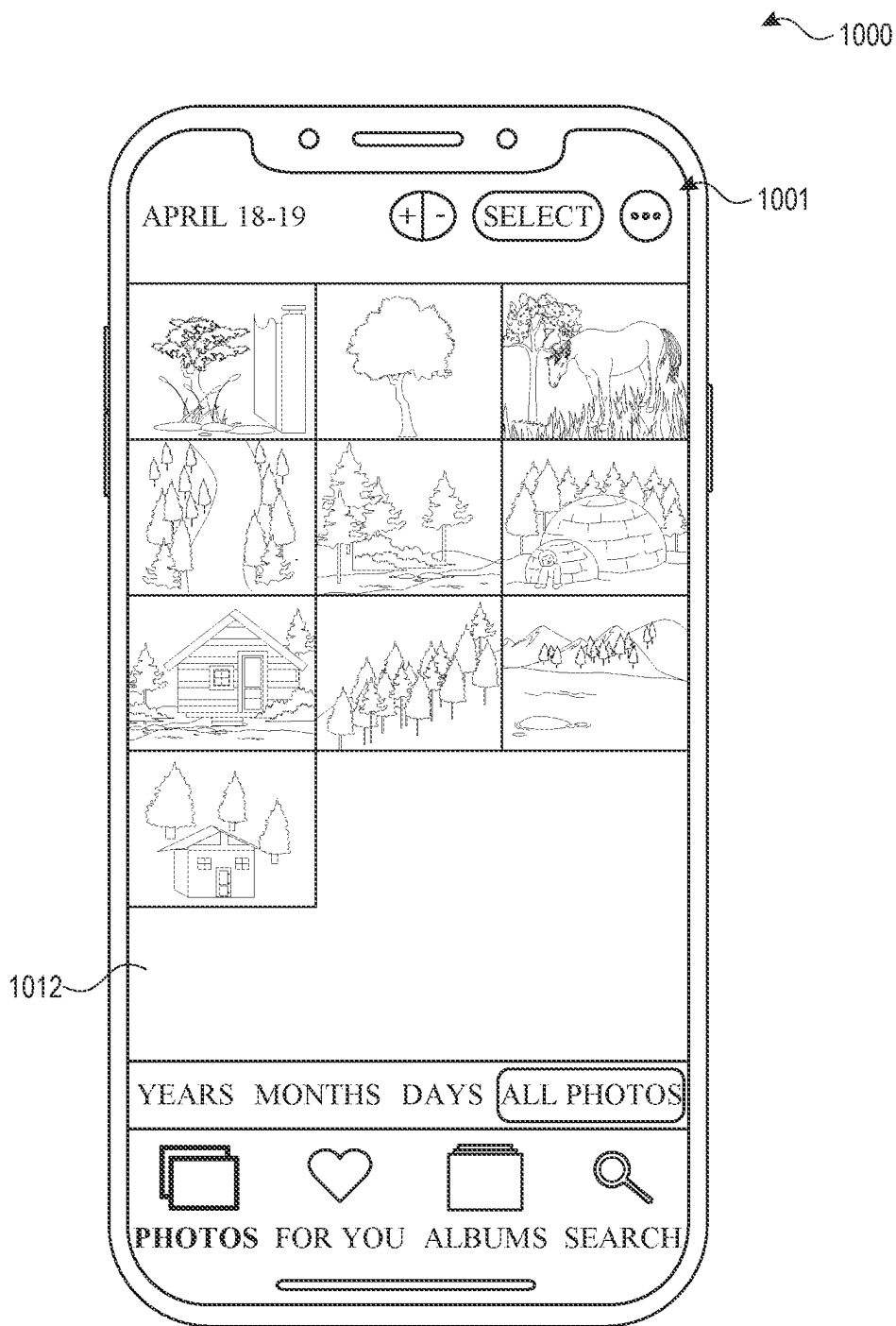


FIG. 10B

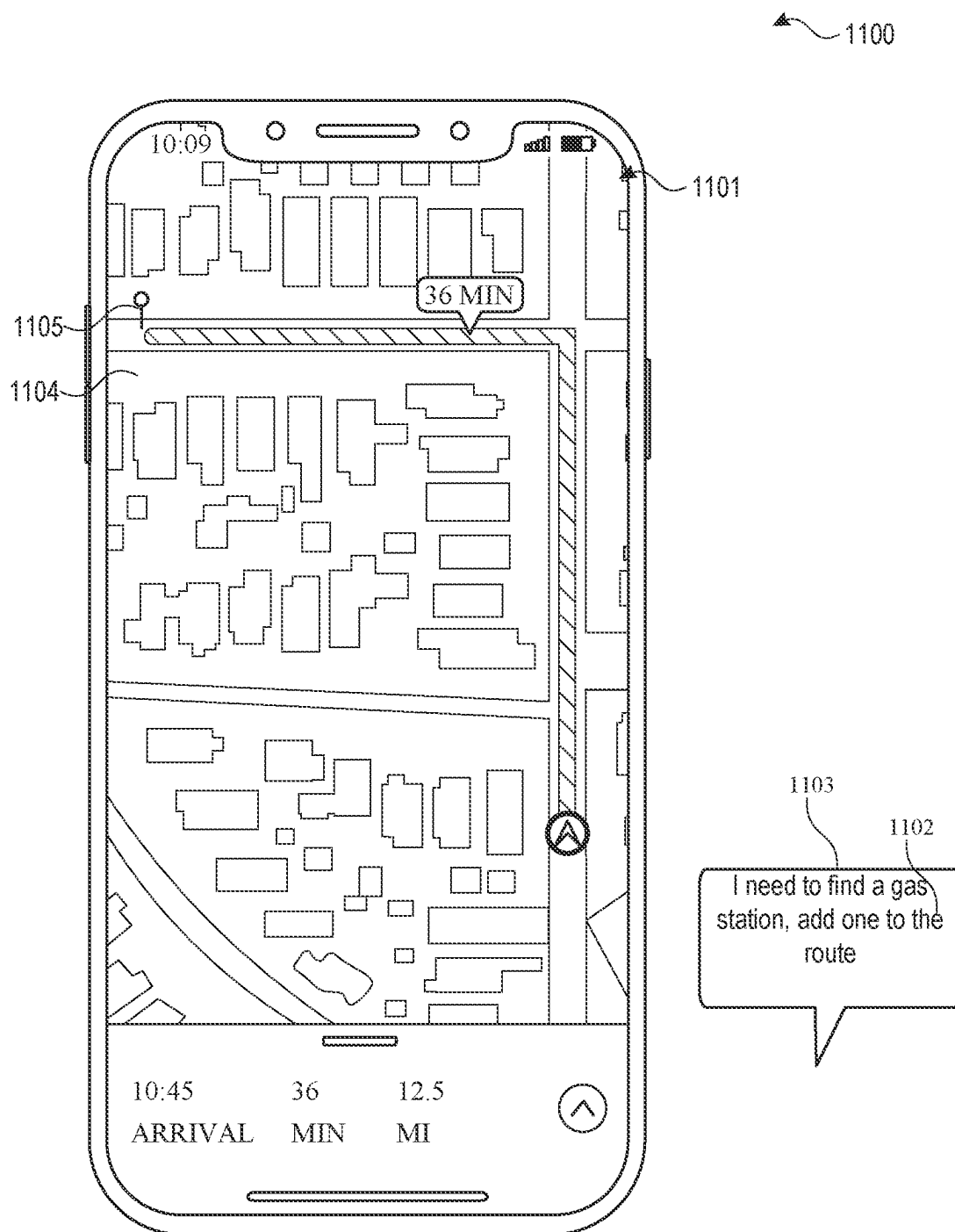


FIG. 11A

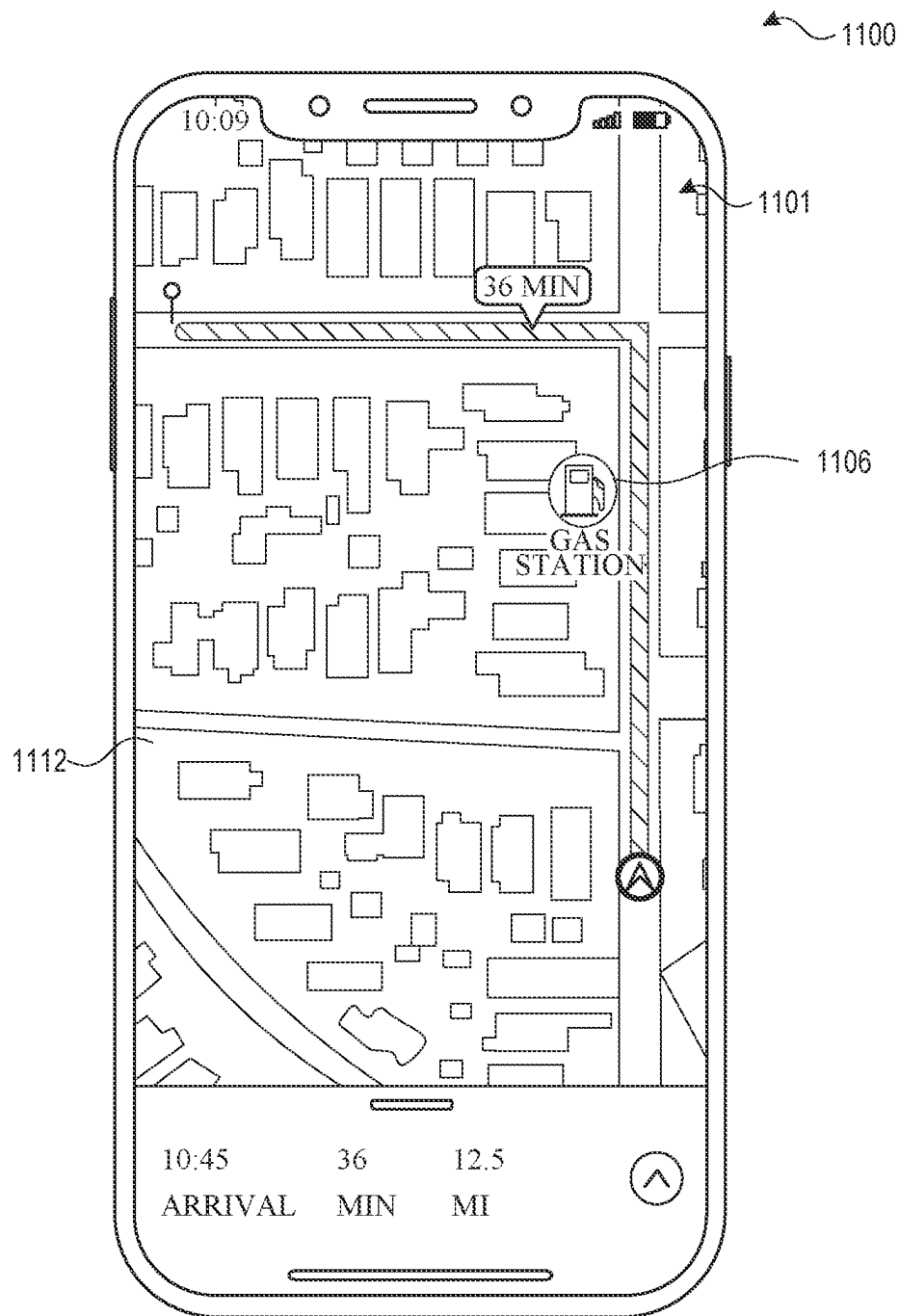
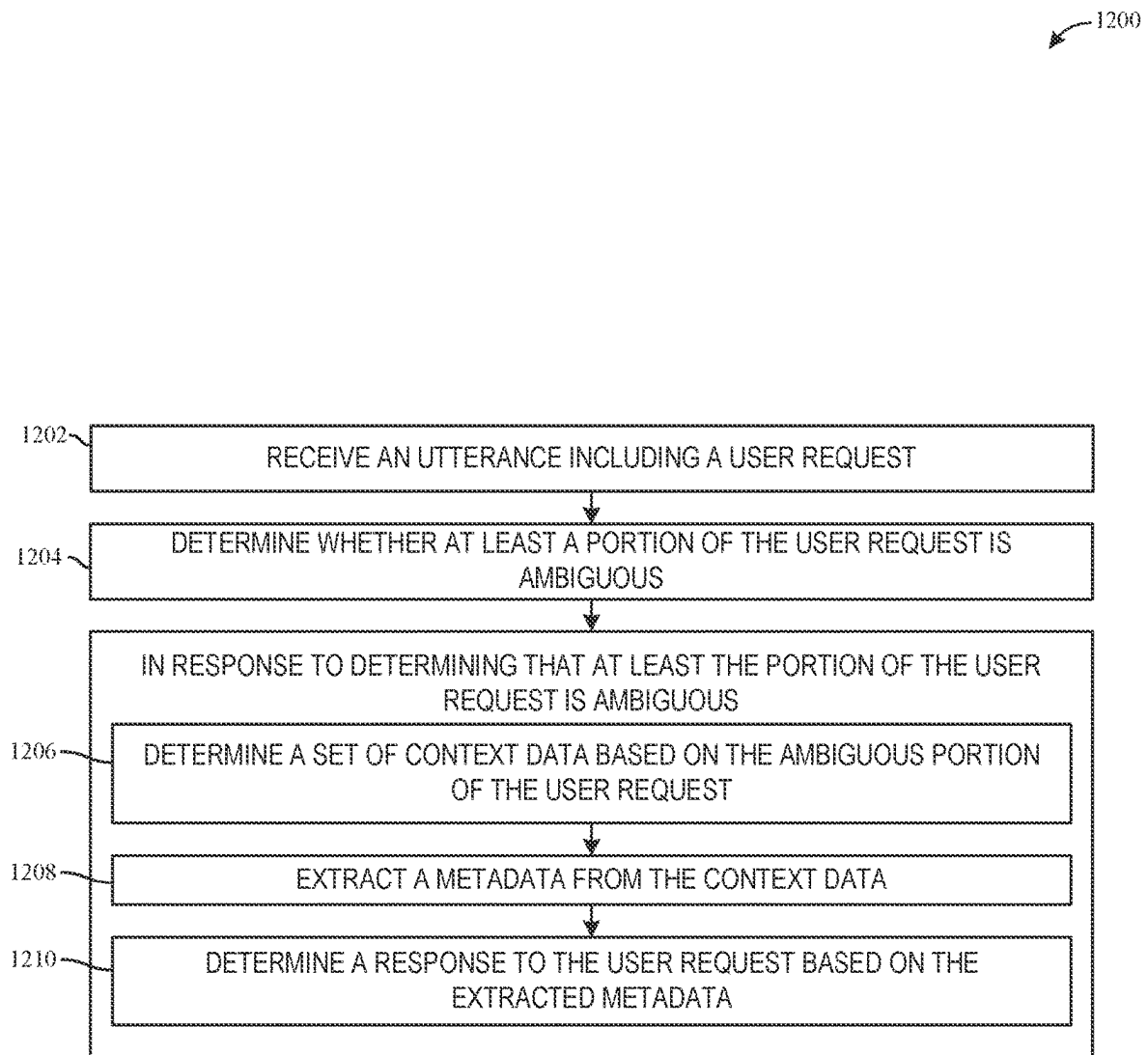


FIG. 11B

*FIG. 12*

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MULTIMODAL ASSISTANT UNDERSTANDING USING ON-SCREEN AND DEVICE CONTEXT

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Patent Application No. 63/230,564, entitled "MULTIMODAL ASSISTANT UNDERSTANDING USING ON-SCREEN AND DEVICE CONTEXT," filed on Aug. 6, 2021, the content of which is hereby incorporated by reference in its entirety.

FIELD

This relates generally to intelligent automated assistants and, more specifically, to improving assistant understanding based on screen and device context.

BACKGROUND

Intelligent automated assistants (or digital assistants) can provide a beneficial interface between human users and electronic devices. Such assistants can allow users to interact with devices or systems using natural language in spoken and/or text forms. For example, a user can provide a speech input containing a user request to a digital assistant operating on an electronic device. The digital assistant can interpret the user's intent from the speech input and operationalize the user's intent into tasks. The tasks can then be performed by executing one or more services of the electronic device, and a relevant output responsive to the user request can be returned to the user. In some cases, a user may provide a request that is ambiguous or references an object ambiguously that is difficult for the digital assistant to determine an appropriate response to. Accordingly, efficient and effective ways of resolving these ambiguities using various context data are desired.

SUMMARY

Example methods are disclosed herein. An example method includes, at an electronic device having one or more processors, and memory, receiving an utterance including a user request and determining whether at least a portion of the user request is ambiguous. If at least the portion of the user request is ambiguous then a set of context data based on the ambiguous portion of the user request is determined, metadata is extracted from the context data, and a response to the user request is determined based on the extracted metadata.

Example non-transitory computer-readable media are disclosed herein. An example non-transitory computer-readable storage medium stores one or more programs. The one or more programs comprise instructions, which when executed by one or more processors of an electronic device, cause the electronic device to receive an utterance including a user request and determine whether at least a portion of the user request is ambiguous. If at least the portion of the user request is ambiguous then a set of context data based on the ambiguous portion of the user request is determined, metadata is extracted from the context data, and a response to the user request is determined based on the extracted metadata.

Example electronic devices are disclosed herein. An example electronic device comprises one or more processors; a memory; and one or more programs, where the one or more programs are stored in the memory and configured

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to be executed by the one or more processors, the one or more programs including instructions for receiving an utterance including a user request and determining whether at least a portion of the user request is ambiguous. If at least the portion of the user request is ambiguous then a set of context data based on the ambiguous portion of the user request is determined, metadata is extracted from the context data, and a response to the user request is determined based on the extracted metadata.

An example electronic device comprises means for receiving an utterance including a user request and means for determining whether at least a portion of the user request is ambiguous. The electronic device further comprises, in response to determining at least the portion of the user request is ambiguous, means for determining a set of context data based on the ambiguous portion of the user request, means for extracting a metadata from the context data, and means for determining a response to the user request based on the extracted metadata.

Determining a set of context data based on the ambiguous portion of the user request, extracting a metadata from the context data, and determining a response to the user request based on the extracted metadata allows a digital assistant to efficiently resolve ambiguities and provide a response to the user. For example, retrieving context data and extracting metadata in this way allows the digital assistant to selectively pull context data as needed. Thus, this provides for more efficient user of the electronic device which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating a system and environment for implementing a digital assistant, according to various examples.

FIG. 2A is a block diagram illustrating a portable multifunction device implementing the client-side portion of a digital assistant, according to various examples.

FIG. 2B is a block diagram illustrating exemplary components for event handling, according to various examples.

FIG. 3 illustrates a portable multifunction device implementing the client-side portion of a digital assistant, according to various examples.

FIG. 4 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface, according to various examples.

FIG. 5A illustrates an exemplary user interface for a menu of applications on a portable multifunction device, according to various examples.

FIG. 5B illustrates an exemplary user interface for a multifunction device with a touch-sensitive surface that is separate from the display, according to various examples.

FIG. 6A illustrates a personal electronic device, according to various examples.

FIG. 6B is a block diagram illustrating a personal electronic device, according to various examples.

FIG. 7A is a block diagram illustrating a digital assistant system or a server portion thereof, according to various examples.

FIG. 7B illustrates the functions of the digital assistant shown in FIG. 7A, according to various examples.

FIG. 7C illustrates a portion of an ontology, according to various examples.

FIG. 8 is a block diagram illustrating a digital assistant for responding to a user request based on context data, according to various examples.

FIGS. 9A-9B illustrate exemplary electronic devices for responding to a user request based on context data, according to various examples.

FIGS. 10A-10B illustrate exemplary electronic devices for responding to a user request based on context data, according to various examples.

FIGS. 11A-11B illustrate exemplary electronic devices for responding to a user request based on context data, according to various examples.

FIG. 12 illustrates a method for responding to a user request based on context data, according to various examples.

DETAILED DESCRIPTION

In the following description of examples, reference is made to the accompanying drawings in which are shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the various examples.

Although the following description uses terms “first,” “second,” etc. to describe various elements, these elements should not be limited by the terms. These terms are only used to distinguish one element from another. For example, a first input could be termed a second input, and, similarly, a second input could be termed a first input, without departing from the scope of the various described examples. The first input and the second input are both inputs and, in some cases, are separate and different inputs.

The terminology used in the description of the various described examples herein is for the purpose of describing particular examples only and is not intended to be limiting. As used in the description of the various described examples and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” may be construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

1. System and Environment

FIG. 1 illustrates a block diagram of system 100 according to various examples. In some examples, system 100 implements a digital assistant. The terms “digital assistant,” “virtual assistant,” “intelligent automated assistant,” or “automatic digital assistant” refer to any information processing system that interprets natural language input in spoken and/or textual form to infer user intent, and performs actions based on the inferred user intent. For example, to act

on an inferred user intent, the system performs one or more of the following: identifying a task flow with steps and parameters designed to accomplish the inferred user intent, inputting specific requirements from the inferred user intent into the task flow; executing the task flow by invoking programs, methods, services, APIs, or the like; and generating output responses to the user in an audible (e.g., speech) and/or visual form.

Specifically, a digital assistant is capable of accepting a user request at least partially in the form of a natural language command, request, statement, narrative, and/or inquiry. Typically, the user request seeks either an informational answer or performance of a task by the digital assistant. A satisfactory response to the user request includes a provision of the requested informational answer, a performance of the requested task, or a combination of the two. For example, a user asks the digital assistant a question, such as “Where am I right now?” Based on the user’s current location, the digital assistant answers, “You are in Central Park near the west gate.” The user also requests the performance of a task, for example, “Please invite my friends to my girlfriend’s birthday party next week.” In response, the digital assistant can acknowledge the request by saying “Yes, right away,” and then send a suitable calendar invite on behalf of the user to each of the user’s friends listed in the user’s electronic address book. During performance of a requested task, the digital assistant sometimes interacts with the user in a continuous dialogue involving multiple exchanges of information over an extended period of time. There are numerous other ways of interacting with a digital assistant to request information or performance of various tasks. In addition to providing verbal responses and taking programmed actions, the digital assistant also provides responses in other visual or audio forms, e.g., as text, alerts, music, videos, animations, etc.

As shown in FIG. 1, in some examples, a digital assistant is implemented according to a client-server model. The digital assistant includes client-side portion 102 (hereafter “DA client 102”) executed on user device 104 and server-side portion 106 (hereafter “DA server 106”) executed on server system 108. DA client 102 communicates with DA server 106 through one or more networks 110. DA client 102 provides client-side functionalities such as user-facing input and output processing and communication with DA server 106. DA server 106 provides server-side functionalities for any number of DA clients 102 each residing on a respective user device 104.

In some examples, DA server 106 includes client-facing I/O interface 112, one or more processing modules 114, data and models 116, and I/O interface to external services 118. The client-facing I/O interface 112 facilitates the client-facing input and output processing for DA server 106. One or more processing modules 114 utilize data and models 116 to process speech input and determine the user’s intent based on natural language input. Further, one or more processing modules 114 perform task execution based on inferred user intent. In some examples, DA server 106 communicates with external services 120 through network(s) 110 for task completion or information acquisition. I/O interface to external services 118 facilitates such communications.

User device 104 can be any suitable electronic device. In some examples, user device 104 is a portable multifunctional device (e.g., device 200, described below with reference to FIG. 2A), a multifunctional device (e.g., device 400, described below with reference to FIG. 4), or a personal electronic device (e.g., device 600, described below with reference to FIGS. 6A-B). A portable multifunctional device

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is, for example, a mobile telephone that also contains other functions, such as PDA and/or music player functions. Specific examples of portable multifunction devices include the Apple Watch®, iPhone®, iPod Touch®, and iPad® devices from Apple Inc. of Cupertino, Calif. Other examples of portable multifunction devices include, without limitation, earphones/headphones, speakers, and laptop or tablet computers. Further, in some examples, user device **104** is a non-portable multifunctional device. In particular, user device **104** is a desktop computer, a game console, a speaker, a television, or a television set-top box. In some examples, user device **104** includes a touch-sensitive surface (e.g., touch screen displays and/or touchpads). Further, user device **104** optionally includes one or more other physical user-interface devices, such as a physical keyboard, a mouse, and/or a joystick. Various examples of electronic devices, such as multifunctional devices, are described below in greater detail.

Examples of communication network(s) **110** include local area networks (LAN) and wide area networks (WAN), e.g., the Internet. Communication network(s) **110** is implemented using any known network protocol, including various wired or wireless protocols, such as, for example, Ethernet, Universal Serial Bus (USB), FIREWIRE, Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Wi-Fi, voice over Internet Protocol (VoIP), Wi-MAX, or any other suitable communication protocol.

Server system **108** is implemented on one or more standalone data processing apparatus or a distributed network of computers. In some examples, server system **108** also employs various virtual devices and/or services of third-party service providers (e.g., third-party cloud service providers) to provide the underlying computing resources and/or infrastructure resources of server system **108**.

In some examples, user device **104** communicates with DA server **106** via second user device **122**. Second user device **122** is similar or identical to user device **104**. For example, second user device **122** is similar to devices **200**, **400**, or **600** described below with reference to FIGS. 2A, 4, and 6A-B. User device **104** is configured to communicatively couple to second user device **122** via a direct communication connection, such as Bluetooth, NFC, BTLE, or the like, or via a wired or wireless network, such as a local Wi-Fi network. In some examples, second user device **122** is configured to act as a proxy between user device **104** and DA server **106**. For example, DA client **102** of user device **104** is configured to transmit information (e.g., a user request received at user device **104**) to DA server **106** via second user device **122**. DA server **106** processes the information and returns relevant data (e.g., data content responsive to the user request) to user device **104** via second user device **122**.

In some examples, user device **104** is configured to communicate abbreviated requests for data to second user device **122** to reduce the amount of information transmitted from user device **104**. Second user device **122** is configured to determine supplemental information to add to the abbreviated request to generate a complete request to transmit to DA server **106**. This system architecture can advantageously allow user device **104** having limited communication capabilities and/or limited battery power (e.g., a watch or a similar compact electronic device) to access services provided by DA server **106** by using second user device **122**, having greater communication capabilities and/or battery power (e.g., a mobile phone, laptop computer, tablet com-

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puter, or the like), as a proxy to DA server **106**. While only two user devices **104** and **122** are shown in FIG. 1, it should be appreciated that system **100**, in some examples, includes any number and type of user devices configured in this proxy configuration to communicate with DA server system **106**.

Although the digital assistant shown in FIG. 1 includes both a client-side portion (e.g., DA client **102**) and a server-side portion (e.g., DA server **106**), in some examples, the functions of a digital assistant are implemented as a standalone application installed on a user device. In addition, the divisions of functionalities between the client and server portions of the digital assistant can vary in different implementations. For instance, in some examples, the DA client is a thin-client that provides only user-facing input and output processing functions, and delegates all other functionalities of the digital assistant to a backend server.

2. Electronic Devices

Attention is now directed toward embodiments of electronic devices for implementing the client-side portion of a digital assistant. FIG. 2A is a block diagram illustrating portable multifunction device **200** with touch-sensitive display system **212** in accordance with some embodiments. Touch-sensitive display **212** is sometimes called a “touch screen” for convenience and is sometimes known as or called a “touch-sensitive display system.” Device **200** includes memory **202** (which optionally includes one or more computer-readable storage mediums), memory controller **222**, one or more processing units (CPUs) **220**, peripherals interface **218**, RF circuitry **208**, audio circuitry **210**, speaker **211**, microphone **213**, input/output (I/O) subsystem **206**, other input control devices **216**, and external port **224**. Device **200** optionally includes one or more optical sensors **264**. Device **200** optionally includes one or more contact intensity sensors **265** for detecting intensity of contacts on device **200** (e.g., a touch-sensitive surface such as touch-sensitive display system **212** of device **200**). Device **200** optionally includes one or more tactile output generators **267** for generating tactile outputs on device **200** (e.g., generating tactile outputs on a touch-sensitive surface such as touch-sensitive display system **212** of device **200** or touchpad **455** of device **400**). These components optionally communicate over one or more communication buses or signal lines **203**.

As used in the specification and claims, the term “intensity” of a contact on a touch-sensitive surface refers to the force or pressure (force per unit area) of a contact (e.g., a finger contact) on the touch-sensitive surface, or to a substitute (proxy) for the force or pressure of a contact on the touch-sensitive surface. The intensity of a contact has a range of values that includes at least four distinct values and more typically includes hundreds of distinct values (e.g., at least 256). Intensity of a contact is, optionally, determined (or measured) using various approaches and various sensors or combinations of sensors. For example, one or more force sensors underneath or adjacent to the touch-sensitive surface are, optionally, used to measure force at various points on the touch-sensitive surface. In some implementations, force measurements from multiple force sensors are combined (e.g., a weighted average) to determine an estimated force of a contact. Similarly, a pressure-sensitive tip of a stylus is, optionally, used to determine a pressure of the stylus on the touch-sensitive surface. Alternatively, the size of the contact area detected on the touch-sensitive surface and/or changes thereto, the capacitance of the touch-sensitive surface proximate to the contact and/or changes thereto, and/or the resistance of the touch-sensitive surface proximate to the contact and/or changes thereto are, optionally, used as a

substitute for the force or pressure of the contact on the touch-sensitive surface. In some implementations, the substitute measurements for contact force or pressure are used directly to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is described in units corresponding to the substitute measurements). In some implementations, the substitute measurements for contact force or pressure are converted to an estimated force or pressure, and the estimated force or pressure is used to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is a pressure threshold measured in units of pressure). Using the intensity of a contact as an attribute of a user input allows for user access to additional device functionality that may otherwise not be accessible by the user on a reduced-size device with limited real estate for displaying affordances (e.g., on a touch-sensitive display) and/or receiving user input (e.g., via a touch-sensitive display, a touch-sensitive surface, or a physical/mechanical control such as a knob or a button).

As used in the specification and claims, the term “tactile output” refers to physical displacement of a device relative to a previous position of the device, physical displacement of a component (e.g., a touch-sensitive surface) of a device relative to another component (e.g., housing) of the device, or displacement of the component relative to a center of mass of the device that will be detected by a user with the user’s sense of touch. For example, in situations where the device or the component of the device is in contact with a surface of a user that is sensitive to touch (e.g., a finger, palm, or other part of a user’s hand), the tactile output generated by the physical displacement will be interpreted by the user as a tactile sensation corresponding to a perceived change in physical characteristics of the device or the component of the device. For example, movement of a touch-sensitive surface (e.g., a touch-sensitive display or trackpad) is, optionally, interpreted by the user as a “down click” or “up click” of a physical actuator button. In some cases, a user will feel a tactile sensation such as an “down click” or “up click” even when there is no movement of a physical actuator button associated with the touch-sensitive surface that is physically pressed (e.g., displaced) by the user’s movements. As another example, movement of the touch-sensitive surface is, optionally, interpreted or sensed by the user as “roughness” of the touch-sensitive surface, even when there is no change in smoothness of the touch-sensitive surface. While such interpretations of touch by a user will be subject to the individualized sensory perceptions of the user, there are many sensory perceptions of touch that are common to a large majority of users. Thus, when a tactile output is described as corresponding to a particular sensory perception of a user (e.g., an “up click,” a “down click,” “roughness”), unless otherwise stated, the generated tactile output corresponds to physical displacement of the device or a component thereof that will generate the described sensory perception for a typical (or average) user.

It should be appreciated that device 200 is only one example of a portable multifunction device, and that device 200 optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. 2A are implemented in hardware, software, or a combination of both hardware and software, including one or more signal processing and/or application-specific integrated circuits.

Memory 202 includes one or more computer-readable storage mediums. The computer-readable storage mediums are, for example, tangible and non-transitory. Memory 202

includes high-speed random access memory and also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Memory controller 222 controls access to memory 202 by other components of device 200.

In some examples, a non-transitory computer-readable storage medium of memory 202 is used to store instructions (e.g., for performing aspects of processes described below) for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. In other examples, the instructions (e.g., for performing aspects of the processes described below) are stored on a non-transitory computer-readable storage medium (not shown) of the server system 108 or are divided between the non-transitory computer-readable storage medium of memory 202 and the non-transitory computer-readable storage medium of server system 108.

Peripherals interface 218 is used to couple input and output peripherals of the device to CPU 220 and memory 202. The one or more processors 220 run or execute various software programs and/or sets of instructions stored in memory 202 to perform various functions for device 200 and to process data. In some embodiments, peripherals interface 218, CPU 220, and memory controller 222 are implemented on a single chip, such as chip 204. In some other embodiments, they are implemented on separate chips.

RF (radio frequency) circuitry 208 receives and sends RF signals, also called electromagnetic signals. RF circuitry 208 converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry 208 optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. RF circuitry 208 optionally communicates with networks, such as the Internet, also referred to as the World Wide Web (WWW), an intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other devices by wireless communication. The RF circuitry 208 optionally includes well-known circuitry for detecting near field communication (NFC) fields, such as by a short-range communication radio. The wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Evolution, Data-Only (EV-DO), HSPA, HSPA+, Dual-Cell HSPA (DC-HSPA), long term evolution (LTE), near field communication (NFC), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Bluetooth Low Energy (BTLE), Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, and/or IEEE 802.11ac), voice over Internet Protocol (VoIP), Wi-MAX, a protocol for e mail (e.g., Internet message access protocol (IMAP) and/or post office protocol (POP)), instant messaging (e.g., extensible messaging and presence protocol (XMPP), Session Initiation Protocol for Instant

Messaging and Presence Leveraging Extensions (SIMPLE), Instant Messaging and Presence Service (IMPS)), and/or Short Message Service (SMS), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

Audio circuitry **210**, speaker **211**, and microphone **213** provide an audio interface between a user and device **200**. Audio circuitry **210** receives audio data from peripherals interface **218**, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **211**. Speaker **211** converts the electrical signal to human-audible sound waves. Audio circuitry **210** also receives electrical signals converted by microphone **213** from sound waves. Audio circuitry **210** converts the electrical signal to audio data and transmits the audio data to peripherals interface **218** for processing. Audio data are retrieved from and/or transmitted to memory **202** and/or RF circuitry **208** by peripherals interface **218**. In some embodiments, audio circuitry **210** also includes a headset jack (e.g., **312**, FIG. 3). The headset jack provides an interface between audio circuitry **210** and removable audio input/output peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

I/O subsystem **206** couples input/output peripherals on device **200**, such as touch screen **212** and other input control devices **216**, to peripherals interface **218**. I/O subsystem **206** optionally includes display controller **256**, optical sensor controller **258**, intensity sensor controller **259**, haptic feedback controller **261**, and one or more input controllers **260** for other input or control devices. The one or more input controllers **260** receive/send electrical signals from/to other input control devices **216**. The other input control devices **216** optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, and so forth. In some alternate embodiments, input controller(s) **260** are, optionally, coupled to any (or none) of the following: a keyboard, an infrared port, a USB port, and a pointer device such as a mouse. The one or more buttons (e.g., **308**, FIG. 3) optionally include an up/down button for volume control of speaker **211** and/or microphone **213**. The one or more buttons optionally include a push button (e.g., **306**, FIG. 3).

A quick press of the push button disengages a lock of touch screen **212** or begin a process that uses gestures on the touch screen to unlock the device, as described in U.S. patent application Ser. No. 11/322,549, "Unlocking a Device by Performing Gestures on an Unlock Image," filed Dec. 23, 2005, U.S. Pat. No. 7,657,849, which is hereby incorporated by reference in its entirety. A longer press of the push button (e.g., **306**) turns power to device **200** on or off. The user is able to customize a functionality of one or more of the buttons. Touch screen **212** is used to implement virtual or soft buttons and one or more soft keyboards.

Touch-sensitive display **212** provides an input interface and an output interface between the device and a user. Display controller **256** receives and/or sends electrical signals from/to touch screen **212**. Touch screen **212** displays visual output to the user. The visual output includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodiments, some or all of the visual output correspond to user-interface objects.

Touch screen **212** has a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. Touch screen **212** and display controller **256** (along with any associated modules and/or sets of instructions in memory **202**) detect contact (and any

movement or breaking of the contact) on touch screen **212** and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on touch screen **212**. In an exemplary embodiment, a point of contact between touch screen **212** and the user corresponds to a finger of the user.

Touch screen **212** uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, or LED (light emitting diode) technology, although other display technologies may be used in other embodiments. Touch screen **212** and display controller **256** detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with touch screen **212**. In an exemplary embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPhone® and iPod Touch® from Apple Inc. of Cupertino, California.

A touch-sensitive display in some embodiments of touch screen **212** is analogous to the multi-touch sensitive touchpads described in the following U.S. Pat. No. 6,323,846 (Westerman et al.), U.S. Pat. No. 6,570,557 (Westerman et al.), and/or U.S. Pat. No. 6,677,932 (Westerman), and/or U.S. Patent Publication 2002/0015024A1, each of which is hereby incorporated by reference in its entirety. However, touch screen **212** displays visual output from device **200**, whereas touch-sensitive touchpads do not provide visual output.

A touch-sensitive display in some embodiments of touch screen **212** is as described in the following applications: (1) U.S. patent application Ser. No. 11/381,313, "Multipoint Touch Surface Controller," filed May 2, 2006; (2) U.S. patent application Ser. No. 10/840,862, "Multipoint Touchscreen," filed May 6, 2004; (3) U.S. patent application Ser. No. 10/903,964, "Gestures For Touch Sensitive Input Devices," filed Jul. 30, 2004; (4) U.S. patent application Ser. No. 11/048,264, "Gestures For Touch Sensitive Input Devices," filed Jan. 31, 2005; (5) U.S. patent application Ser. No. 11/038,590, "Mode-Based Graphical User Interfaces For Touch Sensitive Input Devices," filed Jan. 18, 2005; (6) U.S. patent application Ser. No. 11/228,758, "Virtual Input Device Placement On A Touch Screen User Interface," filed Sep. 16, 2005; (7) U.S. patent application Ser. No. 11/228,700, "Operation Of A Computer With A Touch Screen Interface," filed Sep. 16, 2005; (8) U.S. patent application Ser. No. 11/228,737, "Activating Virtual Keys Of A Touch-Screen Virtual Keyboard," filed Sep. 16, 2005; and (9) U.S. patent application Ser. No. 11/367,749, "Multi-Functional Hand-Held Device," filed Mar. 3, 2006. All of these applications are incorporated by reference herein in their entirety.

Touch screen **212** has, for example, a video resolution in excess of 100 dpi. In some embodiments, the touch screen has a video resolution of approximately 160 dpi. The user makes contact with touch screen **212** using any suitable object or appendage, such as a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures, which can be less precise than stylus-based input due to the larger area of contact of a finger on the touch screen. In some embodiments, the device translates the rough finger-based input into a precise pointer/cursor position or command for performing the actions desired by the user.

In some embodiments, in addition to the touch screen, device **200** includes a touchpad (not shown) for activating or

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deactivating particular functions. In some embodiments, the touchpad is a touch-sensitive area of the device that, unlike the touch screen, does not display visual output. The touchpad is a touch-sensitive surface that is separate from touch screen 212 or an extension of the touch-sensitive surface formed by the touch screen.

Device 200 also includes power system 262 for powering the various components. Power system 262 includes a power management system, one or more power sources (e.g., battery, alternating current (AC)), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in portable devices.

Device 200 also includes one or more optical sensors 264. FIG. 2A shows an optical sensor coupled to optical sensor controller 258 in I/O subsystem 206. Optical sensor 264 includes charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. Optical sensor 264 receives light from the environment, projected through one or more lenses, and converts the light to data representing an image. In conjunction with imaging module 243 (also called a camera module), optical sensor 264 captures still images or video. In some embodiments, an optical sensor is located on the back of device 200, opposite touch screen display 212 on the front of the device so that the touch screen display is used as a viewfinder for still and/or video image acquisition. In some embodiments, an optical sensor is located on the front of the device so that the user's image is obtained for video conferencing while the user views the other video conference participants on the touch screen display. In some embodiments, the position of optical sensor 264 can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a single optical sensor 264 is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

Device 200 optionally also includes one or more contact intensity sensors 265. FIG. 2A shows a contact intensity sensor coupled to intensity sensor controller 259 in I/O subsystem 206. Contact intensity sensor 265 optionally includes one or more piezoresistive strain gauges, capacitive force sensors, electric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor 265 receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system 212). In some embodiments, at least one contact intensity sensor is located on the back of device 200, opposite touch screen display 212, which is located on the front of device 200.

Device 200 also includes one or more proximity sensors 266. FIG. 2A shows proximity sensor 266 coupled to peripherals interface 218. Alternately, proximity sensor 266 is coupled to input controller 260 in I/O subsystem 206. Proximity sensor 266 is performed as described in U.S. patent application Ser. No. 11/241,839, "Proximity Detector In Handheld Device"; Ser. No. 11/240,788, "Proximity Detector In Handheld Device"; Ser. No. 11/620,702, "Using Ambient Light Sensor To Augment Proximity Sensor Output"; Ser. No. 11/586,862, "Automated Response To And Sensing Of User Activity In Portable Devices"; and Ser. No.

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11/638,251, "Methods And Systems For Automatic Configuration Of Peripherals," which are hereby incorporated by reference in their entirety. In some embodiments, the proximity sensor turns off and disables touch screen 212 when the multifunction device is placed near the user's ear (e.g., when the user is making a phone call).

Device 200 optionally also includes one or more tactile output generators 267. FIG. 2A shows a tactile output generator coupled to haptic feedback controller 261 in I/O subsystem 206. Tactile output generator 267 optionally includes one or more electroacoustic devices such as speakers or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity sensor 265 receives tactile feedback generation instructions from haptic feedback module 233 and generates tactile outputs on device 200 that are capable of being sensed by a user of device 200. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system 212) and, optionally, generates a tactile output by moving the touch-sensitive surface vertically (e.g., in/out of a surface of device 200) or laterally (e.g., back and forth in the same plane as a surface of device 200). In some embodiments, at least one tactile output generator sensor is located on the back of device 200, opposite touch screen display 212, which is located on the front of device 200.

Device 200 also includes one or more accelerometers 268. FIG. 2A shows accelerometer 268 coupled to peripherals interface 218. Alternately, accelerometer 268 is coupled to an input controller 260 in I/O subsystem 206. Accelerometer 268 performs, for example, as described in U.S. Patent Publication No. 20050190059, "Acceleration-based Theft Detection System for Portable Electronic Devices," and U.S. Patent Publication No. 20060017692, "Methods And Apparatuses For Operating A Portable Device Based On An Accelerometer," both of which are incorporated by reference herein in their entirety. In some embodiments, information is displayed on the touch screen display in a portrait view or a landscape view based on an analysis of data received from the one or more accelerometers. Device 200 optionally includes, in addition to accelerometer(s) 268, a magnetometer (not shown) and a GPS (or GLONASS or other global navigation system) receiver (not shown) for obtaining information concerning the location and orientation (e.g., portrait or landscape) of device 200.

In some embodiments, the software components stored in memory 202 include operating system 226, communication module (or set of instructions) 228, contact/motion module (or set of instructions) 230, graphics module (or set of instructions) 232, text input module (or set of instructions) 234, Global Positioning System (GPS) module (or set of instructions) 235, Digital Assistant Client Module 229, and applications (or sets of instructions) 236. Further, memory 202 stores data and models, such as user data and models 231. Furthermore, in some embodiments, memory 202 (FIG. 2A) or 470 (FIG. 4) stores device/global internal state 257, as shown in FIGS. 2A and 4. Device/global internal state 257 includes one or more of: active application state, indicating which applications, if any, are currently active; display state, indicating what applications, views or other information occupy various regions of touch screen display 212; sensor state, including information obtained from the device's

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various sensors and input control devices **216**; and location information concerning the device's location and/or attitude.

Operating system **226** (e.g., Darwin, RTXC, LINUX, UNIX, OS X, iOS, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

Communication module **228** facilitates communication with other devices over one or more external ports **224** and also includes various software components for handling data received by RF circuitry **208** and/or external port **224**. External port **224** (e.g., Universal Serial Bus (USB), FIRE-WIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.). In some embodiments, the external port is a multi-pin (e.g., 30-pin) connector that is the same as, or similar to and/or compatible with, the 30-pin connector used on iPod® (trademark of Apple Inc.) devices.

Contact/motion module **230** optionally detects contact with touch screen **212** (in conjunction with display controller **256**) and other touch-sensitive devices (e.g., a touchpad or physical click wheel). Contact/motion module **230** includes various software components for performing various operations related to detection of contact, such as determining if contact has occurred (e.g., detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and determining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module **230** receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., "multitouch"/multiple finger contacts). In some embodiments, contact/motion module **230** and display controller **256** detect contact on a touchpad.

In some embodiments, contact/motion module **230** uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has "clicked" on an icon). In some embodiments, at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation thresholds of particular physical actuators and can be adjusted without changing the physical hardware of device **200**). For example, a mouse "click" threshold of a trackpad or touch screen display can be set to any of a large range of predefined threshold values without changing the trackpad or touch screen display hardware. Additionally, in some implementations, a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click "intensity" parameter).

Contact/motion module **230** optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts).

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Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap gesture includes detecting a finger-down event followed by detecting a finger-up (liftoff) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and subsequently followed by detecting a finger-up (liftoff) event.

Graphics module **232** includes various known software components for rendering and displaying graphics on touch screen **212** or other display, including components for changing the visual impact (e.g., brightness, transparency, saturation, contrast, or other visual property) of graphics that are displayed. As used herein, the term "graphics" includes any object that can be displayed to a user, including, without limitation, text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, animations, and the like.

In some embodiments, graphics module **232** stores data representing graphics to be used. Each graphic is, optionally, assigned a corresponding code. Graphics module **232** receives, from applications etc., one or more codes specifying graphics to be displayed along with, if necessary, coordinate data and other graphic property data, and then generates screen image data to output to display controller **256**.

Haptic feedback module **233** includes various software components for generating instructions used by tactile output generator(s) **267** to produce tactile outputs at one or more locations on device **200** in response to user interactions with device **200**.

Text input module **234**, which is, in some examples, a component of graphics module **232**, provides soft keyboards for entering text in various applications (e.g., contacts module **237**, email e-mail client module **240**, instant messaging (IM) module **241**, browser module **247**, and any other application that needs text input).

GPS module **235** determines the location of the device and provides this information for use in various applications (e.g., to telephone module **238** for use in location-based dialing; to camera module **243** as picture/video metadata; and to applications that provide location-based services such as weather widgets, local yellow page widgets, and map/navigation widgets).

Digital assistant client module **229** includes various client-side digital assistant instructions to provide the client-side functionalities of the digital assistant. For example, digital assistant client module **229** is capable of accepting voice input (e.g., speech input), text input, touch input, and/or gestural input through various user interfaces (e.g., microphone **213**, accelerometer(s) **268**, touch-sensitive display system **212**, optical sensor(s) **264**, other input control devices **216**, etc.) of portable multifunction device **200**. Digital assistant client module **229** is also capable of providing output in audio (e.g., speech output), visual, and/or tactile forms through various output interfaces (e.g., speaker **211**, touch-sensitive display system **212**, tactile output generator(s) **267**, etc.) of portable multifunction device **200**. For example, output is provided as voice, sound, alerts, text messages, menus, graphics, videos, animations, vibrations, and/or combinations of two or more of the above. During operation, digital assistant client module **229** communicates with DA server **106** using RF circuitry **208**.

User data and models **231** include various data associated with the user (e.g., user-specific vocabulary data, user pref-

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erence data, user-specified name pronunciations, data from the user's electronic address book, to-do lists, shopping lists, etc.) to provide the client-side functionalities of the digital assistant. Further, user data and models **231** include various models (e.g., speech recognition models, statistical language models, natural language processing models, ontology, task flow models, service models, etc.) for processing user input and determining user intent.

In some examples, digital assistant client module **229** utilizes the various sensors, subsystems, and peripheral devices of portable multifunction device **200** to gather additional information from the surrounding environment of the portable multifunction device **200** to establish a context associated with a user, the current user interaction, and/or the current user input. In some examples, digital assistant client module **229** provides the contextual information or a subset thereof with the user input to DA server **106** to help infer the user's intent. In some examples, the digital assistant also uses the contextual information to determine how to prepare and deliver outputs to the user. Contextual information is referred to as context data.

In some examples, the contextual information that accompanies the user input includes sensor information, e.g., lighting, ambient noise, ambient temperature, images or videos of the surrounding environment, etc. In some examples, the contextual information can also include the physical state of the device, e.g., device orientation, device location, device temperature, power level, speed, acceleration, motion patterns, cellular signals strength, etc. In some examples, information related to the software state of DA server **106**, e.g., running processes, installed programs, past and present network activities, background services, error logs, resources usage, etc., and of portable multifunction device **200** is provided to DA server **106** as contextual information associated with a user input.

In some examples, the digital assistant client module **229** selectively provides information (e.g., user data **231**) stored on the portable multifunction device **200** in response to requests from DA server **106**. In some examples, digital assistant client module **229** also elicits additional input from the user via a natural language dialogue or other user interfaces upon request by DA server **106**. Digital assistant client module **229** passes the additional input to DA server **106** to help DA server **106** in intent deduction and/or fulfillment of the user's intent expressed in the user request.

A more detailed description of a digital assistant is described below with reference to FIGS. 7A-C. It should be recognized that digital assistant client module **229** can include any number of the sub-modules of digital assistant module **726** described below.

Applications **236** include the following modules (or sets of instructions), or a subset or superset thereof:

- Contacts module **237** (sometimes called an address book or contact list);
- Telephone module **238**;
- Video conference module **239**;
- E-mail client module **240**;
- Instant messaging (IM) module **241**;
- Workout support module **242**;
- Camera module **243** for still and/or video images;
- Image management module **244**;
- Video player module;
- Music player module;
- Browser module **247**;
- Calendar module **248**;
- Widget modules **249**, which includes, in some examples, one or more of: weather widget **249-1**, stocks widget

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249-2, calculator widget **249-3**, alarm clock widget **249-4**, dictionary widget **249-5**, and other widgets obtained by the user, as well as user-created widgets **249-6**;

Widget creator module **250** for making user-created widgets **249-6**;

Search module **251**;

Video and music player module **252**, which merges video player module and music player module;

Notes module **253**;

Map module **254**; and/or

Online video module **255**.

Examples of other applications **236** that are stored in memory **202** include other word processing applications, other image editing applications, drawing applications, presentation applications, JAVA-enabled applications, encryption, digital rights management, voice recognition, and voice replication.

In conjunction with touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, contacts module **237** are used to manage an address book or contact list (e.g., stored in application internal state **292** of contacts module **237** in memory **202** or memory **470**), including: adding name(s) to the address book; deleting name(s) from the address book; associating telephone number(s), e-mail address(es), physical address(es) or other information with a name; associating an image with a name; categorizing and sorting names; providing telephone numbers or e-mail addresses to initiate and/or facilitate communications by telephone module **238**, video conference module **239**, e-mail client module **240**, or IM module **241**; and so forth.

In conjunction with RF circuitry **208**, audio circuitry **210**, speaker **211**, microphone **213**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, telephone module **238** are used to enter a sequence of characters corresponding to a telephone number, access one or more telephone numbers in contacts module **237**, modify a telephone number that has been entered, dial a respective telephone number, conduct a conversation, and disconnect or hang up when the conversation is completed. As noted above, the wireless communication uses any of a plurality of communications standards, protocols, and technologies.

In conjunction with RF circuitry **208**, audio circuitry **210**, speaker **211**, microphone **213**, touch screen **212**, display controller **256**, optical sensor **264**, optical sensor controller **258**, contact/motion module **230**, graphics module **232**, text input module **234**, contacts module **237**, and telephone module **238**, video conference module **239** includes executable instructions to initiate, conduct, and terminate a video conference between a user and one or more other participants in accordance with user instructions.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, e-mail client module **240** includes executable instructions to create, send, receive, and manage e-mail in response to user instructions. In conjunction with image management module **244**, e-mail client module **240** makes it very easy to create and send e-mails with still or video images taken with camera module **243**.

In conjunction with RF circuitry **208**, touch screen **212**, display controller **256**, contact/motion module **230**, graphics module **232**, and text input module **234**, the instant messaging module **241** includes executable instructions to enter a sequence of characters corresponding to an instant mes-

sage, to modify previously entered characters, to transmit a respective instant message (for example, using a Short Message Service (SMS) or Multimedia Message Service (MMS) protocol for telephony-based instant messages or using XMPP, SIMPLE, or IMPS for Internet-based instant messages), to receive instant messages, and to view received instant messages. In some embodiments, transmitted and/or received instant messages include graphics, photos, audio files, video files and/or other attachments as are supported in an MMS and/or an Enhanced Messaging Service (EMS). As used herein, "instant messaging" refers to both telephony-based messages (e.g., messages sent using SMS or MMS) and Internet-based messages (e.g., messages sent using XMPP, SIMPLE, or IMPS).

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, GPS module 235, map module 254, and music player module, workout support module 242 includes executable instructions to create workouts (e.g., with time, distance, and/or calorie burning goals); communicate with workout sensors (sports devices); receive workout sensor data; calibrate sensors used to monitor a workout; select and play music for a workout; and display, store, and transmit workout data.

In conjunction with touch screen 212, display controller 256, optical sensor(s) 264, optical sensor controller 258, contact/motion module 230, graphics module 232, and image management module 244, camera module 243 includes executable instructions to capture still images or video (including a video stream) and store them into memory 202, modify characteristics of a still image or video, or delete a still image or video from memory 202.

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, and camera module 243, image management module 244 includes executable instructions to arrange, modify (e.g., edit), or otherwise manipulate, label, delete, present (e.g., in a digital slide show or album), and store still and/or video images.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, browser module 247 includes executable instructions to browse the Internet in accordance with user instructions, including searching, linking to, receiving, and displaying web pages or portions thereof, as well as attachments and other files linked to web pages.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, e-mail client module 240, and browser module 247, calendar module 248 includes executable instructions to create, display, modify, and store calendars and data associated with calendars (e.g., calendar entries, to-do lists, etc.) in accordance with user instructions.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, and browser module 247, widget modules 249 are mini-applications that can be downloaded and used by a user (e.g., weather widget 249-1, stocks widget 249-2, calculator widget 249-3, alarm clock widget 249-4, and dictionary widget 249-5) or created by the user (e.g., user-created widget 249-6). In some embodiments, a widget includes an HTML (Hypertext Markup Language) file, a CSS (Cascading Style Sheets) file, and a JavaScript file. In some embodiments, a widget includes an XML (Extensible Markup Language) file and a JavaScript file (e.g., Yahoo! Widgets).

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, and browser module 247, the widget creator module 250 are used by a user to create widgets (e.g., turning a user-specified portion of a web page into a widget).

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, search module 251 includes executable instructions to search for text, music, sound, image, video, and/or other files in memory 202 that match one or more search criteria (e.g., one or more user-specified search terms) in accordance with user instructions.

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, audio circuitry 210, speaker 211, RF circuitry 208, and browser module 247, video and music player module 252 includes executable instructions that allow the user to download and play back recorded music and other sound files stored in one or more file formats, such as MP3 or AAC files, and executable instructions to display, present, or otherwise play back videos (e.g., on touch screen 212 or on an external, connected display via external port 224). In some embodiments, device 200 optionally includes the functionality of an MP3 player, such as an iPod (trademark of Apple Inc.).

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, notes module 253 includes executable instructions to create and manage notes, to-do lists, and the like in accordance with user instructions.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, GPS module 235, and browser module 247, map module 254 are used to receive, display, modify, and store maps and data associated with maps (e.g., driving directions, data on stores and other points of interest at or near a particular location, and other location-based data) in accordance with user instructions.

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, audio circuitry 210, speaker 211, RF circuitry 208, text input module 234, e-mail client module 240, and browser module 247, online video module 255 includes instructions that allow the user to access, browse, receive (e.g., by streaming and/or download), play back (e.g., on the touch screen or on an external, connected display via external port 224), send an e-mail with a link to a particular online video, and otherwise manage online videos in one or more file formats, such as H.264. In some embodiments, instant messaging module 241, rather than e-mail client module 240, is used to send a link to a particular online video. Additional description of the online video application can be found in U.S. Provisional Patent Application No. 60/936,562, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Jun. 20, 2007, and U.S. patent application Ser. No. 11/968,067, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Dec. 31, 2007, the contents of which are hereby incorporated by reference in their entirety.

Each of the above-identified modules and applications corresponds to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these

modules can be combined or otherwise rearranged in various embodiments. For example, video player module can be combined with music player module into a single module (e.g., video and music player module **252**, FIG. 2A). In some embodiments, memory **202** stores a subset of the modules and data structures identified above. Furthermore, memory **202** stores additional modules and data structures not described above.

In some embodiments, device **200** is a device where operation of a predefined set of functions on the device is performed exclusively through a touch screen and/or a touchpad. By using a touch screen and/or a touchpad as the primary input control device for operation of device **200**, the number of physical input control devices (such as push buttons, dials, and the like) on device **200** is reduced.

The predefined set of functions that are performed exclusively through a touch screen and/or a touchpad optionally include navigation between user interfaces. In some embodiments, the touchpad, when touched by the user, navigates device **200** to a main, home, or root menu from any user interface that is displayed on device **200**. In such embodiments, a “menu button” is implemented using a touchpad. In some other embodiments, the menu button is a physical push button or other physical input control device instead of a touchpad.

FIG. 2B is a block diagram illustrating exemplary components for event handling in accordance with some embodiments. In some embodiments, memory **202** (FIG. 2A) or **470** (FIG. 4) includes event sorter **270** (e.g., in operating system **226**) and a respective application **236-1** (e.g., any of the aforementioned applications **237-251**, **255**, **480-490**).

Event sorter **270** receives event information and determines the application **236-1** and application view **291** of application **236-1** to which to deliver the event information. Event sorter **270** includes event monitor **271** and event dispatcher module **274**. In some embodiments, application **236-1** includes application internal state **292**, which indicates the current application view(s) displayed on touch-sensitive display **212** when the application is active or executing. In some embodiments, device/global internal state **257** is used by event sorter **270** to determine which application(s) is (are) currently active, and application internal state **292** is used by event sorter **270** to determine application views **291** to which to deliver event information.

In some embodiments, application internal state **292** includes additional information, such as one or more of: resume information to be used when application **236-1** resumes execution, user interface state information that indicates information being displayed or that is ready for display by application **236-1**, a state queue for enabling the user to go back to a prior state or view of application **236-1**, and a redo/undo queue of previous actions taken by the user.

Event monitor **271** receives event information from peripherals interface **218**. Event information includes information about a sub-event (e.g., a user touch on touch-sensitive display **212**, as part of a multi-touch gesture). Peripherals interface **218** transmits information it receives from I/O subsystem **206** or a sensor, such as proximity sensor **266**, accelerometer(s) **268**, and/or microphone **213** (through audio circuitry **210**). Information that peripherals interface **218** receives from I/O subsystem **206** includes information from touch-sensitive display **212** or a touch-sensitive surface.

In some embodiments, event monitor **271** sends requests to the peripherals interface **218** at predetermined intervals. In response, peripherals interface **218** transmits event infor-

mation. In other embodiments, peripherals interface **218** transmits event information only when there is a significant event (e.g., receiving an input above a predetermined noise threshold and/or for more than a predetermined duration).

In some embodiments, event sorter **270** also includes a hit view determination module **272** and/or an active event recognizer determination module **273**.

Hit view determination module **272** provides software procedures for determining where a sub-event has taken place within one or more views when touch-sensitive display **212** displays more than one view. Views are made up of controls and other elements that a user can see on the display.

Another aspect of the user interface associated with an application is a set of views, sometimes herein called application views or user interface windows, in which information is displayed and touch-based gestures occur. The application views (of a respective application) in which a touch is detected correspond to programmatic levels within a programmatic or view hierarchy of the application. For example, the lowest level view in which a touch is detected is called the hit view, and the set of events that are recognized as proper inputs is determined based, at least in part, on the hit view of the initial touch that begins a touch-based gesture.

Hit view determination module **272** receives information related to sub events of a touch-based gesture. When an application has multiple views organized in a hierarchy, hit view determination module **272** identifies a hit view as the lowest view in the hierarchy which should handle the sub-event. In most circumstances, the hit view is the lowest level view in which an initiating sub-event occurs (e.g., the first sub-event in the sequence of sub-events that form an event or potential event). Once the hit view is identified by the hit view determination module **272**, the hit view typically receives all sub-events related to the same touch or input source for which it was identified as the hit view.

Active event recognizer determination module **273** determines which view or views within a view hierarchy should receive a particular sequence of sub-events. In some embodiments, active event recognizer determination module **273** determines that only the hit view should receive a particular sequence of sub-events. In other embodiments, active event recognizer determination module **273** determines that all views that include the physical location of a sub-event are actively involved views, and therefore determines that all actively involved views should receive a particular sequence of sub-events. In other embodiments, even if touch sub-events were entirely confined to the area associated with one particular view, views higher in the hierarchy would still remain as actively involved views.

Event dispatcher module **274** dispatches the event information to an event recognizer (e.g., event recognizer **280**). In embodiments including active event recognizer determination module **273**, event dispatcher module **274** delivers the event information to an event recognizer determined by active event recognizer determination module **273**. In some embodiments, event dispatcher module **274** stores in an event queue the event information, which is retrieved by a respective event receiver **282**.

In some embodiments, operating system **226** includes event sorter **270**. Alternatively, application **236-1** includes event sorter **270**. In yet other embodiments, event sorter **270** is a stand-alone module, or a part of another module stored in memory **202**, such as contact/motion module **230**.

In some embodiments, application **236-1** includes a plurality of event handlers **290** and one or more application

views **291**, each of which includes instructions for handling touch events that occur within a respective view of the application's user interface. Each application view **291** of the application **236-1** includes one or more event recognizers **280**. Typically, a respective application view **291** includes a plurality of event recognizers **280**. In other embodiments, one or more of event recognizers **280** are part of a separate module, such as a user interface kit (not shown) or a higher level object from which application **236-1** inherits methods and other properties. In some embodiments, a respective event handler **290** includes one or more of: data updater **276**, object updater **277**, GUI updater **278**, and/or event data **279** received from event sorter **270**. Event handler **290** utilizes or calls data updater **276**, object updater **277**, or GUI updater **278** to update the application internal state **292**. Alternatively, one or more of the application views **291** include one or more respective event handlers **290**. Also, in some embodiments, one or more of data updater **276**, object updater **277**, and GUI updater **278** are included in a respective application view **291**.

A respective event recognizer **280** receives event information (e.g., event data **279**) from event sorter **270** and identifies an event from the event information. Event recognizer **280** includes event receiver **282** and event comparator **284**. In some embodiments, event recognizer **280** also includes at least a subset of: metadata **283**, and event delivery instructions **288** (which include sub-event delivery instructions).

Event receiver **282** receives event information from event sorter **270**. The event information includes information about a sub-event, for example, a touch or a touch movement. Depending on the sub-event, the event information also includes additional information, such as location of the sub-event. When the sub-event concerns motion of a touch, the event information also includes speed and direction of the sub-event. In some embodiments, events include rotation of the device from one orientation to another (e.g., from a portrait orientation to a landscape orientation, or vice versa), and the event information includes corresponding information about the current orientation (also called device attitude) of the device.

Event comparator **284** compares the event information to predefined event or sub-event definitions and, based on the comparison, determines an event or sub event, or determines or updates the state of an event or sub-event. In some embodiments, event comparator **284** includes event definitions **286**. Event definitions **286** contain definitions of events (e.g., predefined sequences of sub-events), for example, event **1** (**287-1**), event **2** (**287-2**), and others. In some embodiments, sub-events in an event (**287**) include, for example, touch begin, touch end, touch movement, touch cancellation, and multiple touching. In one example, the definition for event **1** (**287-1**) is a double tap on a displayed object. The double tap, for example, comprises a first touch (touch begin) on the displayed object for a predetermined phase, a first liftoff (touch end) for a predetermined phase, a second touch (touch begin) on the displayed object for a predetermined phase, and a second liftoff (touch end) for a predetermined phase. In another example, the definition for event **2** (**287-2**) is a dragging on a displayed object. The dragging, for example, comprises a touch (or contact) on the displayed object for a predetermined phase, a movement of the touch across touch-sensitive display **212**, and liftoff of the touch (touch end). In some embodiments, the event also includes information for one or more associated event handlers **290**.

In some embodiments, event definition **287** includes a definition of an event for a respective user-interface object. In some embodiments, event comparator **284** performs a hit test to determine which user-interface object is associated with a sub-event. For example, in an application view in which three user-interface objects are displayed on touch-sensitive display **212**, when a touch is detected on touch-sensitive display **212**, event comparator **284** performs a hit test to determine which of the three user-interface objects is associated with the touch (sub-event). If each displayed object is associated with a respective event handler **290**, the event comparator uses the result of the hit test to determine which event handler **290** should be activated. For example, event comparator **284** selects an event handler associated with the sub-event and the object triggering the hit test.

In some embodiments, the definition for a respective event (**287**) also includes delayed actions that delay delivery of the event information until after it has been determined whether the sequence of sub-events does or does not correspond to the event recognizer's event type.

When a respective event recognizer **280** determines that the series of sub-events do not match any of the events in event definitions **286**, the respective event recognizer **280** enters an event impossible, event failed, or event ended state, after which it disregards subsequent sub-events of the touch-based gesture. In this situation, other event recognizers, if any, that remain active for the hit view continue to track and process sub-events of an ongoing touch-based gesture.

In some embodiments, a respective event recognizer **280** includes metadata **283** with configurable properties, flags, and/or lists that indicate how the event delivery system should perform sub-event delivery to actively involved event recognizers. In some embodiments, metadata **283** includes configurable properties, flags, and/or lists that indicate how event recognizers interact, or are enabled to interact, with one another. In some embodiments, metadata **283** includes configurable properties, flags, and/or lists that indicate whether sub-events are delivered to varying levels in the view or programmatic hierarchy.

In some embodiments, a respective event recognizer **280** activates event handler **290** associated with an event when one or more particular sub-events of an event are recognized. In some embodiments, a respective event recognizer **280** delivers event information associated with the event to event handler **290**. Activating an event handler **290** is distinct from sending (and deferred sending) sub-events to a respective hit view. In some embodiments, event recognizer **280** throws a flag associated with the recognized event, and event handler **290** associated with the flag catches the flag and performs a predefined process.

In some embodiments, event delivery instructions **288** include sub-event delivery instructions that deliver event information about a sub-event without activating an event handler. Instead, the sub-event delivery instructions deliver event information to event handlers associated with the series of sub-events or to actively involved views. Event handlers associated with the series of sub-events or with actively involved views receive the event information and perform a predetermined process.

In some embodiments, data updater **276** creates and updates data used in application **236-1**. For example, data updater **276** updates the telephone number used in contacts module **237**, or stores a video file used in video player module. In some embodiments, object updater **277** creates and updates objects used in application **236-1**. For example, object updater **277** creates a new user-interface object or

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updates the position of a user-interface object. GUI updater 278 updates the GUI. For example, GUI updater 278 prepares display information and sends it to graphics module 232 for display on a touch-sensitive display.

In some embodiments, event handler(s) 290 includes or has access to data updater 276, object updater 277, and GUI updater 278. In some embodiments, data updater 276, object updater 277, and GUI updater 278 are included in a single module of a respective application 236-1 or application view 291. In other embodiments, they are included in two or more software modules.

It shall be understood that the foregoing discussion regarding event handling of user touches on touch-sensitive displays also applies to other forms of user inputs to operate multifunction devices 200 with input devices, not all of which are initiated on touch screens. For example, mouse movement and mouse button presses, optionally coordinated with single or multiple keyboard presses or holds; contact movements such as taps, drags, scrolls, etc. on touchpads; pen stylus inputs; movement of the device; oral instructions; detected eye movements; biometric inputs; and/or any combination thereof are optionally utilized as inputs corresponding to sub-events which define an event to be recognized.

FIG. 3 illustrates a portable multifunction device 200 having a touch screen 212 in accordance with some embodiments. The touch screen optionally displays one or more graphics within user interface (UI) 300. In this embodiment, as well as others described below, a user is enabled to select one or more of the graphics by making a gesture on the graphics, for example, with one or more fingers 302 (not drawn to scale in the figure) or one or more styluses 303 (not drawn to scale in the figure). In some embodiments, selection of one or more graphics occurs when the user breaks contact with the one or more graphics. In some embodiments, the gesture optionally includes one or more taps, one or more swipes (from left to right, right to left, upward and/or downward), and/or a rolling of a finger (from right to left, left to right, upward and/or downward) that has made contact with device 200. In some implementations or circumstances, inadvertent contact with a graphic does not select the graphic. For example, a swipe gesture that sweeps over an application icon optionally does not select the corresponding application when the gesture corresponding to selection is a tap.

Device 200 also includes one or more physical buttons, such as "home" or menu button 304. As described previously, menu button 304 is used to navigate to any application 236 in a set of applications that is executed on device 200. Alternatively, in some embodiments, the menu button is implemented as a soft key in a GUI displayed on touch screen 212.

In one embodiment, device 200 includes touch screen 212, menu button 304, push button 306 for powering the device on/off and locking the device, volume adjustment button(s) 308, subscriber identity module (SIM) card slot 310, headset jack 312, and docking/charging external port 224. Push button 306 is, optionally, used to turn the power on/off on the device by depressing the button and holding the button in the depressed state for a predefined time interval; to lock the device by depressing the button and releasing the button before the predefined time interval has elapsed; and/or to unlock the device or initiate an unlock process. In an alternative embodiment, device 200 also accepts verbal input for activation or deactivation of some functions through microphone 213. Device 200 also, optionally, includes one or more contact intensity sensors 265 for detecting intensity of contacts on touch screen 212 and/or

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one or more tactile output generators 267 for generating tactile outputs for a user of device 200.

FIG. 4 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments. Device 400 need not be portable. In some embodiments, device 400 is a laptop computer, a desktop computer, a tablet computer, a multimedia player device, a navigation device, an educational device (such as a child's learning toy), a gaming system, or a control device (e.g., a home or industrial controller). Device 400 typically includes one or more processing units (CPUs) 410, one or more network or other communications interfaces 460, memory 470, and one or more communication buses 420 for interconnecting these components. Communication buses 420 optionally include circuitry (sometimes called a chipset) that interconnects and controls communications between system components. Device 400 includes input/output (I/O) interface 430 comprising display 440, which is typically a touch screen display. I/O interface 430 also optionally includes a keyboard and/or mouse (or other pointing device) 450 and touchpad 455, tactile output generator 457 for generating tactile outputs on device 400 (e.g., similar to tactile output generator(s) 267 described above with reference to FIG. 2A), sensors 459 (e.g., optical, acceleration, proximity, touch-sensitive, and/or contact intensity sensors similar to contact intensity sensor(s) 265 described above with reference to FIG. 2A). Memory 470 includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and optionally includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory 470 optionally includes one or more storage devices remotely located from CPU(s) 410. In some embodiments, memory 470 stores programs, modules, and data structures analogous to the programs, modules, and data structures stored in memory 202 of portable multifunction device 200 (FIG. 2A), or a subset thereof. Furthermore, memory 470 optionally stores additional programs, modules, and data structures not present in memory 202 of portable multifunction device 200. For example, memory 470 of device 400 optionally stores drawing module 480, presentation module 482, word processing module 484, website creation module 486, disk authoring module 488, and/or spreadsheet module 490, while memory 202 of portable multifunction device 200 (FIG. 2A) optionally does not store these modules.

Each of the above-identified elements in FIG. 4 is, in some examples, stored in one or more of the previously mentioned memory devices. Each of the above-identified modules corresponds to a set of instructions for performing a function described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are combined or otherwise rearranged in various embodiments. In some embodiments, memory 470 stores a subset of the modules and data structures identified above. Furthermore, memory 470 stores additional modules and data structures not described above.

Attention is now directed towards embodiments of user interfaces that can be implemented on, for example, portable multifunction device 200.

FIG. 5A illustrates an exemplary user interface for a menu of applications on portable multifunction device 200 in accordance with some embodiments. Similar user interfaces

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are implemented on device **400**. In some embodiments, user interface **500** includes the following elements, or a subset or superset thereof:

Signal strength indicator(s) **502** for wireless communication(s), such as cellular and Wi-Fi signals;

Time **504**;

Bluetooth indicator **505**;

Battery status indicator **506**;

Tray **508** with icons for frequently used applications, such as:

Icon **516** for telephone module **238**, labeled "Phone," which optionally includes an indicator **514** of the number of missed calls or voicemail messages;

Icon **518** for e-mail client module **240**, labeled "Mail," which optionally includes an indicator **510** of the number of unread e-mails;

Icon **520** for browser module **247**, labeled "Browser;" and

Icon **522** for video and music player module **252**, also referred to as iPod (trademark of Apple Inc.) module **252**, labeled "iPod;" and

Icons for other applications, such as:

Icon **524** for IM module **241**, labeled "Messages;"

Icon **526** for calendar module **248**, labeled "Calendar;"

Icon **528** for image management module **244**, labeled "Photos;"

Icon **530** for camera module **243**, labeled "Camera;"

Icon **532** for online video module **255**, labeled "Online Video;"

Icon **534** for stocks widget **249-2**, labeled "Stocks;"

Icon **536** for map module **254**, labeled "Maps;"

Icon **538** for weather widget **249-1**, labeled "Weather;"

Icon **540** for alarm clock widget **249-4**, labeled "Clock;"

Icon **542** for workout support module **242**, labeled "Workout Support;"

Icon **544** for notes module **253**, labeled "Notes;" and

Icon **546** for a settings application or module, labeled "Settings;" which provides access to settings for device **200** and its various applications **236**.

It should be noted that the icon labels illustrated in FIG. 5A are merely exemplary. For example, icon **522** for video and music player module **252** is optionally labeled "Music" or "Music Player." Other labels are, optionally, used for various application icons. In some embodiments, a label for a respective application icon includes a name of an application corresponding to the respective application icon. In some embodiments, a label for a particular application icon is distinct from a name of an application corresponding to the particular application icon.

FIG. 5B illustrates an exemplary user interface on a device (e.g., device **400**, FIG. 4) with a touch-sensitive surface **551** (e.g., a tablet or touchpad **455**, FIG. 4) that is separate from the display **550** (e.g., touch screen display **212**). Device **400** also, optionally, includes one or more contact intensity sensors (e.g., one or more of sensors **459**) for detecting intensity of contacts on touch-sensitive surface **551** and/or one or more tactile output generators **457** for generating tactile outputs for a user of device **400**.

Although some of the examples which follow will be given with reference to inputs on touch screen display **212** (where the touch-sensitive surface and the display are combined), in some embodiments, the device detects inputs on a touch-sensitive surface that is separate from the display, as shown in FIG. 5B. In some embodiments, the touch-sensitive surface (e.g., **551** in FIG. 5B) has a primary axis (e.g., **552** in FIG. 5B) that corresponds to a primary axis (e.g., **553**

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in FIG. 5B) on the display (e.g., **550**). In accordance with these embodiments, the device detects contacts (e.g., contacts **560** and **562** in FIG. 5B) with the touch-sensitive surface **551** at locations that correspond to respective locations on the display (e.g., in FIG. 5B, contact **560** corresponds to **568** and contact **562** corresponds to **570**). In this way, user inputs (e.g., contacts **560** and **562**, and movements thereof) detected by the device on the touch-sensitive surface (e.g., **551** in FIG. 5B) are used by the device to manipulate the user interface on the display (e.g., **550** in FIG. 5B) of the multifunction device when the touch-sensitive surface is separate from the display. It should be understood that similar methods are, optionally, used for other user interfaces described herein.

Additionally, while the following examples are given primarily with reference to finger inputs (e.g., finger contacts, finger tap gestures, finger swipe gestures), it should be understood that, in some embodiments, one or more of the finger inputs are replaced with input from another input device (e.g., a mouse-based input or stylus input). For example, a swipe gesture is, optionally, replaced with a mouse click (e.g., instead of a contact) followed by movement of the cursor along the path of the swipe (e.g., instead of movement of the contact). As another example, a tap gesture is, optionally, replaced with a mouse click while the cursor is located over the location of the tap gesture (e.g., instead of detection of the contact followed by ceasing to detect the contact). Similarly, when multiple user inputs are simultaneously detected, it should be understood that multiple computer mice are, optionally, used simultaneously, or a mouse and finger contacts are, optionally, used simultaneously.

FIG. 6A illustrates exemplary personal electronic device **600**. Device **600** includes body **602**. In some embodiments, device **600** includes some or all of the features described with respect to devices **200** and **400** (e.g., FIGS. 2A-4). In some embodiments, device **600** has touch-sensitive display screen **604**, hereafter touch screen **604**. Alternatively, or in addition to touch screen **604**, device **600** has a display and a touch-sensitive surface. As with devices **200** and **400**, in some embodiments, touch screen **604** (or the touch-sensitive surface) has one or more intensity sensors for detecting intensity of contacts (e.g., touches) being applied. The one or more intensity sensors of touch screen **604** (or the touch-sensitive surface) provide output data that represents the intensity of touches. The user interface of device **600** responds to touches based on their intensity, meaning that touches of different intensities can invoke different user interface operations on device **600**.

Techniques for detecting and processing touch intensity are found, for example, in related applications: International Patent Application Serial No. PCT/US2013/040061, titled "Device, Method, and Graphical User Interface for Displaying User Interface Objects Corresponding to an Application," filed May 8, 2013, and International Patent Application Serial No. PCT/US2013/069483, titled "Device, Method, and Graphical User Interface for Transitioning Between Touch Input to Display Output Relationships," filed Nov. 11, 2013, each of which is hereby incorporated by reference in their entirety.

In some embodiments, device **600** has one or more input mechanisms **606** and **608**. Input mechanisms **606** and **608**, if included, are physical. Examples of physical input mechanisms include push buttons and rotatable mechanisms. In some embodiments, device **600** has one or more attachment mechanisms. Such attachment mechanisms, if included, can permit attachment of device **600** with, for example, hats,

eyewear, earrings, necklaces, shirts, jackets, bracelets, watch straps, chains, trousers, belts, shoes, purses, backpacks, and so forth. These attachment mechanisms permit device 600 to be worn by a user.

FIG. 6B depicts exemplary personal electronic device 600. In some embodiments, device 600 includes some or all of the components described with respect to FIGS. 2A, 2B, and 4. Device 600 has bus 612 that operatively couples I/O section 614 with one or more computer processors 616 and memory 618. I/O section 614 is connected to display 604, which can have touch-sensitive component 622 and, optionally, touch-intensity sensitive component 624. In addition, I/O section 614 is connected with communication unit 630 for receiving application and operating system data, using Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques. Device 600 includes input mechanisms 606 and/or 608. Input mechanism 606 is a rotatable input device or a depressible and rotatable input device, for example. Input mechanism 608 is a button, in some examples.

Input mechanism 608 is a microphone, in some examples. Personal electronic device 600 includes, for example, various sensors, such as GPS sensor 632, accelerometer 634, directional sensor 640 (e.g., compass), gyroscope 636, motion sensor 638, and/or a combination thereof, all of which are operatively connected to I/O section 614.

Memory 618 of personal electronic device 600 is a non-transitory computer-readable storage medium, for storing computer-executable instructions, which, when executed by one or more computer processors 616, for example, cause the computer processors to perform the techniques and processes described below. The computer-executable instructions, for example, are also stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. Personal electronic device 600 is not limited to the components and configuration of FIG. 6B, but can include other or additional components in multiple configurations.

As used here, the term “affordance” refers to a user-interactive graphical user interface object that is, for example, displayed on the display screen of devices 200, 400, 600, 900, 1000, and/or 1100 (FIGS. 2A, 4, 6A-6B, 9A-9B, 10A-10B, 11A-11B). For example, an image (e.g., icon), a button, and text (e.g., hyperlink) each constitutes an affordance.

As used herein, the term “focus selector” refers to an input element that indicates a current part of a user interface with which a user is interacting. In some implementations that include a cursor or other location marker, the cursor acts as a “focus selector” so that when an input (e.g., a press input) is detected on a touch-sensitive surface (e.g., touchpad 455 in FIG. 4 or touch-sensitive surface 551 in FIG. 5B) while the cursor is over a particular user interface element (e.g., a button, window, slider or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations that include a touch screen display (e.g., touch-sensitive display system 212 in FIG. 2A or touch screen 212 in FIG. 5A) that enables direct interaction with user interface elements on the touch screen display, a detected contact on the touch screen acts as a “focus selector” so that when an input (e.g., a press input by the contact) is detected on the touch screen display at a location of a particular user interface element (e.g., a

button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations, focus is moved from one region of a user interface to another region of the user interface without corresponding movement of a cursor or movement of a contact on a touch screen display (e.g., by using a tab key or arrow keys to move focus from one button to another button); in these implementations, the focus selector moves in accordance with movement of focus between different regions of the user interface. Without regard to the specific form taken by the focus selector, the focus selector is generally the user interface element (or contact on a touch screen display) that is controlled by the user so as to communicate the user’s intended interaction with the user interface (e.g., by indicating, to the device, the element of the user interface with which the user is intending to interact). For example, the location of a focus selector (e.g., a cursor, a contact, or a selection box) over a respective button while a press input is detected on the touch-sensitive surface (e.g., a touchpad or touch screen) will indicate that the user is intending to activate the respective button (as opposed to other user interface elements shown on a display of the device).

As used in the specification and claims, the term “characteristic intensity” of a contact refers to a characteristic of the contact based on one or more intensities of the contact. In some embodiments, the characteristic intensity is based on multiple intensity samples. The characteristic intensity is, optionally, based on a predefined number of intensity samples, or a set of intensity samples collected during a predetermined time period (e.g., 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10 seconds) relative to a predefined event (e.g., after detecting the contact, prior to detecting liftoff of the contact, before or after detecting a start of movement of the contact, prior to detecting an end of the contact, before or after detecting an increase in intensity of the contact, and/or before or after detecting a decrease in intensity of the contact). A characteristic intensity of a contact is, optionally based on one or more of: a maximum value of the intensities of the contact, a mean value of the intensities of the contact, an average value of the intensities of the contact, a top 10 percentile value of the intensities of the contact, a value at the half maximum of the intensities of the contact, a value at the 90 percent maximum of the intensities of the contact, or the like. In some embodiments, the duration of the contact is used in determining the characteristic intensity (e.g., when the characteristic intensity is an average of the intensity of the contact over time). In some embodiments, the characteristic intensity is compared to a set of one or more intensity thresholds to determine whether an operation has been performed by a user. For example, the set of one or more intensity thresholds includes a first intensity threshold and a second intensity threshold. In this example, a contact with a characteristic intensity that does not exceed the first threshold results in a first operation, a contact with a characteristic intensity that exceeds the first intensity threshold and does not exceed the second intensity threshold results in a second operation, and a contact with a characteristic intensity that exceeds the second threshold results in a third operation. In some embodiments, a comparison between the characteristic intensity and one or more thresholds is used to determine whether or not to perform one or more operations (e.g., whether to perform a respective operation or forgo performing the respective operation) rather than being used to determine whether to perform a first operation or a second operation.

In some embodiments, a portion of a gesture is identified for purposes of determining a characteristic intensity. For example, a touch-sensitive surface receives a continuous swipe contact transitioning from a start location and reaching an end location, at which point the intensity of the contact increases. In this example, the characteristic intensity of the contact at the end location is based on only a portion of the continuous swipe contact, and not the entire swipe contact (e.g., only the portion of the swipe contact at the end location). In some embodiments, a smoothing algorithm is applied to the intensities of the swipe contact prior to determining the characteristic intensity of the contact. For example, the smoothing algorithm optionally includes one or more of: an unweighted sliding-average smoothing algorithm, a triangular smoothing algorithm, a median filter smoothing algorithm, and/or an exponential smoothing algorithm. In some circumstances, these smoothing algorithms eliminate narrow spikes or dips in the intensities of the swipe contact for purposes of determining a characteristic intensity.

The intensity of a contact on the touch-sensitive surface is characterized relative to one or more intensity thresholds, such as a contact-detection intensity threshold, a light press intensity threshold, a deep press intensity threshold, and/or one or more other intensity thresholds. In some embodiments, the light press intensity threshold corresponds to an intensity at which the device will perform operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, the deep press intensity threshold corresponds to an intensity at which the device will perform operations that are different from operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, when a contact is detected with a characteristic intensity below the light press intensity threshold (e.g., and above a nominal contact-detection intensity threshold below which the contact is no longer detected), the device will move a focus selector in accordance with movement of the contact on the touch-sensitive surface without performing an operation associated with the light press intensity threshold or the deep press intensity threshold. Generally, unless otherwise stated, these intensity thresholds are consistent between different sets of user interface figures.

An increase of characteristic intensity of the contact from an intensity below the light press intensity threshold to an intensity between the light press intensity threshold and the deep press intensity threshold is sometimes referred to as a “light press” input. An increase of characteristic intensity of the contact from an intensity below the deep press intensity threshold to an intensity above the deep press intensity threshold is sometimes referred to as a “deep press” input. An increase of characteristic intensity of the contact from an intensity below the contact-detection intensity threshold to an intensity between the contact-detection intensity threshold and the light press intensity threshold is sometimes referred to as detecting the contact on the touch-surface. A decrease of characteristic intensity of the contact from an intensity above the contact-detection intensity threshold to an intensity below the contact-detection intensity threshold is sometimes referred to as detecting liftoff of the contact from the touch-surface. In some embodiments, the contact-detection intensity threshold is zero. In some embodiments, the contact-detection intensity threshold is greater than zero.

In some embodiments described herein, one or more operations are performed in response to detecting a gesture that includes a respective press input or in response to detecting the respective press input performed with a respec-

tive contact (or a plurality of contacts), where the respective press input is detected based at least in part on detecting an increase in intensity of the contact (or plurality of contacts) above a press-input intensity threshold. In some embodiments, the respective operation is performed in response to detecting the increase in intensity of the respective contact above the press-input intensity threshold (e.g., a “down stroke” of the respective press input). In some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the press-input threshold (e.g., an “up stroke” of the respective press input).

In some embodiments, the device employs intensity hysteresis to avoid accidental inputs sometimes termed “jitter,” where the device defines or selects a hysteresis intensity threshold with a predefined relationship to the press-input intensity threshold (e.g., the hysteresis intensity threshold is X intensity units lower than the press-input intensity threshold or the hysteresis intensity threshold is 75%, 90%, or some reasonable proportion of the press-input intensity threshold). Thus, in some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the hysteresis intensity threshold that corresponds to the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the hysteresis intensity threshold (e.g., an “up stroke” of the respective press input). Similarly, in some embodiments, the press input is detected only when the device detects an increase in intensity of the contact from an intensity at or below the hysteresis intensity threshold to an intensity at or above the press-input intensity threshold and, optionally, a subsequent decrease in intensity of the contact to an intensity at or below the hysteresis intensity threshold, and the respective operation is performed in response to detecting the press input (e.g., the increase in intensity of the contact or the decrease in intensity of the contact, depending on the circumstances).

For ease of explanation, the descriptions of operations performed in response to a press input associated with a press-input intensity threshold or in response to a gesture including the press input are, optionally, triggered in response to detecting either: an increase in intensity of a contact above the press-input intensity threshold, an increase in intensity of a contact from an intensity below the hysteresis intensity threshold to an intensity above the press-input intensity threshold, a decrease in intensity of the contact below the press-input intensity threshold, and/or a decrease in intensity of the contact below the hysteresis intensity threshold corresponding to the press-input intensity threshold. Additionally, in examples where an operation is described as being performed in response to detecting a decrease in intensity of a contact below the press-input intensity threshold, the operation is, optionally, performed in response to detecting a decrease in intensity of the contact below a hysteresis intensity threshold corresponding to, and lower than, the press-input intensity threshold.

3. Digital Assistant System

FIG. 7A illustrates a block diagram of digital assistant system 700 in accordance with various examples. In some examples, digital assistant system 700 is implemented on a standalone computer system. In some examples, digital

assistant system **700** is distributed across multiple computers. In some examples, some of the modules and functions of the digital assistant are divided into a server portion and a client portion, where the client portion resides on one or more user devices (e.g., devices **104**, **122**, **200**, **400**, **600**, **900**, **1000**, and **1100**) and communicates with the server portion (e.g., server system **108**) through one or more networks, e.g., as shown in FIG. **1**. In some examples, digital assistant system **700** is an implementation of server system **108** (and/or DA server **106**) shown in FIG. **1**. It should be noted that digital assistant system **700** is only one example of a digital assistant system, and that digital assistant system **700** can have more or fewer components than shown, can combine two or more components, or can have a different configuration or arrangement of the components. The various components shown in FIG. **7A** are implemented in hardware, software instructions for execution by one or more processors, firmware, including one or more signal processing and/or application specific integrated circuits, or a combination thereof.

Digital assistant system **700** includes memory **702**, one or more processors **704**, input/output (I/O) interface **706**, and network communications interface **708**. These components can communicate with one another over one or more communication buses or signal lines **710**.

In some examples, memory **702** includes a non-transitory computer-readable medium, such as high-speed random access memory and/or a non-volatile computer-readable storage medium (e.g., one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices).

In some examples, I/O interface **706** couples input/output devices **716** of digital assistant system **700**, such as displays, keyboards, touch screens, and microphones, to user interface module **722**. I/O interface **706**, in conjunction with user interface module **722**, receives user inputs (e.g., voice input, keyboard inputs, touch inputs, etc.) and processes them accordingly. In some examples, e.g., when the digital assistant is implemented on a standalone user device, digital assistant system **700** includes any of the components and I/O communication interfaces described with respect to devices **200**, **400**, **600**, **900**, **1000**, or **1100** in FIGS. **2A**, **4**, **6A-6B**, **9A-9B**, **10A-10B**, **11A-11B**, respectively. In some examples, digital assistant system **700** represents the server portion of a digital assistant implementation, and can interact with the user through a client-side portion residing on a user device (e.g., devices **104**, **200**, **400**, **600**, **900**, **1000**, or **1100**).

In some examples, the network communications interface **708** includes wired communication port(s) **712** and/or wireless transmission and reception circuitry **714**. The wired communication port(s) receives and send communication signals via one or more wired interfaces, e.g., Ethernet, Universal Serial Bus (USB), FIREWIRE, etc. The wireless circuitry **714** receives and sends RF signals and/or optical signals from/to communications networks and other communications devices. The wireless communications use any of a plurality of communications standards, protocols, and technologies, such as GSM, EDGE, CDMA, TDMA, Bluetooth, Wi-Fi, VoIP, Wi-MAX, or any other suitable communication protocol. Network communications interface **708** enables communication between digital assistant system **700** with networks, such as the Internet, an intranet, and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN), and/or a metropolitan area network (MAN), and other devices.

In some examples, memory **702**, or the computer-readable storage media of memory **702**, stores programs, modules,

instructions, and data structures including all or a subset of: operating system **718**, communications module **720**, user interface module **722**, one or more applications **724**, and digital assistant module **726**. In particular, memory **702**, or the computer-readable storage media of memory **702**, stores instructions for performing the processes described below. One or more processors **704** execute these programs, modules, and instructions, and reads/writes from/to the data structures.

Operating system **718** (e.g., Darwin, RTXC, LINUX, UNIX, iOS, OS X, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communications between various hardware, firmware, and software components.

Communications module **720** facilitates communications between digital assistant system **700** with other devices over network communications interface **708**. For example, communications module **720** communicates with RF circuitry **208** of electronic devices such as devices **200**, **400**, and **600** shown in FIGS. **2A**, **4**, **6A-B**, respectively. Communications module **720** also includes various components for handling data received by wireless circuitry **714** and/or wired communications port **712**.

User interface module **722** receives commands and/or inputs from a user via I/O interface **706** (e.g., from a keyboard, touch screen, pointing device, controller, and/or microphone), and generate user interface objects on a display. User interface module **722** also prepares and delivers outputs (e.g., speech, sound, animation, text, icons, vibrations, haptic feedback, light, etc.) to the user via the I/O interface **706** (e.g., through displays, audio channels, speakers, touch-pads, etc.).

Applications **724** include programs and/or modules that are configured to be executed by one or more processors **704**. For example, if the digital assistant system is implemented on a standalone user device, applications **724** include user applications, such as games, a calendar application, a navigation application, or an email application. If digital assistant system **700** is implemented on a server, applications **724** include resource management applications, diagnostic applications, or scheduling applications, for example.

Memory **702** also stores digital assistant module **726** (or the server portion of a digital assistant). In some examples, digital assistant module **726** includes the following sub-modules, or a subset or superset thereof: input/output processing module **728**, speech-to-text (STT) processing module **730**, natural language processing module **732**, dialogue flow processing module **734**, task flow processing module **736**, service processing module **738**, and speech synthesis processing module **740**. Each of these modules has access to one or more of the following systems or data and models of the digital assistant module **726**, or a subset or superset thereof: ontology **760**, vocabulary index **744**, user data **748**, task flow models **754**, service models **756**, and ASR systems **758**.

In some examples, using the processing modules, data, and models implemented in digital assistant module **726**, the digital assistant can perform at least some of the following: converting speech input into text; identifying a user's intent expressed in a natural language input received from the user; actively eliciting and obtaining information needed to fully infer the user's intent (e.g., by disambiguating words,

games, intentions, etc.); determining the task flow for fulfilling the inferred intent; and executing the task flow to fulfill the inferred intent.

In some examples, as shown in FIG. 7B, I/O processing module 728 interacts with the user through I/O devices 716 in FIG. 7A or with a user device (e.g., devices 104, 200, 400, or 600) through network communications interface 708 in FIG. 7A to obtain user input (e.g., a speech input) and to provide responses (e.g., as speech outputs) to the user input. I/O processing module 728 optionally obtains contextual information associated with the user input from the user device, along with or shortly after the receipt of the user input. The contextual information includes user-specific data, vocabulary, and/or preferences relevant to the user input. In some examples, the contextual information also includes software and hardware states of the user device at the time the user request is received, and/or information related to the surrounding environment of the user at the time that the user request was received. In some examples, I/O processing module 728 also sends follow-up questions to, and receive answers from, the user regarding the user request. When a user request is received by I/O processing module 728 and the user request includes speech input, I/O processing module 728 forwards the speech input to STT processing module 730 (or speech recognizer) for speech-to-text conversions.

STT processing module 730 includes one or more ASR systems 758. The one or more ASR systems 758 can process the speech input that is received through I/O processing module 728 to produce a recognition result. Each ASR system 758 includes a front-end speech pre-processor. The front-end speech pre-processor extracts representative features from the speech input. For example, the front-end speech pre-processor performs a Fourier transform on the speech input to extract spectral features that characterize the speech input as a sequence of representative multi-dimensional vectors. Further, each ASR system 758 includes one or more speech recognition models (e.g., acoustic models and/or language models) and implements one or more speech recognition engines. Examples of speech recognition models include Hidden Markov Models, Gaussian-Mixture Models, Deep Neural Network Models, n-gram language models, and other statistical models. Examples of speech recognition engines include the dynamic time warping based engines and weighted finite-state transducers (WFST) based engines. The one or more speech recognition models and the one or more speech recognition engines are used to process the extracted representative features of the front-end speech pre-processor to produce intermediate recognitions results (e.g., phonemes, phonemic strings, and sub-words), and ultimately, text recognition results (e.g., words, word strings, or sequence of tokens). In some examples, the speech input is processed at least partially by a third-party service or on the user's device (e.g., device 104, 200, 400, or 600) to produce the recognition result. Once STT processing module 730 produces recognition results containing a text string (e.g., words, or sequence of words, or sequence of tokens), the recognition result is passed to natural language processing module 732 for intent deduction. In some examples, STT processing module 730 produces multiple candidate text representations of the speech input. Each candidate text representation is a sequence of words or tokens corresponding to the speech input. In some examples, each candidate text representation is associated with a speech recognition confidence score. Based on the speech recognition confidence scores, STT processing module 730 ranks the candidate text representations and provides the n-best (e.g., n

highest ranked) candidate text representation(s) to natural language processing module 732 for intent deduction, where n is a predetermined integer greater than zero. For example, in one example, only the highest ranked (n=1) candidate text representation is passed to natural language processing module 732 for intent deduction. In another example, the five highest ranked (n=5) candidate text representations are passed to natural language processing module 732 for intent deduction.

More details on the speech-to-text processing are described in U.S. Utility application Ser. No. 13/236,942 for "Consolidating Speech Recognition Results," filed on Sep. 20, 2011, the entire disclosure of which is incorporated herein by reference.

In some examples, STT processing module 730 includes and/or accesses a vocabulary of recognizable words via phonetic alphabet conversion module 731. Each vocabulary word is associated with one or more candidate pronunciations of the word represented in a speech recognition phonetic alphabet. In particular, the vocabulary of recognizable words includes a word that is associated with a plurality of candidate pronunciations. For example, the vocabulary includes the word "tomato" that is associated with the candidate pronunciations of /tə'meɪroʊ/ and /tə'matoʊ/. Further, vocabulary words are associated with custom candidate pronunciations that are based on previous speech inputs from the user. Such custom candidate pronunciations are stored in STT processing module 730 and are associated with a particular user via the user's profile on the device. In some examples, the candidate pronunciations for words are determined based on the spelling of the word and one or more linguistic and/or phonetic rules. In some examples, the candidate pronunciations are manually generated, e.g., based on known canonical pronunciations.

In some examples, the candidate pronunciations are ranked based on the commonness of the candidate pronunciation. For example, the candidate pronunciation /tə'meɪroʊ/ is ranked higher than /tə'matoʊ/, because the former is a more commonly used pronunciation (e.g., among all users, for users in a particular geographical region, or for any other appropriate subset of users). In some examples, candidate pronunciations are ranked based on whether the candidate pronunciation is a custom candidate pronunciation associated with the user. For example, custom candidate pronunciations are ranked higher than canonical candidate pronunciations. This can be useful for recognizing proper nouns having a unique pronunciation that deviates from canonical pronunciation. In some examples, candidate pronunciations are associated with one or more speech characteristics, such as geographic origin, nationality, or ethnicity. For example, the candidate pronunciation /tə'meɪroʊ/ is associated with the United States, whereas the candidate pronunciation /tə'matoʊ/ is associated with Great Britain. Further, the rank of the candidate pronunciation is based on one or more characteristics (e.g., geographic origin, nationality, ethnicity, etc.) of the user stored in the user's profile on the device. For example, it can be determined from the user's profile that the user is associated with the United States. Based on the user being associated with the United States, the candidate pronunciation /tə'meɪroʊ/ (associated with the United States) is ranked higher than the candidate pronunciation /tə'matoʊ/ (associated with Great Britain). In some examples, one of the ranked candidate pronunciations is selected as a predicted pronunciation (e.g., the most likely pronunciation).

When a speech input is received, STT processing module 730 is used to determine the phonemes corresponding to the speech input (e.g., using an acoustic model), and then

attempt to determine words that match the phonemes (e.g., using a language model). For example, if STT processing module 730 first identifies the sequence of phonemes /tə'meɪroʊ/ corresponding to a portion of the speech input, it can then determine, based on vocabulary index 744, that this sequence corresponds to the word "tomato."

In some examples, STT processing module 730 uses approximate matching techniques to determine words in an utterance. Thus, for example, the STT processing module 730 determines that the sequence of phonemes /tə'meɪroʊ/ corresponds to the word "tomato," even if that particular sequence of phonemes is not one of the candidate sequence of phonemes for that word.

Natural language processing module 732 ("natural language processor") of the digital assistant takes the n-best candidate text representation(s) ("word sequence(s)" or "token sequence(s)") generated by STT processing module 730, and attempts to associate each of the candidate text representations with one or more "actionable intents" recognized by the digital assistant. An "actionable intent" (or "user intent") represents a task that can be performed by the digital assistant, and can have an associated task flow implemented in task flow models 754. The associated task flow is a series of programmed actions and steps that the digital assistant takes in order to perform the task. The scope of a digital assistant's capabilities is dependent on the number and variety of task flows that have been implemented and stored in task flow models 754, or in other words, on the number and variety of "actionable intents" that the digital assistant recognizes. The effectiveness of the digital assistant, however, also depends on the assistant's ability to infer the correct "actionable intent(s)" from the user request expressed in natural language.

In some examples, in addition to the sequence of words or tokens obtained from STT processing module 730, natural language processing module 732 also receives contextual information associated with the user request, e.g., from I/O processing module 728. The natural language processing module 732 optionally uses the contextual information to clarify, supplement, and/or further define the information contained in the candidate text representations received from STT processing module 730. The contextual information includes, for example, user preferences, hardware, and/or software states of the user device, sensor information collected before, during, or shortly after the user request, prior interactions (e.g., dialogue) between the digital assistant and the user, and the like. As described herein, contextual information is, in some examples, dynamic, and changes with time, location, content of the dialogue, and other factors.

In some examples, the natural language processing is based on, e.g., ontology 760. Ontology 760 is a hierarchical structure containing many nodes, each node representing either an "actionable intent" or a "property" relevant to one or more of the "actionable intents" or other "properties." As noted above, an "actionable intent" represents a task that the digital assistant is capable of performing, i.e., it is "actionable" or can be acted on. A "property" represents a parameter associated with an actionable intent or a sub-aspect of another property. A linkage between an actionable intent node and a property node in ontology 760 defines how a parameter represented by the property node pertains to the task represented by the actionable intent node.

In some examples, ontology 760 is made up of actionable intent nodes and property nodes. Within ontology 760, each actionable intent node is linked to one or more property nodes either directly or through one or more intermediate

property nodes. Similarly, each property node is linked to one or more actionable intent nodes either directly or through one or more intermediate property nodes. For example, as shown in FIG. 7C, ontology 760 includes a "restaurant reservation" node (i.e., an actionable intent node). Property nodes "restaurant," "date/time" (for the reservation), and "party size" are each directly linked to the actionable intent node (i.e., the "restaurant reservation" node).

In addition, property nodes "cuisine," "price range," "phone number," and "location" are sub-nodes of the property node "restaurant," and are each linked to the "restaurant reservation" node (i.e., the actionable intent node) through the intermediate property node "restaurant." For another example, as shown in FIG. 7C, ontology 760 also includes a "set reminder" node (i.e., another actionable intent node). Property nodes "date/time" (for setting the reminder) and "subject" (for the reminder) are each linked to the "set reminder" node. Since the property "date/time" is relevant to both the task of making a restaurant reservation and the task of setting a reminder, the property node "date/time" is linked to both the "restaurant reservation" node and the "set reminder" node in ontology 760.

An actionable intent node, along with its linked property nodes, is described as a "domain." In the present discussion, each domain is associated with a respective actionable intent, and refers to the group of nodes (and the relationships there between) associated with the particular actionable intent. For example, ontology 760 shown in FIG. 7C includes an example of restaurant reservation domain 762 and an example of reminder domain 764 within ontology 760. The restaurant reservation domain includes the actionable intent node "restaurant reservation," property nodes "restaurant," "date/time," and "party size," and sub-property nodes "cuisine," "price range," "phone number," and "location." Reminder domain 764 includes the actionable intent node "set reminder," and property nodes "subject" and "date/time." In some examples, ontology 760 is made up of many domains. Each domain shares one or more property nodes with one or more other domains. For example, the "date/time" property node is associated with many different domains (e.g., a scheduling domain, a travel reservation domain, a movie ticket domain, etc.), in addition to restaurant reservation domain 762 and reminder domain 764.

While FIG. 7C illustrates two example domains within ontology 760, other domains include, for example, "find a movie," "initiate a phone call," "find directions," "schedule a meeting," "send a message," and "provide an answer to a question," "read a list," "providing navigation instructions," "provide instructions for a task" and so on. A "send a message" domain is associated with a "send a message" actionable intent node, and further includes property nodes such as "recipient(s)," "message type," and "message body." The property node "recipient" is further defined, for example, by the sub-property nodes such as "recipient name" and "message address."

In some examples, ontology 760 includes all the domains (and hence actionable intents) that the digital assistant is capable of understanding and acting upon. In some examples, ontology 760 is modified, such as by adding or removing entire domains or nodes, or by modifying relationships between the nodes within the ontology 760.

In some examples, nodes associated with multiple related actionable intents are clustered under a "super domain" in ontology 760. For example, a "travel" super-domain includes a cluster of property nodes and actionable intent nodes related to travel. The actionable intent nodes related to

travel includes “airline reservation,” “hotel reservation,” “car rental,” “get directions,” “find points of interest,” and so on. The actionable intent nodes under the same super domain (e.g., the “travel” super domain) have many property nodes in common. For example, the actionable intent nodes for “airline reservation,” “hotel reservation,” “car rental,” “get directions,” and “find points of interest” share one or more of the property nodes “start location,” “destination,” “departure date/time,” “arrival date/time,” and “party size.”

In some examples, each node in ontology 760 is associated with a set of words and/or phrases that are relevant to the property or actionable intent represented by the node. The respective set of words and/or phrases associated with each node are the so-called “vocabulary” associated with the node. The respective set of words and/or phrases associated with each node are stored in vocabulary index 744 in association with the property or actionable intent represented by the node. For example, returning to FIG. 7B, the vocabulary associated with the node for the property of “restaurant” includes words such as “food,” “drinks,” “cuisine,” “hungry,” “eat,” “pizza,” “fast food,” “meal,” and so on. For another example, the vocabulary associated with the node for the actionable intent of “initiate a phone call” includes words and phrases such as “call,” “phone,” “dial,” “ring,” “call this number,” “make a call to,” and so on. The vocabulary index 744 optionally includes words and phrases in different languages.

Natural language processing module 732 receives the candidate text representations (e.g., text string(s) or token sequence(s)) from STT processing module 730, and for each candidate representation, determines what nodes are implicated by the words in the candidate text representation. In some examples, if a word or phrase in the candidate text representation is found to be associated with one or more nodes in ontology 760 (via vocabulary index 744), the word or phrase “triggers” or “activates” those nodes. Based on the quantity and/or relative importance of the activated nodes, natural language processing module 732 selects one of the actionable intents as the task that the user intended the digital assistant to perform. In some examples, the domain that has the most “triggered” nodes is selected. In some examples, the domain having the highest confidence value (e.g., based on the relative importance of its various triggered nodes) is selected. In some examples, the domain is selected based on a combination of the number and the importance of the triggered nodes. In some examples, additional factors are considered in selecting the node as well, such as whether the digital assistant has previously correctly interpreted a similar request from a user.

User data 748 includes user-specific information, such as user-specific vocabulary, user preferences, user address, user’s default and secondary languages, user’s contact list, and other short-term or long-term information for each user. In some examples, natural language processing module 732 uses the user-specific information to supplement the information contained in the user input to further define the user intent. For example, for a user request “invite my friends to my birthday party,” natural language processing module 732 is able to access user data 748 to determine who the “friends” are and when and where the “birthday party” would be held, rather than requiring the user to provide such information explicitly in his/her request.

It should be recognized that in some examples, natural language processing module 732 is implemented using one or more machine learning mechanisms (e.g., neural networks). In particular, the one or more machine learning

mechanisms are configured to receive a candidate text representation and contextual information associated with the candidate text representation. Based on the candidate text representation and the associated contextual information, the one or more machine learning mechanisms are configured to determine intent confidence scores over a set of candidate actionable intents. Natural language processing module 732 can select one or more candidate actionable intents from the set of candidate actionable intents based on the determined intent confidence scores. In some examples, an ontology (e.g., ontology 760) is also used to select the one or more candidate actionable intents from the set of candidate actionable intents.

Other details of searching an ontology based on a token string are described in U.S. Utility application Ser. No. 12/341,743 for “Method and Apparatus for Searching Using An Active Ontology,” filed Dec. 22, 2008, the entire disclosure of which is incorporated herein by reference.

In some examples, once natural language processing module 732 identifies an actionable intent (or domain) based on the user request, natural language processing module 732 generates a structured query to represent the identified actionable intent. In some examples, the structured query includes parameters for one or more nodes within the domain for the actionable intent, and at least some of the parameters are populated with the specific information and requirements specified in the user request. For example, the user says “Make me a dinner reservation at a sushi place at 7.” In this case, natural language processing module 732 is able to correctly identify the actionable intent to be “restaurant reservation” based on the user input. According to the ontology, a structured query for a “restaurant reservation” domain includes parameters such as {Cuisine}, {Time}, {Date}, {Party Size}, and the like. In some examples, based on the speech input and the text derived from the speech input using STT processing module 730, natural language processing module 732 generates a partial structured query for the restaurant reservation domain, where the partial structured query includes the parameters {Cuisine=“Sushi”} and {Time=“7 pm”}. However, in this example, the user’s utterance contains insufficient information to complete the structured query associated with the domain. Therefore, other necessary parameters such as {Party Size} and {Date} are not specified in the structured query based on the information currently available. In some examples, natural language processing module 732 populates some parameters of the structured query with received contextual information. For example, in some examples, if the user requested a sushi restaurant “near me,” natural language processing module 732 populates a {location} parameter in the structured query with GPS coordinates from the user device.

In some examples, natural language processing module 732 identifies multiple candidate actionable intents for each candidate text representation received from STT processing module 730. Further, in some examples, a respective structured query (partial or complete) is generated for each identified candidate actionable intent. Natural language processing module 732 determines an intent confidence score for each candidate actionable intent and ranks the candidate actionable intents based on the intent confidence scores. In some examples, natural language processing module 732 passes the generated structured query (or queries), including any completed parameters, to task flow processing module 736 (“task flow processor”). In some examples, the structured query (or queries) for the m-best (e.g., m highest ranked) candidate actionable intents are provided to task flow processing module 736, where m is a predetermined

integer greater than zero. In some examples, the structured query (or queries) for the m-best candidate actionable intents are provided to task flow processing module 736 with the corresponding candidate text representation(s).

Other details of inferring a user intent based on multiple candidate actionable intents determined from multiple candidate text representations of a speech input are described in U.S. Utility application Ser. No. 14/298,725 for “System and Method for Inferring User Intent From Speech Inputs,” filed Jun. 6, 2014, the entire disclosure of which is incorporated herein by reference.

Task flow processing module 736 is configured to receive the structured query (or queries) from natural language processing module 732, complete the structured query, if necessary, and perform the actions required to “complete” the user’s ultimate request. In some examples, the various procedures necessary to complete these tasks are provided in task flow models 754. In some examples, task flow models 754 include procedures for obtaining additional information from the user and task flows for performing actions associated with the actionable intent.

As described above, in order to complete a structured query, task flow processing module 736 needs to initiate additional dialogue with the user in order to obtain additional information, and/or disambiguate potentially ambiguous utterances. When such interactions are necessary, task flow processing module 736 invokes dialogue flow processing module 734 to engage in a dialogue with the user. In some examples, dialogue flow processing module 734 determines how (and/or when) to ask the user for the additional information and receives and processes the user responses. The questions are provided to and answers are received from the users through I/O processing module 728. In some examples, dialogue flow processing module 734 presents dialogue output to the user via audio and/or visual output, and receives input from the user via spoken or physical (e.g., clicking) responses. Continuing with the example above, when task flow processing module 736 invokes dialogue flow processing module 734 to determine the “party size” and “date” information for the structured query associated with the domain “restaurant reservation,” dialogue flow processing module 734 generates questions such as “For how many people?” and “On which day?” to pass to the user. Once answers are received from the user, dialogue flow processing module 734 then populates the structured query with the missing information, or pass the information to task flow processing module 736 to complete the missing information from the structured query.

Once task flow processing module 736 has completed the structured query for an actionable intent, task flow processing module 736 proceeds to perform the ultimate task associated with the actionable intent. Accordingly, task flow processing module 736 executes the steps and instructions in the task flow model according to the specific parameters contained in the structured query. For example, the task flow model for the actionable intent of “restaurant reservation” includes steps and instructions for contacting a restaurant and actually requesting a reservation for a particular party size at a particular time. For example, using a structured query such as: {restaurant reservation, restaurant=ABC Café, date=3/12/2012, time=7 pm, party size=5}, task flow processing module 736 performs the steps of: (1) logging onto a server of the ABC Café or a restaurant reservation system such as OPENTABLE®, (2) entering the date, time, and party size information in a form on the website, (3) submitting the form, and (4) making a calendar entry for the reservation in the user’s calendar.

In some examples, task flow processing module 736 employs the assistance of service processing module 738 (“service processing module”) to complete a task requested in the user input or to provide an informational answer requested in the user input. For example, service processing module 738 acts on behalf of task flow processing module 736 to make a phone call, set a calendar entry, invoke a map search, invoke or interact with other user applications installed on the user device, and invoke or interact with third-party services (e.g., a restaurant reservation portal, a social networking website, a banking portal, etc.). In some examples, the protocols and application programming interfaces (API) required by each service are specified by a respective service model among service models 756. Service processing module 738 accesses the appropriate service model for a service and generates requests for the service in accordance with the protocols and APIs required by the service according to the service model.

For example, if a restaurant has enabled an online reservation service, the restaurant submits a service model specifying the necessary parameters for making a reservation and the APIs for communicating the values of the necessary parameter to the online reservation service. When requested by task flow processing module 736, service processing module 738 establishes a network connection with the online reservation service using the web address stored in the service model, and sends the necessary parameters of the reservation (e.g., time, date, party size) to the online reservation interface in a format according to the API of the online reservation service.

In some examples, natural language processing module 732, dialogue flow processing module 734, and task flow processing module 736 are used collectively and iteratively to infer and define the user’s intent, obtain information to further clarify and refine the user intent, and finally generate a response (i.e., an output to the user, or the completion of a task) to fulfill the user’s intent. The generated response is a dialogue response to the speech input that at least partially fulfills the user’s intent. Further, in some examples, the generated response is output as a speech output. In these examples, the generated response is sent to speech synthesis processing module 740 (e.g., speech synthesizer) where it can be processed to synthesize the dialogue response in speech form. In yet other examples, the generated response is data content relevant to satisfying a user request in the speech input.

In examples where task flow processing module 736 receives multiple structured queries from natural language processing module 732, task flow processing module 736 initially processes the first structured query of the received structured queries to attempt to complete the first structured query and/or execute one or more tasks or actions represented by the first structured query. In some examples, the first structured query corresponds to the highest ranked actionable intent. In other examples, the first structured query is selected from the received structured queries based on a combination of the corresponding speech recognition confidence scores and the corresponding intent confidence scores. In some examples, if task flow processing module 736 encounters an error during processing of the first structured query (e.g., due to an inability to determine a necessary parameter), the task flow processing module 736 can proceed to select and process a second structured query of the received structured queries that corresponds to a lower ranked actionable intent. The second structured query is selected, for example, based on the speech recognition confidence score of the corresponding candidate text repre-

sentation, the intent confidence score of the corresponding candidate actionable intent, a missing necessary parameter in the first structured query, or any combination thereof.

Speech synthesis processing module **740** is configured to synthesize speech outputs for presentation to the user. Speech synthesis processing module **740** synthesizes speech outputs based on text provided by the digital assistant. For example, the generated dialogue response is in the form of a text string. Speech synthesis processing module **740** converts the text string to an audible speech output. Speech synthesis processing module **740** uses any appropriate speech synthesis technique in order to generate speech outputs from text, including, but not limited to, concatenative synthesis, unit selection synthesis, diphone synthesis, domain-specific synthesis, formant synthesis, articulatory synthesis, hidden Markov model (HMM) based synthesis, and sinewave synthesis. In some examples, speech synthesis processing module **740** is configured to synthesize individual words based on phonemic strings corresponding to the words. For example, a phonemic string is associated with a word in the generated dialogue response. The phonemic string is stored in metadata associated with the word. Speech synthesis processing module **740** is configured to directly process the phonemic string in the metadata to synthesize the word in speech form.

In some examples, instead of (or in addition to) using speech synthesis processing module **740**, speech synthesis is performed on a remote device (e.g., the server system **108**), and the synthesized speech is sent to the user device for output to the user. For example, this can occur in some implementations where outputs for a digital assistant are generated at a server system. And because server systems generally have more processing power or resources than a user device, it is possible to obtain higher quality speech outputs than would be practical with client-side synthesis.

Additional details on digital assistants can be found in the U.S. Utility application Ser. No. 12/987,982, entitled "Intelligent Automated Assistant," filed Jan. 10, 2011, and U.S. Utility application Ser. No. 13/251,088, entitled "Generating and Processing Task Items That Represent Tasks to Perform," filed Sep. 30, 2011, the entire disclosures of which are incorporated herein by reference.

4. System and Process for Responding to User Requests

FIG. **8** illustrates a block diagram of digital assistant **800** for responding to user requests, according to various examples. In some examples, digital assistant **800** is implemented on one or more electronic devices (e.g., devices **104**, **122**, **200**, **400**, or **600**) and the modules and functions of digital assistant **800** may be distributed in any manner between the devices. In some examples, some of the modules and functions of digital assistant **800** are divided into a server portion and a client portion, where the client portion resides on one or more user devices (e.g., devices **104**, **122**, **200**, **400**, or **600**) and communicates with the server portion (e.g., server system **108**) through one or more networks, e.g., as shown in FIG. **1**. Digital assistant **800** is implemented using hardware, software, or a combination of hardware and software to carry out the principles discussed herein.

In some examples, the sub-modules or a subset or superset of the sub-modules described with regard to digital assistant **800** may include one or more of the components discussed above including components for automatic speech recognition, natural language processing, or speech to text capabilities.

It should be noted that digital assistant **800** is exemplary, and thus digital assistant **800** can have more or fewer components than shown, can combine two or more compo-

nents, or can have a different configuration or arrangement of the components. Further, although the below discussion describes functions being performed at a single component of digital assistant **800**, it is to be understood that such functions can be performed at other components of digital assistant **800** and that such functions can be performed at more than one component of digital assistant **800**.

FIGS. **9A**, **9B**, **10A**, **10B**, **11A**, and **11B** illustrate exemplary electronic devices **900**, **1000**, and **1100** for responding to a user request with digital assistant **800** based on context data, according to various examples. Each of FIGS. **9A**, **9B**, **10A**, **10B**, **11A**, and **11B** will be discussed alongside digital assistant **800**.

As shown in FIG. **8** digital assistant **800** includes natural language understanding (NLU) module **804**, and context gatherers **806** and **808**. Digital assistant **800** receives an utterance including user request **802** and provides user request **802** to NLU module **804** for processing to determine response **810** to user request **802**. In some examples, the utterance only includes user request **802**. For example, as shown in FIG. **9A**, utterance **903** of "show me pictures of my father and sister" is also user request **902** because the entirety of utterance **903** is user request **902**.

In some examples, the utterance includes speech inputs other than user request **802**. In some examples, the speech inputs other than user request **802** include a trigger phrase or other way to get the attention of digital assistant **800**. For example, as shown in FIG. **10A**, utterance **1003** of "Hey Siri, show me more like this," includes user request **1002** of "show me more like this," as well as the trigger phrase "Hey Siri." As another example, as shown in FIG. **11A**, utterance **1103** of "I need to find a gas station, add one to the route," includes user request **1102** of "add one to the route," as well as the speech input "I need to find a gas station," which signals digital assistant **800** that the user requires assistance.

After receiving user request **802**, NLU module **804** determines whether at least a portion of user request **802** is ambiguous. In some examples, determining whether at least a portion of user request **802** is ambiguous includes determining whether one or more words of the portion of user request **802** does not have an associated meaning. In particular, NLU module **804** determines whether one or more words of user request **802** cannot be linked to an object or assigned a particular value.

For example, when request **902** of "show me pictures of my father and sister" is received NLU module **804** may determine that while the words "father" and "sister" are understood, NLU module **804** (and possibly digital assistant **800**) does not understand who the user's father and sister are and thus determines that "father and sister" is ambiguous. Similarly, when request **1002** of "show me more like this," is received, NLU module **804** may not have a meaning associated with the term "this" and thus can determine that it is ambiguous. As another example, when request **1102** of "add one to the route," NLU module **804** may determine that "one" and "route" do not have associated meanings without further processing and thus are ambiguous.

In some examples, NLU module **804** receives user request **802** and determines that no portion of user request **802** is ambiguous. In particular, NLU module **804** may determine an associated meaning for every word of user request **802**. For example, when NLU module **804** receives the user request "Book me a reservation at La Grande Burrito at 7 PM this evening for two people," NLU module **804** determines a meaning for every term of the user request and indeed can determine the intent of the entire user request without any further processing. Accordingly, NLU module

804 does not determine that at least a portion of user request **802** and therefore does not request (e.g., retrieve) any context data.

In some examples, determining whether at least a portion of user request **802** is ambiguous includes determining whether the portion of user request **802** includes a deictic reference. A deictic reference is a word or phrase that ambiguously references something like an object, time, person, or place. Exemplary deictic references include but are not limited to that, this, here, there, then, those, them, he, she, etc. particularly when used with a question such as the questions “what is this?,” “where is that?,” and “who is he?” or with a request like “get me that,” “show me these,” or “find some of that.” Accordingly, as described above, NLU module **804** may evaluate request **1002** and determine that “this” in “show me more like this” is a deictic reference and thus is ambiguous.

In some examples, determining whether at least a portion of user request **802** is ambiguous includes determining whether the deictic reference has an associated meaning. In particular, some deictic references may have an associated meaning based on previous interactions between the user and digital assistant **800**, as discussed further below, or may have an associated meaning based on a portion of the utterance not included in user request **802**. For example, when utterance **1103** of “I need to find a gas station, add one to the route,” digital assistant **800** and NLU module **804** determines that “one” in user request **1102** likely refers to “gas station” based on utterance **1103** as a whole.

In some examples, the portion of user request **802** that is ambiguous is a word. For example, when analyzing user request **902**, the individual words “father” and “sister” may be ambiguous. Similarly, the word “this” of **1002** and the word “route” of **1102** may be ambiguous. In some examples, the portion of user request **802** that is ambiguous is a phrase. For example, while “one” may have an associated meaning as discussed above, “add one to this route” may be ambiguous without understanding the meaning of route. Similarly, while “father” and “sister” may be individually ambiguous they may also be ambiguous as the phrase “father and sister” which may have a different or additional meaning.

In some examples, the portion of user request **802** that is ambiguous is determined with a classifier of NLU module **804**. In particular, NLU module **804** may incorporate various classifiers within a unified architecture to handle discreet tasks using rule-based processing. Accordingly, one classifier of NLU module **804** can determine whether portions of user request **802** are ambiguous using rules that incorporate the determinations discussed above such as determining whether the portion includes a deictic reference and/or a word or phrase that does not have an associated meaning.

In response to determining that at least a portion of user request **802** is ambiguous, NLU module **804** determines a set of context data based on the ambiguous portion of user request **802**. In particular, NLU module **804** may determine a set of context data to retrieve based on factors such as the ease of obtaining the set of context data, the likelihood that the context data will resolve the ambiguity, whether similar context data has resolved similar ambiguities previously, and particular rules incorporated in NLU module **804** or a component of NLU module **804** (e.g., a classifier) that provide a hierarchy of context data. Accordingly, NLU module **804** may determine a first set of context data that will be helpful in resolving the ambiguity and retrieve the set of context data from context gatherers **806** and **808** as discussed further below.

Context data **807** and **809** is retrieved by context gatherers **806** and **808** and provided to NLU module **804** when NLU module **804** requests the context data. Context gatherers **806** and **808** can gather context data **807** and **809** continuously in real time and be ready to provide it to NLU module **804**, when NLU module **804** requests it. Accordingly, the processing time required to seek out specific context data is reduced and NLU module **804** may request one or more sets of context data from a variety of context data that context gatherers **806** and **808** have determined including the types of context data discussed below.

In some examples, context data **807** and **809** includes data determined from a screen of an electronic device. For example, context data **807** and **809** can include data representing pictures **904**, **906**, and **908**, as well as any other pictures being displayed on screen **901** of electronic device **900**. Further, context data **807** and **809** can include data representing objects of the individual pictures being displayed on screen **901** of electronic device **900**. Thus, context data **907** can include data representing objects **905**, **907**, **909**, and **911** (e.g., the people that are the subject of pictures **904**, **906**, and **908**). Similarly, context data **807** and **809** can include data representing picture **1004** displayed on screen **1001** of electronic device **1000** and object **1005** included in picture **1004**. As yet another example, context data **807** and **809** can include data representing map **1104** displayed on screen **1101** of electronic device **1100** and location **1105** included in map **1104**.

In some examples, context data **807** and **809** includes data determined based on an interaction history between the user and digital assistant **800**. For example, context data **807** and **809** can include a prior interaction between the user and digital assistant **800** in which the user requested that digital assistant **800** send a text message to their sister and the user provided the contact information of their sister. Thus, context data **807** and **809** can include data representing the contact information of the user’s sister as well as the relationship between the user and their sister.

Similarly, context data **807** and **809** can include a prior interaction between the user and digital assistant **800** in which the user provided the request “what kind of tree is this?” and digital assistant **800** determined and provided the answer “A Norwegian spruce tree.” Thus, context data **807** and **809** can include data representing the Norwegian spruce tree as well as identifiers indicating that object **1005** in picture **1004** is a Norwegian spruce tree.

In some examples, context data **807** and **809** include data determined based on data associated with one or more applications of the electronic device. For example, context data **807** and **809** can include data from the maps application that is creating and displaying map **1104** including route data, location data, and data of points of interest near electronic device **1100**. Context data **807** and **809** can also include data associated with other applications that are not being displayed by the electronic device such as a messaging application, a contacts application, a sports application etc. Thus, context data **807** and **809** can include data representing conversations between the user and other people, contact information, sports scores, or any other type of data that is associated with an application of the electronic device, whether the application is displayed or is not displayed.

In some examples, context data **807** and **809** include data determined based on a profile associated with the user. For example, context data **807** and **809** can include the name of the user, user preferences, relationships between the user and other people, and other information that the user has stored in a profile for digital assistant **800** to reference. Thus,

context data **807** and **809** can include the identity and/or names of those related to the user, what type of gas station the user prefers, the name of the user, and other similar data.

In some examples, context data **807** and **809** include data determined with one or more sensors of the electronic device. Data determined with the one or more sensors of the electronic device and represented as context data **807** and **809** can include location data such as a GPS location, latitude and longitude, pressure data, elevation data, speed data etc. which can be determined from sensors including GPS sensors, altimeters, barometers, accelerometers, etc.

In some examples, context gatherer **806** is associated with a first source of context data (e.g., context data **807**) and context gatherer **808** is associated with a second source of context data (e.g., context data **809**) different from the first source of context data. For example, context gatherer **806** may be associated with data determined from a screen of the electronic device and thus may be responsible for gathering context data based on the current display while context gatherer **808** may be associated with data determined from sources other than the screen of the electronic device. For example, context gatherer **808** may be associated with data determined based on a profile associated with the user and thus may be responsible for gathering context from the user profile. As another example, context gatherer **808** may be associated with data determined based on the current environment of the electronic device and thus may be responsible for gathering context from one or more sensors.

Thus, in the above examples, context data **807** can include images being provided on the screen of the electronic device, points of interest being displayed on the screen of the electronic device, a current route being displayed on the electronic device, an application being displayed on the electronic device, or any other data that can be displayed and thus accessed by context gatherer **806**. Similarly, context data **808** can include a nickname of the user, relationships associated with the user, context information associated with the user, location data such as a GPS location, latitude and longitude, pressure data, elevation data, speed data, or any other data that could be obtained from the sources described above.

In some examples, the source of context that each of context gatherer **806** and **808** are associated with is predetermined. For example, context gatherer **806** may be programmed to access on-screen data and thus will always focus and prioritize context data gathered from on-screen data. In some examples, the source of context that each of context gatherer **806** and **808** are associated with is dynamic and based on the current needs of the system and context data that is available. Thus, context gatherer **806** may determine that based on the current request on-screen data would likely be most helpful and thus gather context data **807** including data related to what is currently displayed. Similarly, context gatherer **808** may determine that new data related to the interaction history between the user and digital assistant **800** is available and thus gathers and provides context data **809** including the interaction history to the system.

In some examples, context gatherer **806** and context gatherer **808** are associated with the same source of context data and thus, each gather the data they can. Accordingly, context data **807** and context data **809** may include different data from the same source, the same data from the same source, different data from different sources, or any combination of data based on the current availability of context data for context gatherers **806** and **808** as well as the needs of the system.

Further, it will be recognized that the user of two context gatherers in the system is exemplary and the system could include any number of context gatherers needed to adequately process the context available to the system from the various sources. Thus, the system may add or use context gatherers as the processing needs of the request or task require and is not limited to only the number of context gatherers shown. For example, the system could include a third context gatherer associated with data determined from applications of the electronic device and thus may be responsible for monitoring open applications and gathering data from those applications including current tasks, parameters being used for those tasks, or any other data that the application may utilize.

Upon receiving context data **807** and **809** from context gatherers **806** and **808**, NLU module **804** extracts metadata from context data **807** and **809**. In particular, NLU module **804** processes and/or parses context data **807** and **809** to extract different objects and pieces of data as discrete pieces of metadata that can be further analyzed in isolation and in combination. For example, NLU module **804** can extract pictures **904**, **906**, and **908**, as well as objects **905**, **907**, **909**, and **911** from context data **807** and **809** as individual pieces of metadata. Similarly, NLU module **804** can extract a current location, a route, and end location **1105** from context data **807** and **809** as pieces of metadata.

In some examples, extracting metadata from context data **807** and **809** includes extracting metadata from objects of context data **807** and **809**. For example, in addition to extracting objects **905**, **907**, **909**, and **911** from context data **807** and **809**, NLU module **804** may extract metadata associated with those objects such as tags that indicate the identity of each of the objects. As another example, NLU module **804** may extract metadata associated with object **1005** indicating properties of a Norwegian spruce tree such as where the tree grows naturally or how long they typically live for. As yet another example, NLU module **804** may extract metadata associated with pictures **904**, **906**, **908**, and **1004** that indicate a time that the pictures were taken or received or a location at which the picture were taken.

In some examples, the metadata includes data representing a relationship between the user and other users or individuals. For example, as discussed above, the metadata can be extracted from objects **905**, **907**, **909**, and **911** that indicate an identity of the objects and possibly a relationship between the user and the people designated as objects **905**, **907**, **909**, and **911**. Similarly, the relationships between the user and various contacts stored in a phone or contact application can be extracted as metadata, in some cases in combination with extracting the identities of the people in pictures **904**, **906**, and **908**.

In some examples, the metadata includes data representing objects being displayed on a screen of the electronic device. For example, as discussed above, NLU module **804** can extract data representing pictures **904**, **906**, **908**, and **1004**, as well as objects **905**, **907**, **909**, **911**, and **1005**, and data representing map **1104** and location **1105**. Thus NLU module **804** can extract metadata representing each of these different entities being displayed on the displays **901**, **1001**, and **1101** of electronic devices **900**, **1000**, and **1100** for further processing to determine response **810**.

In some examples, the metadata includes data representing applications of the electronic device. For example, the metadata extracted by NLU module **804** can include points of interest near electronic device **1100** provided by the maps application that is installed on electronic device **1100**. Similarly, the metadata extracted by NLU module **804** can

include the contact information of various contacts stored in a contact application and data from messages stored in a text messaging or e-mail application.

In some examples, the metadata includes parameters of a previously requested or executed task. For example, the metadata can include the parameter “Norwegian spruce” provided to the user in response to the user’s prior request related to a type of tree. Similarly, the metadata can include dates determined in response to a request to schedule a meeting, or a size of coffee determined in response to a request to order a coffee.

In some examples, the metadata is extracted from context data **807** and **809** using a classifier of NLU module **804**. In particular, as discussed above, NLU module **804** can include one or more classifiers that implement rule-based determinations to select and extract various pieces of metadata that may be relevant to resolving the ambiguity of user request **802**. Accordingly, the classifier used to extract metadata may be the same classifier used to select context data **807** and **809** or it may be a different classifier.

NLU module **804** then determines response **810** to user request **802** based on the extracted metadata. In particular, NLU module **804** determines a meaning for the ambiguous portion of user request **802** and then determines a response after resolving that ambiguity using one or more pieces of metadata, as discussed further below.

In some examples, determining response **810** includes determining whether a meaning for the ambiguous portion of user request **802** can be determined based on the metadata extracted from context data **807**. For example, upon receiving user request **902** of “show me pictures of my father and sister” NLU module **804** will request context data **807** including data representing objects **905**, **907**, **909**, and **911** displayed on display **901** of electronic device **900**. NLU module **804** then extracts metadata associated with objects **905**, **907**, **909**, and **911** including tags of the identities of the people represented by objects **905**, **907**, **909**, and **911**. Based on this metadata NLU module **804** determines that the identities of the people is not enough to determine a meaning for “father and sister” (e.g., the ambiguous portion of user request **902**) because relationships between the identities and the user is needed in addition to the identities to determine the user’s father and sister.

As another example, upon receiving user request **1002** of “show me more like this” NLU module **804** requests context data **807** including data representing picture **1004** and object **1005** which can include that object **1005** is a Norwegian spruce. Metadata representing the type of tree can be extracted and NLU module **804** determines that this metadata alone is not enough to resolve the deictic reference “this” because there are other trees in picture **1004** and the type of tree alone does not allow NLU module **804** to resolve this ambiguity.

In contrast, in the example where NLU module **804** receives user request **1102** of “add one to the route,” NLU module **804** requests context data **807** including data representing map **1104** displayed on display **1101** of electronic device **1100**. NLU module **804** extracts metadata from map **1104** including metadata representing the current route. Accordingly, NLU module **804** may determine that “route” of user request **1104** is referencing the metadata representing the current route and thus can resolve the ambiguity. However, as discussed further below, NLU module **804** may request further context data (e.g., context data **807** or **809**) even if the ambiguity can be resolved to further determine response **810**.

If NLU module **804** cannot determine a meaning for the ambiguous portion of user request **802** with the metadata extracted from context data **807** then NLU module **804** requests additional context data or sets of context data including context data **809** from context gatherers **808** and extracts metadata from context data **809**. For example, NLU module **804** requests context data **809** including relationships between the user and the identities determined from objects **905**, **907**, **909**, and **911** from a contact application or a profile associated with the user profile. Based on context data **809**, NLU module **804** extracts metadata representing these relationships including the identity of the user’s sister and father. NLU module **804** can then determine whether the identities of people **905**, **907**, **909**, and **911** match the determined identities and use this to determine response **810**.

As another example, NLU module **804** requests context data **809** including conversation history between the user and digital assistant **800** which includes the previous user requests related to object **1005** (e.g., what kind of tree is this?). Accordingly, NLU module **804** extracts metadata representing that the user previously requested information about object **1005** and determines that “this” in user request **1002** is referring to the same tree that the user previously asked about. NLU module then uses this information to determine response **810** as described below.

In some examples, determining the meaning for the ambiguous portion of user request **802** includes determining whether a term or word of the ambiguous portion of user request **802** references an object being displayed on the display of the electronic device. In particular, as discussed above, NLU module **804** determines whether “father and sister” reference any of objects **905**, **907**, **909**, and **911** displayed on display **901**, whether “this” references object **1005** displayed on display **1001**, and whether “route” references map **1104** displayed on display **1101** based on metadata extracted from context data **807** and **809**.

In some examples, NLU module **804** determines whether a second set of context data, such as context data **809**, references a first set of context data, such as context data **807**. For example, NLU module **804** can receive context data **807** including data representing objects **905**, **907**, **909**, and **911** and context data **809** including data representing the relationships between the user and various identities. As discussed above, NLU module **804** then determines that the metadata representing those relationships matches the identities of the people represented as objects **905**, **907**, **909**, and **911**.

When context data **809** references context data **807**, the relationship between context data **809** and context data **807** is extracted as metadata that can then be used to determine response **810**. Thus, the matching of the identities stored as the users “father” and “sister” in context data **809** to the identities of the people represented as objects **905**, **907**, **909**, and **911** is extracted as metadata and used to determine **810**. In particular, the metadata representing these matching identities is used to search for all pictures tagged with these particular identities and provide them in user interface **912**, as shown in FIG. 9B.

In some examples, determining response **810** to user request **802** includes determining a task associated with user request **802** and determining a parameter for the task based on the extracted metadata. For example, as discussed above, when user request **902** of “show me pictures of my father and sister” is received NLU module **804** determines that the task is to search for specific pictures and that the parameter for the task is the user’s father and sister. NLU module **804**

further determines the identity of the user's father and sister based on the extracted metadata representing people tagged in the photos and the relationships between the user and various other people. Accordingly, NLU module **804** executes the task of searching for photos using the identities

As another example, when user request **1002** of "show me more like this" is received NLU module **804** may determine that the task is to search for more of "this." Based on the extracted metadata indicating that object **1005** was previously referenced by the user and digital assistant **800** NLU module **804** determines that object **1005** is the parameter for the searching task. Accordingly, NLU module **804** searches for pictures including Norwegian spruces and provides the pictures in user interface **1012**, as shown in FIG. **10B**.

As yet another example, when user request **1102** of "add one to the route," is received NLU module **804** determines based on the extracted metadata from map **1104** that the route is the current route the user is on. NLU module **804** further determines that the task is to add a gas station to the route based on the other portion of utterance **1103**. However, NLU module **804** requires further context data to find a gas station to add to the route as requested (e.g., the parameter). Accordingly, NLU module **804** extracts metadata from the context data received from the maps application that indicates points of interest along the route. Using this metadata, NLU module **804** determines a gas station along the route to use as the parameter and executes the task to add stop **1106** to user interface **1112** as shown in FIG. **11B**.

In some examples, determining response **810** to user request **802** includes determining relevancy scores for the metadata and ranking the metadata based on the relevancy scores. Thus, a first relevancy score is determined for a first metadata and a second relevancy score is determined for a second metadata. NLU module **804** then determines whether the first relevancy score or the second relevancy score is higher and determines a parameter for a task based on the metadata with the higher relevance score.

For example, NLU module **804** may determine a first relevancy score associated with metadata representing points of interest near the route provided by the maps application that is high because this information will be useful in satisfying the user's request. NLU module **804** further determines a second relevancy score associated with metadata representing gas stations found with a web browser that is low because the metadata does not take into account the user's current location. Accordingly, NLU module **804** will determine that the first relevancy score is higher than the second relevancy score and select the metadata representing points of interest near the user provided by the maps application. Using this selected metadata, NLU module **804** then determines a gas station along the route as discussed above and provides it in user interface **1112** as shown in FIG. **11B**.

In some examples, NLU module **804** determines a difference between the first relevancy score and the second relevancy score. NLU module **804** then determine whether the difference between the first relevancy score and the second relevancy score is below a predetermined threshold and if the difference between the first relevancy score and the second relevancy score is below the predetermined threshold, determines the parameter for the task based on the first metadata and the second metadata.

For example, NLU module **804** may determine that the difference between the first relevancy scores and the second relevancy scores in the example discussed above is relatively small because both provide various gas stations that

the user could navigate to. Accordingly, NLU module **804** determines that the difference is below the predetermined threshold and thus may select a gas station that appears in both sets of metadata, using both the first and second metadata to determine the parameter.

In some examples, determining response **810** to user request **802** includes providing the task and the parameter to an application of the electronic device and executing the task with the application using the parameter. For example, once the gas station is determined by NLU module **804** based on the extracted metadata, NLU module **804** provides the task and the parameter to the maps application and the maps application determines response **810** by executing the mapping task with the gas stations address or location.

In some examples, the application of the electronic device is an application that is open or being displayed on the display of the electronic device. For example, as discussed above, when response **810** is determined and user interface **1112** is shown in the maps application is currently open and in use as the user is driving to another location.

In some examples, determining response **810** to user request **802** includes executing the task with digital assistant **800** using the parameter. For example, when the task is determined to be searching for pictures as described above with reference to FIGS. **9A** and **10A**, digital assistant **800** may perform the search using the parameters determined by NLU module **804**. Accordingly, digital assistant **800** will perform a search for pictures with tags of the identities of the user's father and sister and a search for pictures including Norwegian spruces and provide the results in user interface **912** and user interface **1012**, respectively.

In some examples, response **810** is determined by NLU module **804** using a classifier of NLU module **804**. In particular, as discussed above, NLU module **804** may include one or more classifiers implementing rule-based decision making to determine context data, metadata, etc. Accordingly, one of these classifiers may implement rules to determine which metadata is the most relevant and determine parameters based on the selected metadata, as discussed above.

After response **810** is determined by NLU module **804**, response **810** is provided as an output. In some examples, the output includes the results of the executed task. For example, as discussed above, user interfaces **912**, **1012**, and **1112** are provided as response **810** including the results of searches for specific entities.

In some examples, the output is an audio output. For example, a received utterance may be "I need gas, where is the closest one?" NLU module **804** may process the request "where is the closest one?" the same as the requests discussed above and determine based on metadata extracted from the maps application that the nearest gas station is in 0.54 miles. Accordingly, NLU module **804** determines that response **810** should be an audio output providing the information "the nearest gas station is in 0.54 miles." This audio output is then provided to the user through a speaker of the electronic device.

In some examples, the output is an output displayed on the display of the electronic device. For example, the output can be user interfaces **912**, **1012**, or **1112** displayed on displays **901**, **1001**, and **1101** respectively.

In some examples, the output is both an audio output and displayed on the display of the electronic device. For example, when adding gas station **1106** to the map and providing user interface **1112** digital assistant **800** may also provide an audio output of "Gas station added to your route in 1.3 miles."

FIG. 12 illustrates process 1200 for responding to user requests, according to various examples, according to various examples. Process 1200 is performed, for example, using one or more electronic devices implementing a digital assistant. In some examples, process 1200 is performed using a client-server system (e.g., system 100), and the blocks of process 1200 are divided up in any manner between the server (e.g., DA server 106) and a client device. In other examples, the blocks of process 1200 are divided up between the server and multiple client devices (e.g., a mobile phone and a smart watch). Thus, while portions of process 1200 are described herein as being performed by particular devices of a client-server system, it will be appreciated that process 1200 is not so limited. In other examples, process 1200 is performed using only a client device (e.g., user device 104) or only multiple client devices. In process 1200, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the process 1200.

At step 1202 an utterance (e.g., utterance 903, 1003, 1103) including a user request (e.g., user request 802, 902, 1002, 1102) is received.

At step 1204 whether at least a portion of the user request (e.g., user request 802, 902, 1002, 1102) is ambiguous is determined. In some examples, determining whether at least a portion of the user request is ambiguous further comprises determining whether one or more words of the portion of the user request does not have an associated meaning. In some examples, determining whether at least a portion of the user request is ambiguous further comprises determining whether the portion of the user request includes one or more ambiguous words.

At step 1206, in response to determining that at least a portion of the user request (e.g., user request 802, 902, 1002, 1102) is ambiguous a set of context data (e.g., context data 807, 809) is determined based on the ambiguous portion of the user request. In some examples, the context data includes data determined from a screen (e.g., screen 901, 1001, 1101) of the electronic device (e.g., electronic device 900, 1000, 1100).

In some examples, the set of context data (e.g., context data 807, 809) is a first set of context data and determining the response (e.g., response 810, user interface 912, 1012, 1112) to the user request (e.g., user request 802, 902, 1002, 1102) based on the extracted metadata further comprises

At step 1208, a metadata is extracted from the context data (e.g., context data 807, 809). In some examples, extracting a metadata from the context data further comprises extracting a plurality of metadata from the context data, wherein each metadata of the plurality of metadata is extracted from an object (e.g., object 905, 907, 909, 911, 1005, 1105) of the context data.

In some examples, the metadata includes data representing relationships between a user and other individuals. In some examples, the metadata includes objects being displayed on the screen (e.g., display 901, 1001, 1101) of the electronic device (e.g., electronic device 900, 1000, 1100). In some examples, the metadata includes application data. In some examples, the metadata includes parameters of a previously requested task.

At step 1210, a response (e.g., response 810, user interface 912, 1012, 1112) to the user request (e.g., user request 802, 902, 1002, 1102) is determined based on the extracted metadata. In some examples, determining the response to the user request based on the extracted metadata further comprises determining a task associated with the user request

and determining a parameter for the task based on the extracted metadata. In some examples, determining the response to the user request based on the extracted metadata further comprises providing the task and the parameter to an application and executing the task with the application using the parameter. In some examples, the application is an application that is being displayed on the display of the electronic device.

In some examples, determining the response to the user request (e.g., user request 802, 902, 1002, 1102) based on the extracted metadata further comprises determining a first relevancy score associated with a first metadata of the plurality of metadata, determining a second relevancy score associated with a second meta data of the plurality of metadata, determining whether the first relevancy score is greater than the second relevancy score, and in accordance with a determination that the first relevancy score is greater than the second relevance score, determining the parameter for the task based on the first metadata.

In some examples, a difference between the first relevancy score and the second relevancy score is determined. In some examples, whether the difference between the first relevancy score and the second relevancy score is below a predetermined threshold is determined. In some examples, in accordance with a determination that the difference between the first relevancy score and the second relevancy score is below the predetermined threshold, the parameter for the task is determined based on the first metadata and the second metadata

In some examples, the set of context data (e.g., context data 807, 809) is a first set of context data and determining the response (e.g., response 810, user interface 912, 1012, 1112) to the user request (e.g., user request 802, 902, 1002, 1102) based on the extracted metadata further comprises determining whether a meaning for the ambiguous portion of the user request can be determined based on the metadata extracted from the first set of context data and in accordance with a determination that the meaning for the ambiguous portion of the user request cannot be determined based on the metadata extracted from the first set of context data, determining a second set of context data.

In some examples, the second set of context data (e.g., context data 807, 809) includes a prior interaction history between a user and a digital assistant (e.g., digital assistant 800). In some examples, the second set of context data includes a location of the electronic device (e.g., electronic device 900, 1000, 1100).

In some examples, determining whether a meaning for the ambiguous portion of the user request (e.g., user request 802, 902, 1002, 1102) can be determined based on the first set of context data (e.g., context data 807, 809) further comprises determining whether a term of the ambiguous portion of the user request references an object (e.g., object 905, 907, 909, 911, 1005, 1105) on the display (e.g., display 901, 1001, 1101) of the electronic device (e.g., electronic device 900, 1000, 1100).

In some examples, whether the second set of context data (e.g., context data 807, 809) references the first set of context data (e.g., context data 807, 809) is determined and in accordance with the determination that the second set of context data references the first set of context data, the relationship between the second set of context data and the first set of context data is extracted as metadata.

In some examples, an output of the response (e.g., response 810, user interface 912, 1012, 1112) is provided. In some examples, the output of the response to the user request includes a result of an executed task.

The operations described above with reference to FIG. 12 are optionally implemented by components depicted in FIGS. 1-4, 6A-6B, 7A-7C, 8, 9A-9B, 10A-10B, and 11A-11B. For example, the operations of process 800 may be implemented by digital assistant 800 on electronic devices 900, 1000, and 1100. It would be clear to a person having ordinary skill in the art how other processes are implemented based on the components depicted in FIGS. 1-4, 6A-6B, 7A-7C, 8, 9A-9B, 10A-10B, and 11A-11B.

In accordance with some implementations, a computer-readable storage medium (e.g., a non-transitory computer readable storage medium) is provided, the computer-readable storage medium storing one or more programs for execution by one or more processors of an electronic device, the one or more programs including instructions for performing any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises means for performing any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises a processing unit configured to perform any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises one or more processors and memory storing one or more programs for execution by the one or more processors, the one or more programs including instructions for performing any of the methods or processes described herein.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

Although the disclosure and examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims.

As described above, one aspect of the present technology is the gathering and use of data available from various sources to improve the delivery of responses to user requests. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter IDs, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to resolve ambiguities more efficiently and determine the appropriate response. Accord-

ingly, use of such personal information data enables users to calculated control of the delivered content. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of determining response, the present technology can be configured to allow users to select to "opt in" or "opt out" of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide personal data in context data. In yet another example, users can select to limit the length of personal data is maintained. In addition to providing "opt in" and "opt out" options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer

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needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, content can be selected and delivered to users by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as the content being requested by the device associated with a user, other non-personal information available to the content delivery services, or publicly available information.

What is claimed is:

1. A non-transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of an electronic device, the one or more programs including instructions for:

displaying, via a screen of the electronic device, a plurality of objects;

receiving an utterance including a user request;

determining whether at least a portion of the user request is ambiguous;

in response to determining that at least the portion of the user request is ambiguous:

determining a first set of context data based on the ambiguous portion of the user request, wherein the first set of context data includes data determined by a digital assistant from the plurality of objects displayed via the screen of the electronic device;

extracting a metadata from the first set of context data; and

determining a response to the user request based on the metadata extracted from the first set of context data, wherein determining the response comprises:

determining whether a meaning for the ambiguous portion of the user request can be determined based on the metadata extracted from the first set of context data; and

in accordance with a determination that the meaning for the ambiguous portion of the user request cannot be determined based on the metadata extracted from the first set of context data:

determining a second set of context data, wherein the second set of context data includes data determined by the digital assistant from sources other than the screen of the electronic device; generating the response to the user request based on a comparison of the metadata extracted from the first set of context data to another metadata extracted from the second set of context data; and

updating the display, via the screen of the electronic device, to display an object of the plurality of objects as the response to the user request, wherein the object is identified from the plurality of objects based on the comparison

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of the metadata extracted from the first set of context data to another metadata extracted from the second set of context data.

2. The non-transitory computer-readable storage medium of claim 1, wherein determining whether at least a portion of the user request is ambiguous further comprises determining whether one or more words of the portion of the user request does not have an associated meaning.

3. The non-transitory computer-readable storage medium of claim 1, wherein determining whether at least a portion of the user request is ambiguous further comprises determining whether the portion of the user request includes one or more ambiguous words.

4. The non-transitory computer-readable storage medium of claim 1, wherein the second set of context data includes a prior interaction history between a user and a digital assistant.

5. The non-transitory computer-readable storage medium of claim 1, wherein the second set of context data includes a location of the electronic device.

6. The non-transitory computer-readable storage medium of claim 1, wherein determining whether a meaning for the ambiguous portion of the user request can be determined based on the first set of context data further comprises:

determining whether a term of the ambiguous portion of the user request references an object on a display of the electronic device.

7. The non-transitory computer-readable storage medium of claim 1, the one or more programs further including instructions for:

determining whether the second set of context data references the first set of context data; and

in accordance with the determination that the second set of context data references the first set of context data, extracting the relationship between the second set of context data and the first set of context data as metadata.

8. The non-transitory computer-readable storage medium of claim 1, wherein extracting a metadata from the first set of context data further comprises:

extracting a plurality of metadata from the context data, wherein each metadata of the plurality of metadata is extracted from an object of the context data.

9. The non-transitory computer-readable storage medium of claim 1, wherein the metadata includes data representing relationships between a user and other individuals.

10. The non-transitory computer-readable storage medium of claim 1, wherein the metadata includes objects being displayed on a display of the electronic device.

11. The non-transitory computer-readable storage medium of claim 1, wherein the metadata includes application data.

12. The non-transitory computer-readable storage medium of claim 1, wherein the metadata includes parameters of a previously requested task.

13. The non-transitory computer-readable storage medium of claim 1, wherein determining the response to the user request based on the metadata extracted from the first set of context data further comprises:

determining a task associated with the user request; and determining a parameter for the task based on the extracted metadata.

14. The non-transitory computer-readable storage medium of claim 13, wherein determining the response to the user request based on the metadata extracted from the first set of context data further comprises:

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providing the task and the parameter to an application;
and
executing the task with the application using the parameter.

15. The non-transitory computer-readable storage medium of claim 14, wherein the application is an application that is being displayed on the display of the electronic device.

16. The non-transitory computer-readable storage medium of claim 13, wherein determining the response to the user request based on the metadata extracted from the first set of context data further comprises:

determining a first relevancy score associated with a first metadata of the plurality of metadata;

determining a second relevancy score associated with a second meta data of the plurality of metadata;

determining whether the first relevancy score is greater than the second relevancy score; and

in accordance with a determination that the first relevancy score is greater than the second relevancy score, determining the parameter for the task based on the first metadata.

17. The non-transitory computer-readable storage medium of claim 16, further comprising:

determining a difference between the first relevancy score and the second relevancy score;

determining whether the difference between the first relevancy score and the second relevancy score is below a predetermined threshold; and

in accordance with a determination that the difference between the first relevancy score and the second relevancy score is below the predetermined threshold, determining the parameter for the task based on the first metadata and the second metadata.

18. The non-transitory computer-readable storage medium of claim 1, the one or more programs further including instructions for:

providing an output of the response to the user request.

19. The non-transitory computer-readable storage medium of claim 18, wherein the output of the response to the user request includes a result of an executed task.

20. An electronic device comprising:

one or more processors;
a memory; and

one or more programs, wherein the one or more programs are stored in the memory and configured to be executed by the one or more processors, the one or more programs including instructions for:

displaying, via a screen of the electronic device, a plurality of objects;

receiving an utterance including a user request;

determining whether at least a portion of the user request is ambiguous;

in response to determining that at least the portion of the user request is ambiguous:

determining a first set of context data based on the ambiguous portion of the user request, wherein the first set of context data includes data determined by a digital assistant from the plurality of objects displayed via the screen of the electronic device; extracting a metadata from the first set of context data; and

determining a response to the user request based on the metadata extracted from the first set of context data,

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wherein determining the response comprises:

determining whether a meaning for the ambiguous portion of the user request can be determined based on the metadata extracted from the first set of context data; and

in accordance with a determination that the meaning for the ambiguous portion of the user request cannot be determined based on the metadata extracted from the first set of context data:

determining a second set of context data, wherein the second set of context data includes data determined by the digital assistant from sources other than the screen of the electronic device;

generating the response to the user request based on a comparison of the metadata extracted from the first set of context data to another metadata extracted from the second set of context data; and

updating the display, via the screen of the electronic device, to display an object of the plurality of objects as the response to the user request, wherein the object is identified from the plurality of objects based on the comparison of the metadata extracted from the first set of context data to another metadata extracted from the second set of context data.

21. A method, comprising:

at an electronic device with one or more processors and memory:

displaying, via a screen of the electronic device, a plurality of objects;

receiving an utterance including a user request;

determining whether at least a portion of the user request is ambiguous;

in response to determining that at least the portion of the user request is ambiguous:

determining a first set of context data based on the ambiguous portion of the user request, wherein the first set of context data includes data determined by a digital assistant from the plurality of objects displayed via the screen of the electronic device; extracting a metadata from the first set of context data; and

determining a response to the user request based on the metadata extracted from the first set of context data, wherein determining the response comprises:

determining whether a meaning for the ambiguous portion of the user request can be determined based on the metadata extracted from the first set of context data; and

in accordance with a determination that the meaning for the ambiguous portion of the user request cannot be determined based on the metadata extracted from the first set of context data:

determining a second set of context data, wherein the second set of context data includes data determined by the digital assistant from sources other than the screen of the electronic device;

generating the response to the user request based on a comparison of the metadata extracted from the first set of context data to another metadata extracted from the second set of context data; and

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updating the display, via the screen of the electronic device, to display an object of the plurality of objects as the response to the user request, wherein the object is identified from the plurality of objects based on the comparison of the metadata extracted from the first set of context data to another metadata extracted from the second set of context data.

22. The non-transitory computer-readable storage medium of claim 1, wherein a first classifier is used to determine whether at least the portion of the user request is ambiguous, and wherein a second classifier is used to extract the metadata from the context data.

23. The electronic device of claim 20, wherein determining whether at least a portion of the user request is ambiguous further comprises determining whether one or more words of the portion of the user request does not have an associated meaning.

24. The electronic device of claim 20, wherein determining whether at least a portion of the user request is ambiguous further comprises determining whether the portion of the user request includes one or more ambiguous words.

25. The electronic device of claim 20, wherein the second set of context data includes a prior interaction history between a user and a digital assistant.

26. The electronic device of claim 20, wherein the second set of context data includes a location of the electronic device.

27. The electronic device of claim 20, wherein determining whether a meaning for the ambiguous portion of the user request can be determined based on the first set of context data further comprises:

determining whether a term of the ambiguous portion of the user request references an object on a display of the electronic device.

28. The electronic device of claim 20, the one or more programs further including instructions for:

determining whether the second set of context data references the first set of context data; and

in accordance with the determination that the second set of context data references the first set of context data, extracting the relationship between the second set of context data and the first set of context data as metadata.

29. The electronic device of claim 20, wherein extracting a metadata from the first set of context data further comprises:

extracting a plurality of metadata from the context data, wherein each metadata of the plurality of metadata is extracted from an object of the context data.

30. The electronic device of claim 20, wherein the metadata includes at least one of data representing relationships between a user and other individuals, objects being displayed on a display of the electronic device, application data, and parameters of a previously requested task.

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31. The electronic device of claim 20, wherein determining the response to the user request based on the metadata extracted from the first set of context data further comprises: determining a task associated with the user request; and determining a parameter for the task based on the extracted metadata.

32. The electronic device of claim 20, the one or more programs further including instructions for: providing an output of the response to the user request.

33. The method of claim 21, wherein determining whether at least a portion of the user request is ambiguous further comprises determining whether one or more words of the portion of the user request does not have an associated meaning.

34. The method of claim 21, wherein determining whether at least a portion of the user request is ambiguous further comprises determining whether the portion of the user request includes one or more ambiguous words.

35. The method of claim 21, wherein the second set of context data includes a prior interaction history between a user and a digital assistant.

36. The method of claim 21, wherein the second set of context data includes a location of the electronic device.

37. The method of claim 21, wherein determining whether a meaning for the ambiguous portion of the user request can be determined based on the first set of context data further comprises:

determining whether a term of the ambiguous portion of the user request references an object on a display of the electronic device.

38. The method of claim 21, further comprising: determining whether the second set of context data references the first set of context data; and in accordance with the determination that the second set of context data references the first set of context data, extracting the relationship between the second set of context data and the first set of context data as metadata.

39. The method of claim 21, wherein extracting a metadata from the first set of context data further comprises:

extracting a plurality of metadata from the context data, wherein each metadata of the plurality of metadata is extracted from an object of the context data.

40. The method of claim 21, wherein the metadata includes at least one of data representing relationships between a user and other individuals, objects being displayed on a display of the electronic device, application data, and parameters of a previously requested task.

41. The method of claim 21, wherein determining the response to the user request based on the metadata extracted from the first set of context data further comprises:

determining a task associated with the user request; and determining a parameter for the task based on the extracted metadata.

42. The method of claim 21, further comprising: providing an output of the response to the user request.

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